

(12) **United States Patent**
Lin et al.

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(45) **Date of Patent:** **Aug. 12, 2025**

(54) **SEMICONDUCTOR STRUCTURE WITH
EXTENDED CONTACT STRUCTURE**

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Cheng**, Hsinchu (TW); **Hung-Li
Chiang**, Taipei (TW); **Shih-Syuan
Huang**, Taichung (TW); **Tzu-Chiang
Chen**, Hsinchu (TW); **I-Sheng Chen**,
Taipei (TW); **Sai-Hooi Yeong**, Zhubei
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(73) Assignee: **TAIWAN SEMICONDUCTOR
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LTD.**, Hsinchu (TW)

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patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/429,755**

(22) Filed: **Feb. 1, 2024**

(65) **Prior Publication Data**
US 2024/0170537 A1 May 23, 2024

Related U.S. Application Data
(60) Division of application No. 17/666,051, filed on Feb.
7, 2022, now Pat. No. 11,923,413, which is a division
(Continued)

(51) **Int. Cl.**
H01L 29/76 (2006.01)
H01L 21/02 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **H10D 62/121** (2025.01); **H01L 21/02603**
(2013.01); **H10D 30/024** (2025.01);
(Continued)

(58) **Field of Classification Search**
CPC H01L 21/02603; H01L 21/823431; H01L
21/823821; H01L 27/0886; H01L
27/0924;
(Continued)

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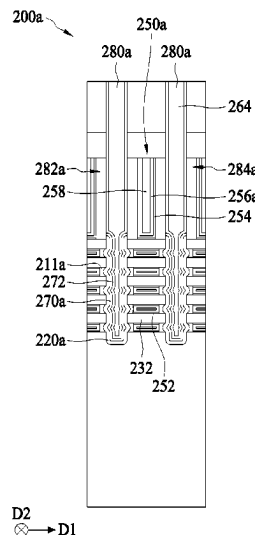
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19, 2021.

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(57) **ABSTRACT**
Semiconductor structures are provided. The semiconductor
structure includes a substrate and nanostructures formed
over the substrate. In addition, the nanostructures includes
channel regions and source/drain regions. The semiconduc-
tor structure further includes a gate structure vertically
sandwiched the channel regions of the nanostructures and a
contact wrapping around and vertically sandwiched between
the source/drain regions of the nanostructures.

20 Claims, 42 Drawing Sheets



Related U.S. Application Data

of application No. 16/868,625, filed on May 7, 2020, now Pat. No. 11,245,005, which is a continuation-in-part of application No. 16/681,097, filed on Nov. 12, 2019, now Pat. No. 11,183,560, which is a continuation of application No. 15/979,123, filed on May 14, 2018, now Pat. No. 10,522,622.

29/66795; H01L 29/785; H01L 21/823814; H01L 29/0653; H01L 29/0847; H01L 29/1079; H01L 29/78696; H01L 21/823412; H01L 21/823418; H01L 29/0669; H01L 29/41791; B82Y

10/00

USPC 257/288

See application file for complete search history.

(51) **Int. Cl.**

H01L 29/94 (2006.01)
H10D 30/01 (2025.01)
H10D 30/62 (2025.01)
H10D 30/67 (2025.01)
H10D 62/10 (2025.01)
H10D 64/01 (2025.01)
H10D 84/01 (2025.01)
H10D 84/03 (2025.01)
H10D 84/83 (2025.01)
H10D 84/85 (2025.01)

(52) **U.S. Cl.**

CPC **H10D 30/62** (2025.01); **H10D 30/6735** (2025.01); **H10D 64/017** (2025.01); **H10D 84/0158** (2025.01); **H10D 84/0193** (2025.01); **H10D 84/038** (2025.01); **H10D 84/834** (2025.01); **H10D 84/853** (2025.01)

(58) **Field of Classification Search**

CPC H01L 29/42392; H01L 29/66545; H01L

(56)

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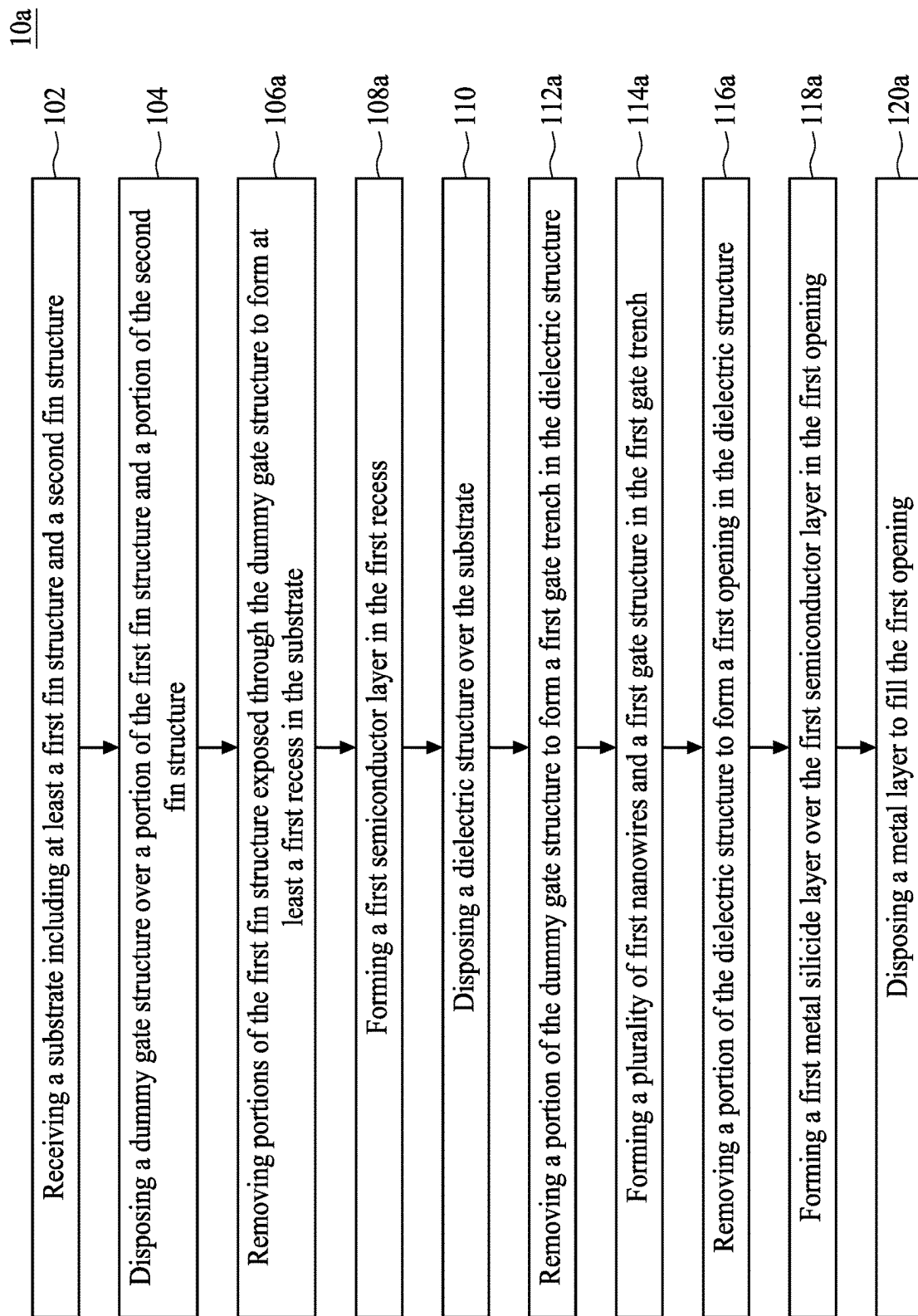


FIG. 1

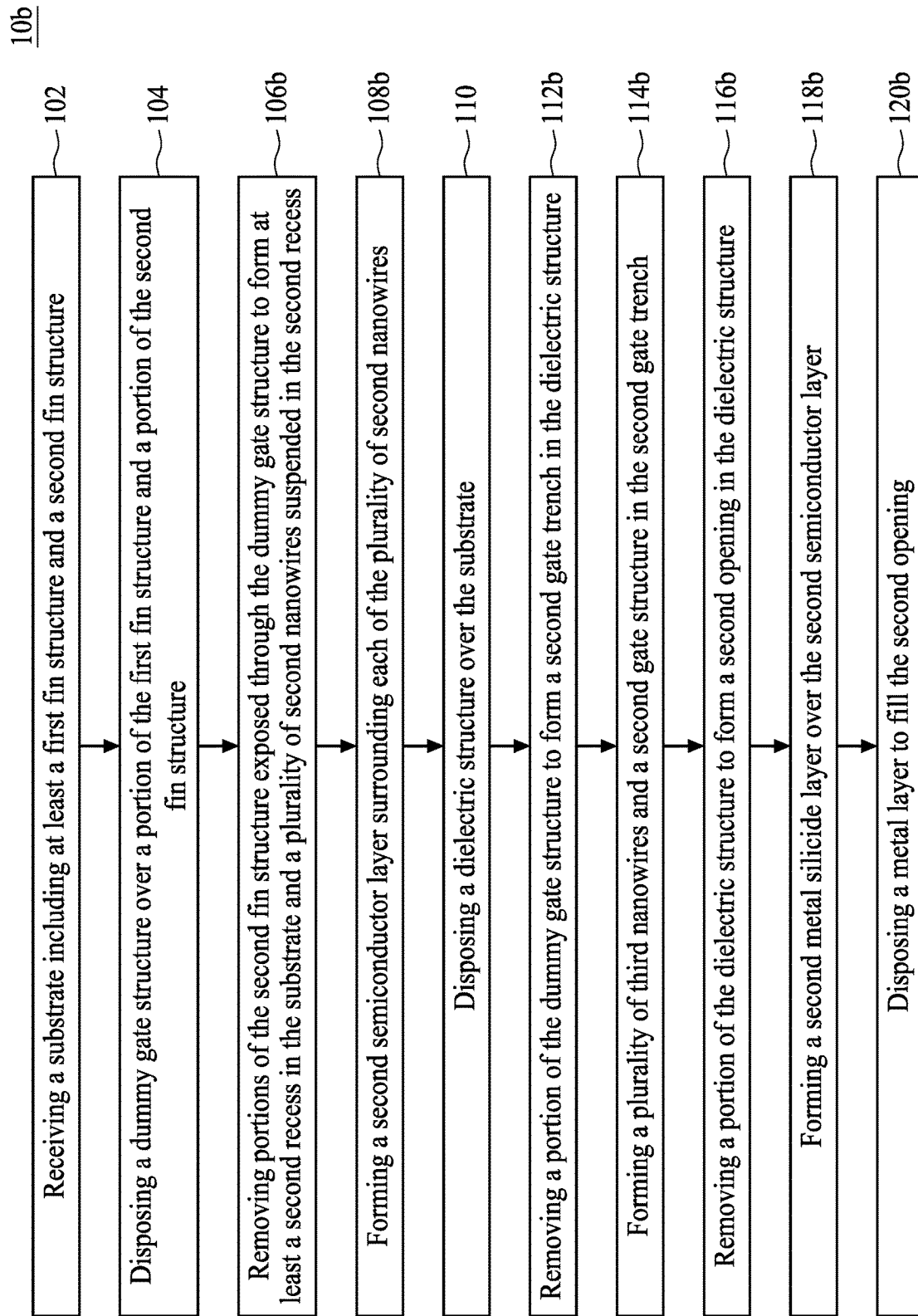


FIG. 2

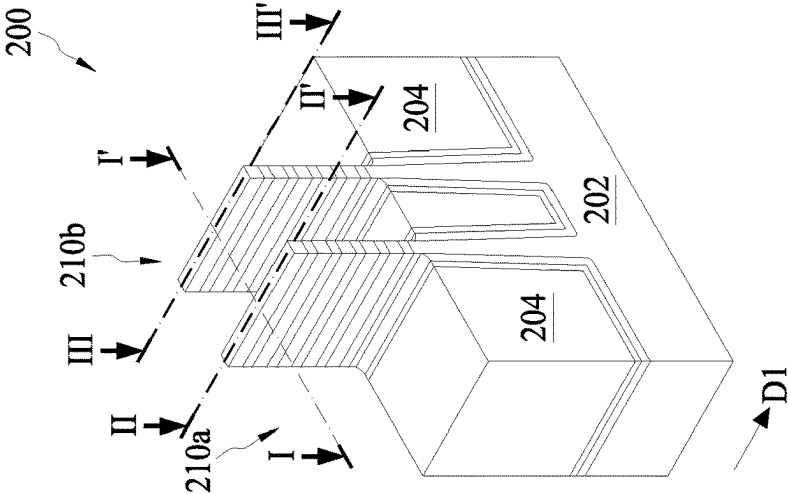


FIG. 3A

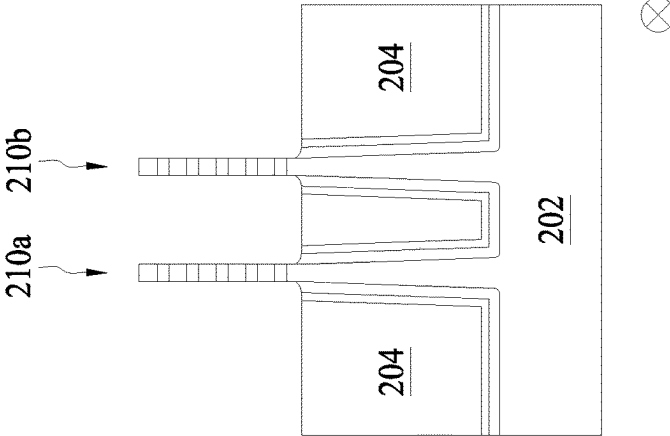


FIG. 3B

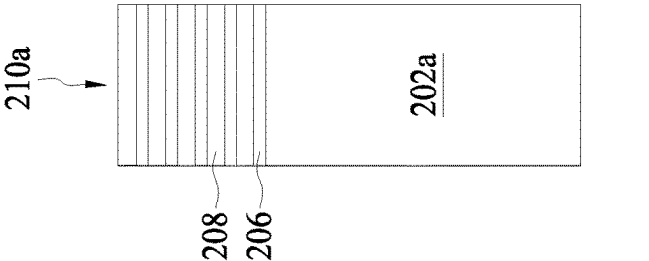


FIG. 3C

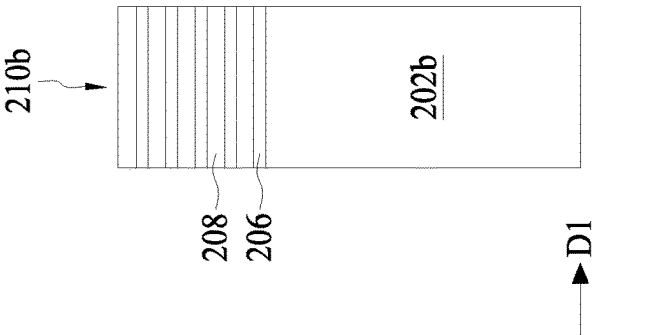
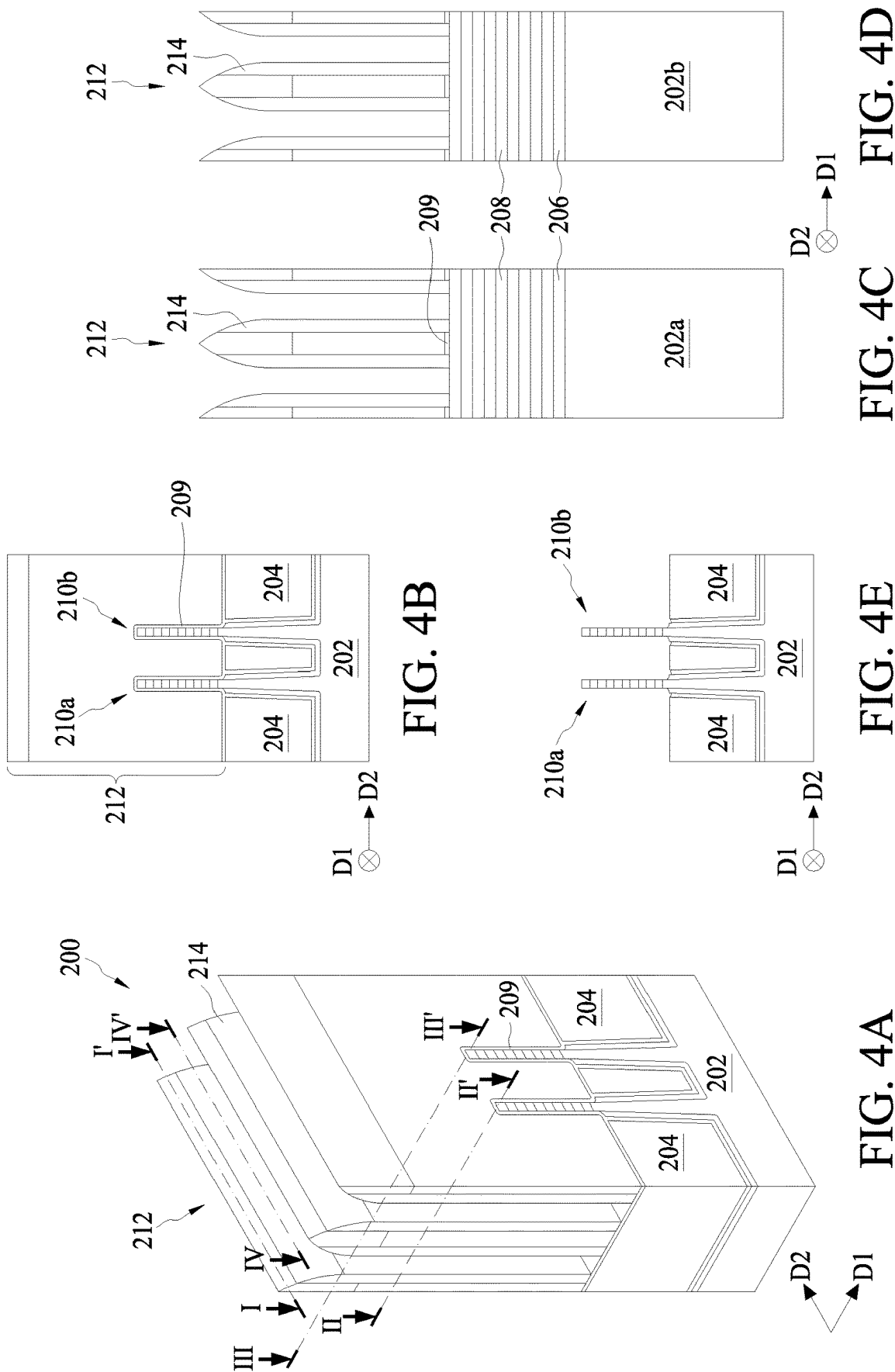
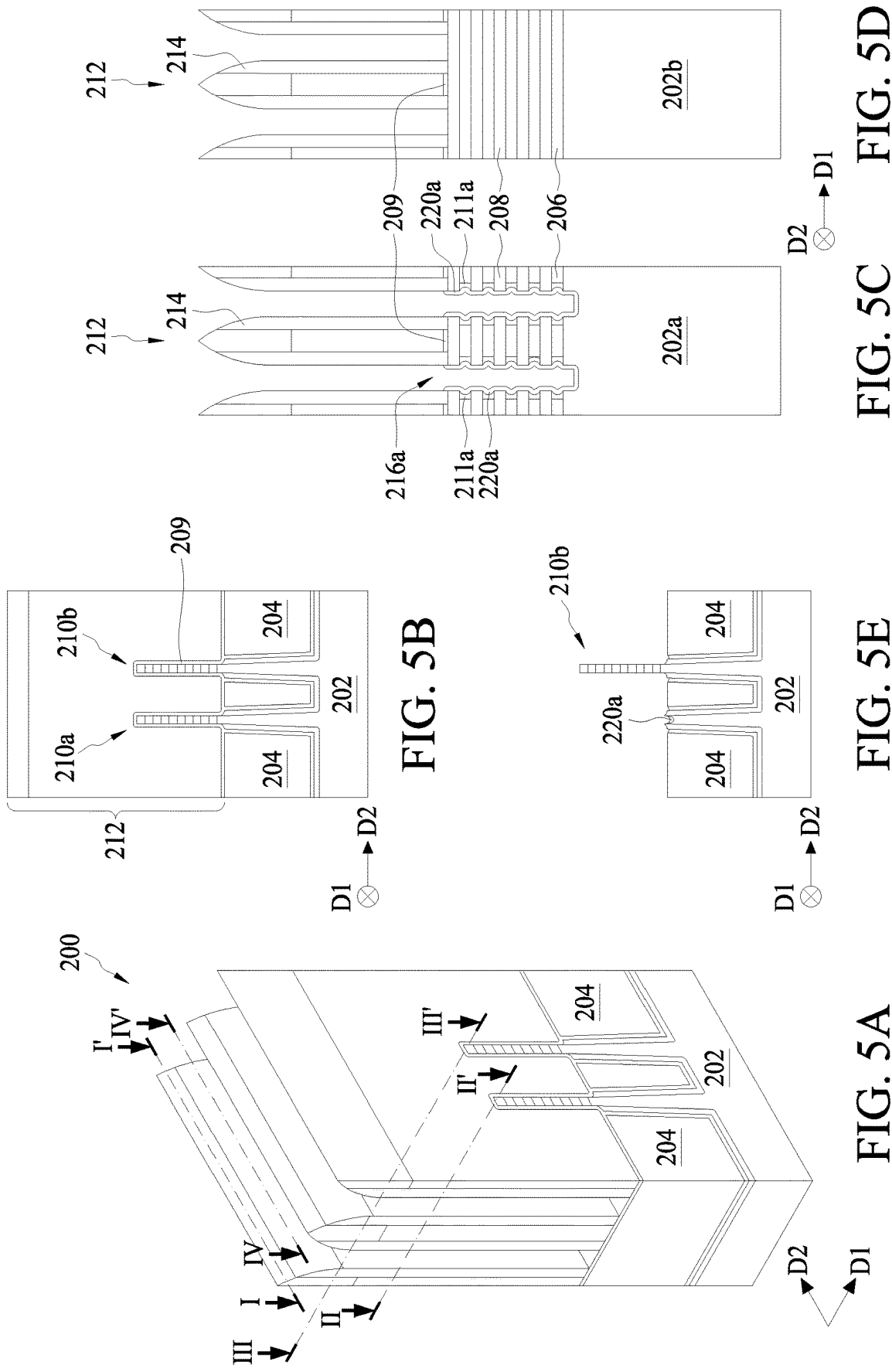


FIG. 3D





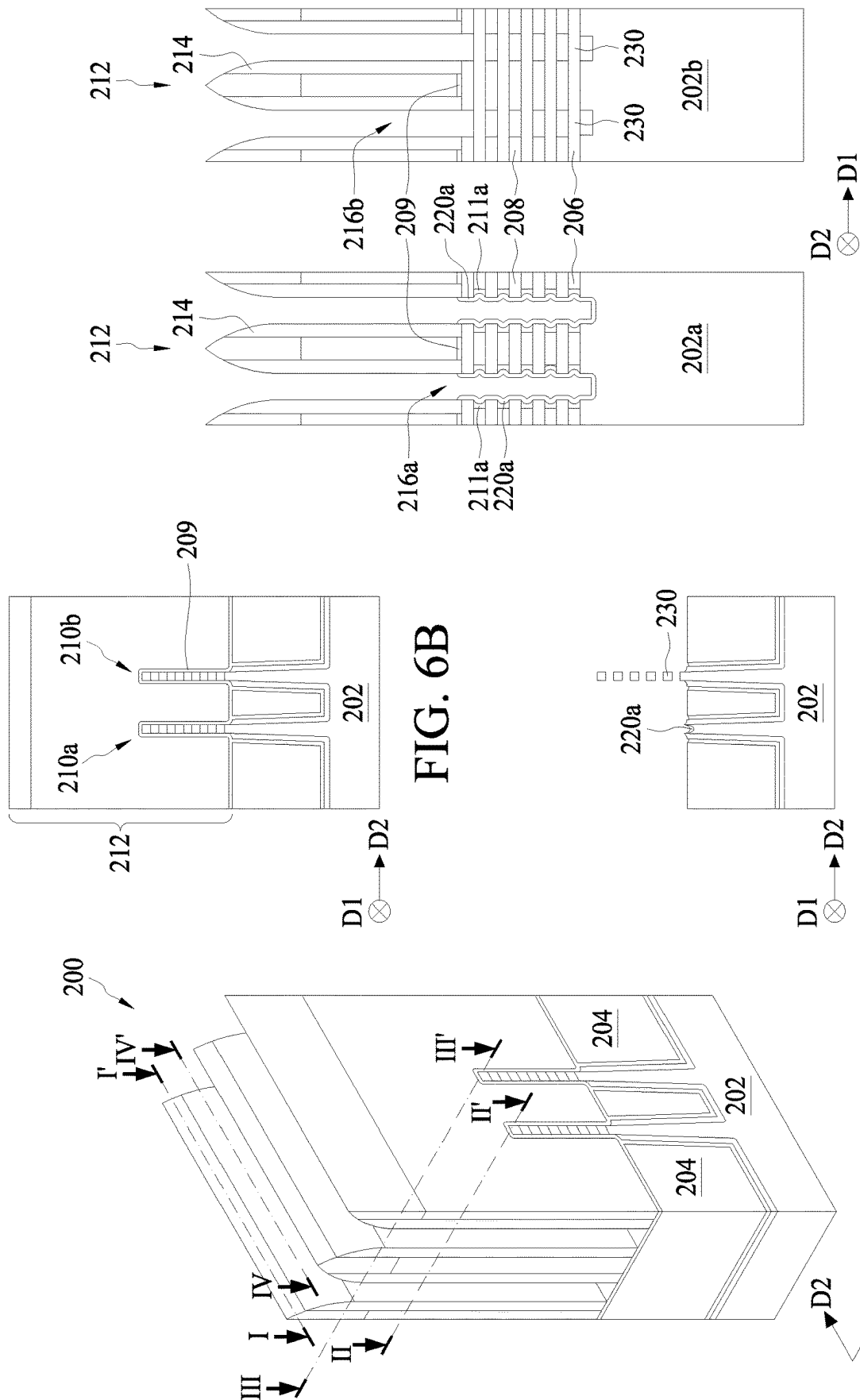
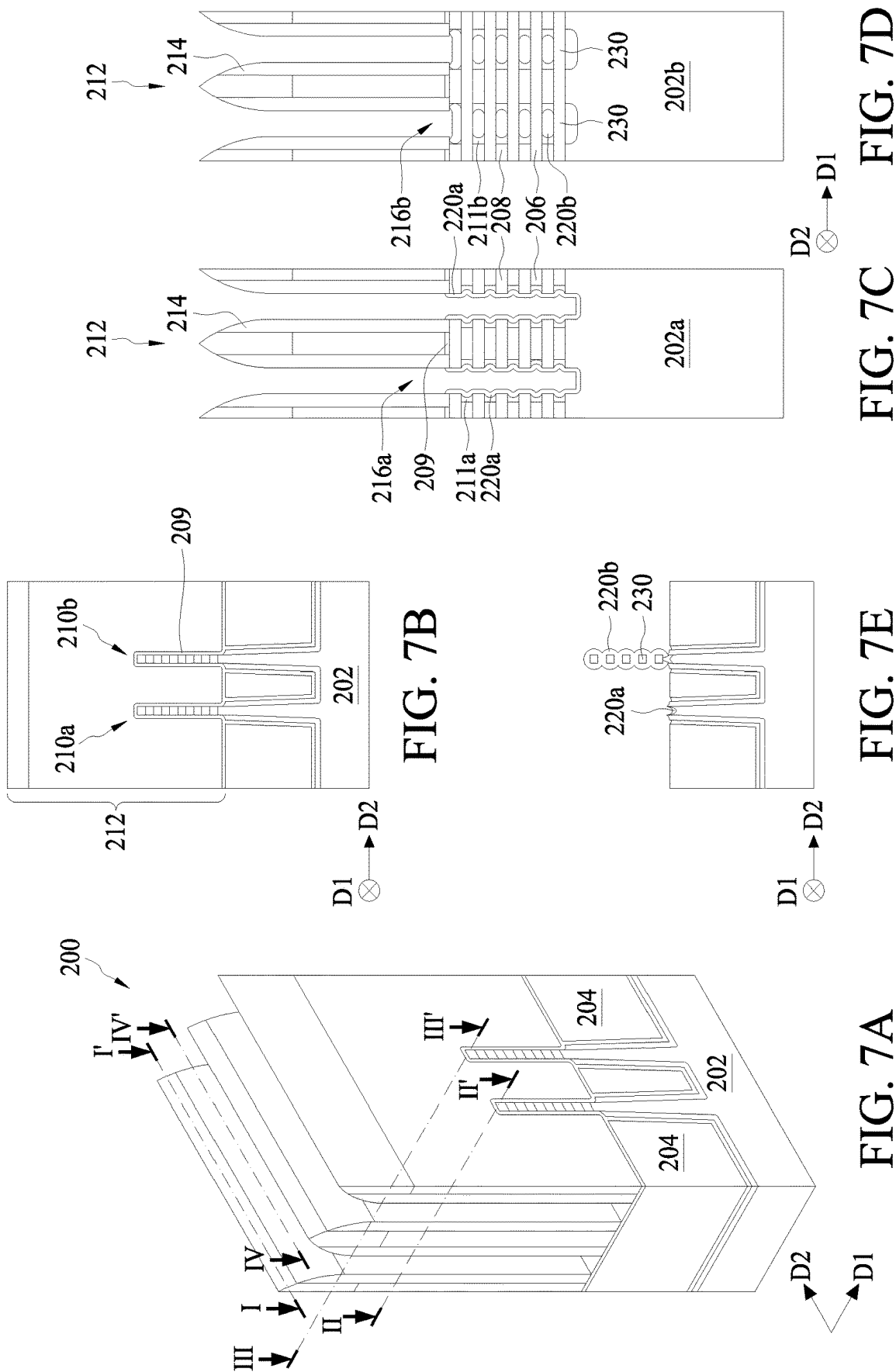


FIG. 6A

FIG. 6E

FIG. 6C

FIG. 6D



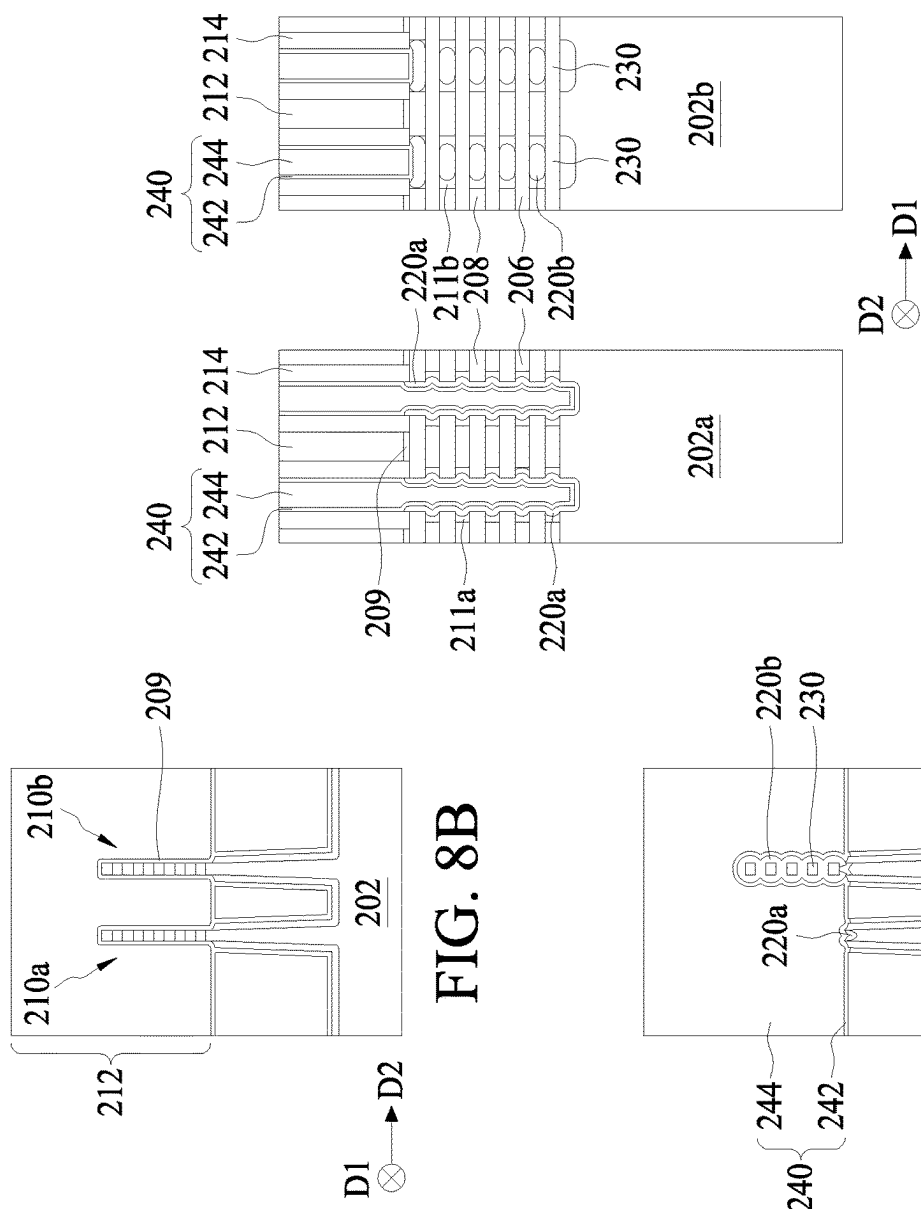


FIG. 8A

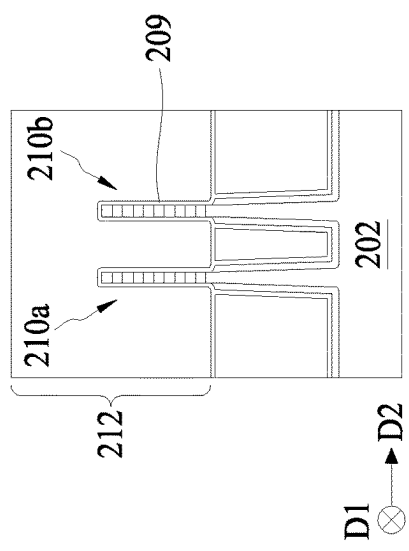


FIG. 8B

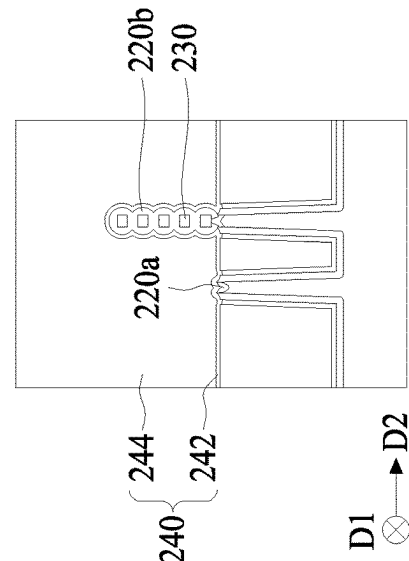


FIG. 8C

FIG. 8E

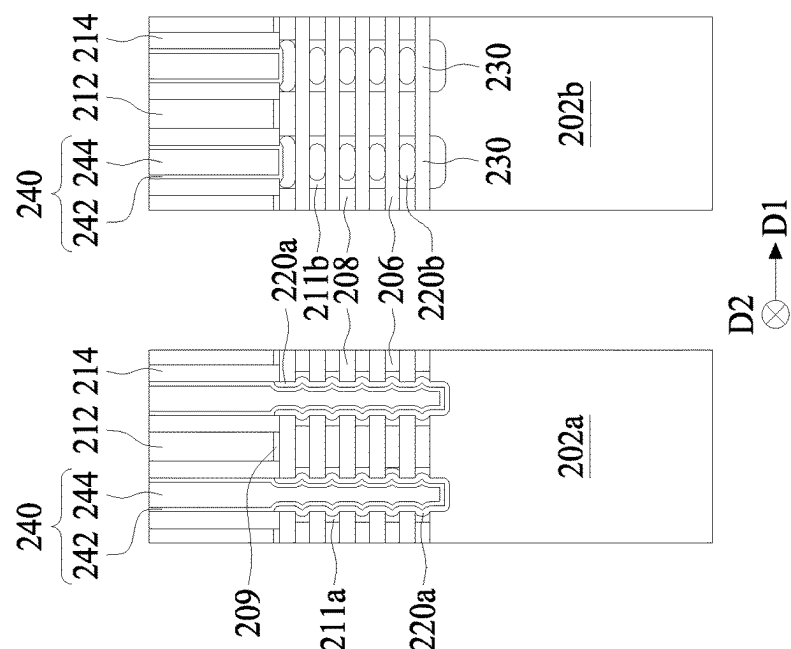
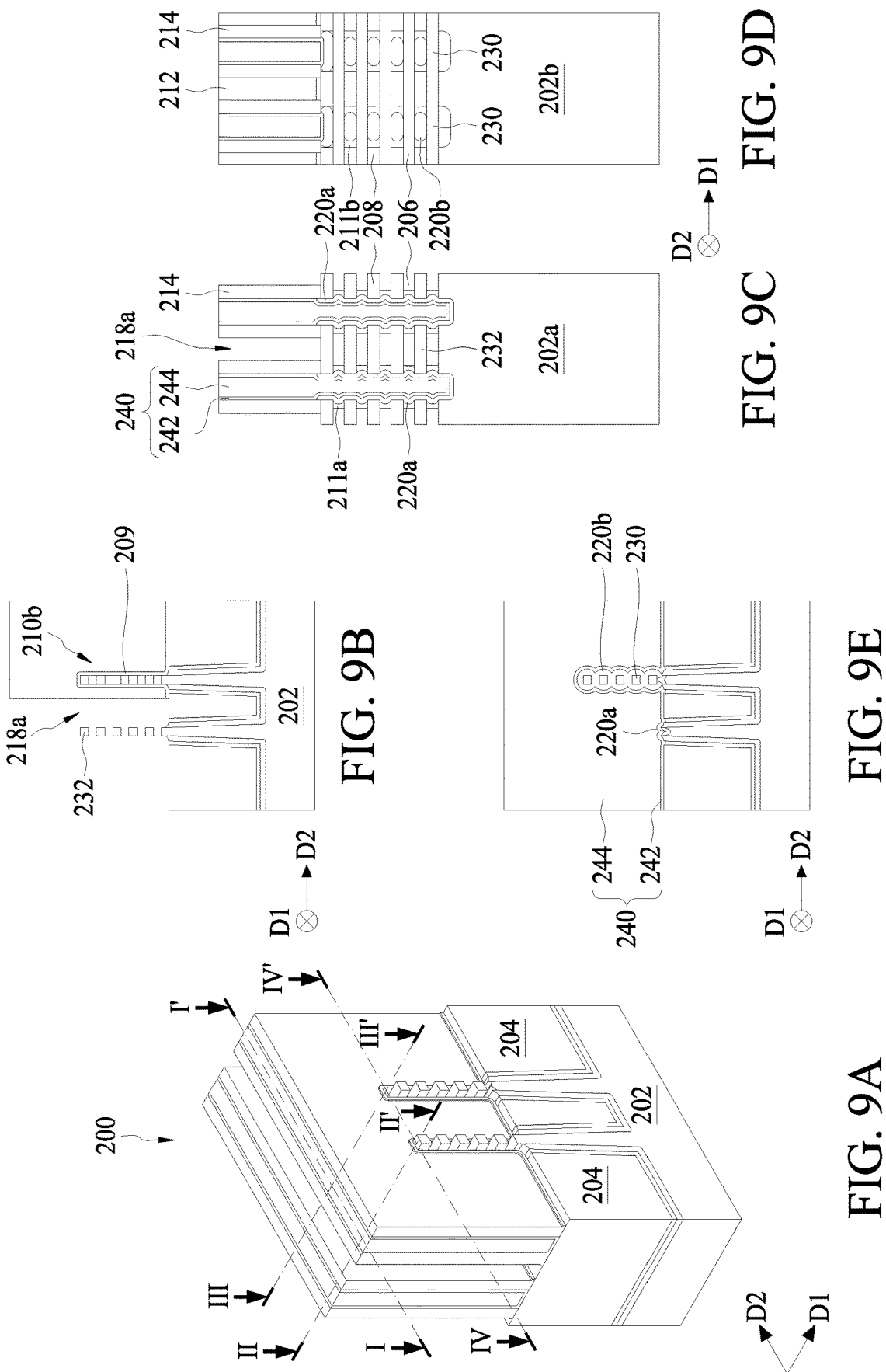
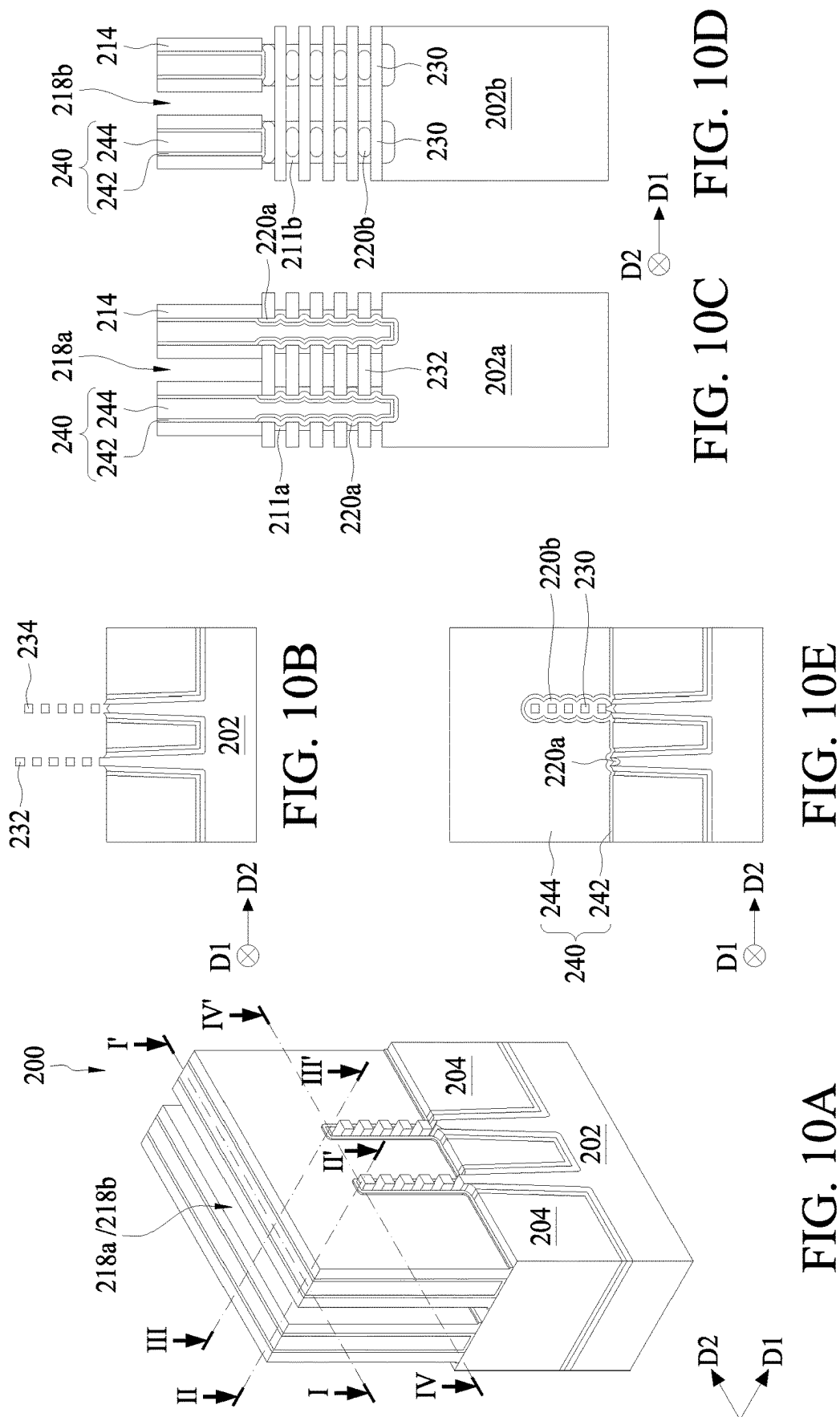


FIG. 8D





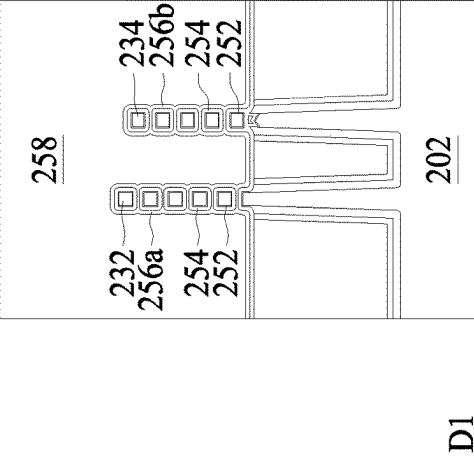


FIG. 11B

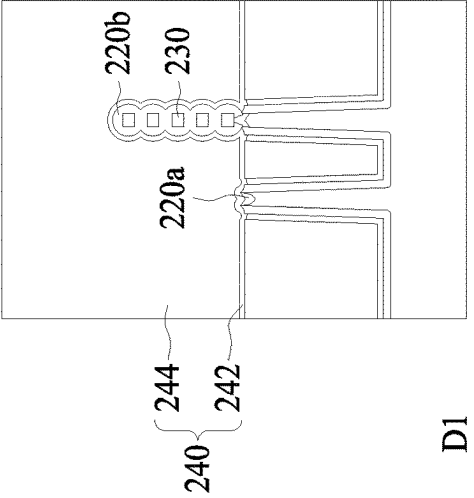


FIG. 11E

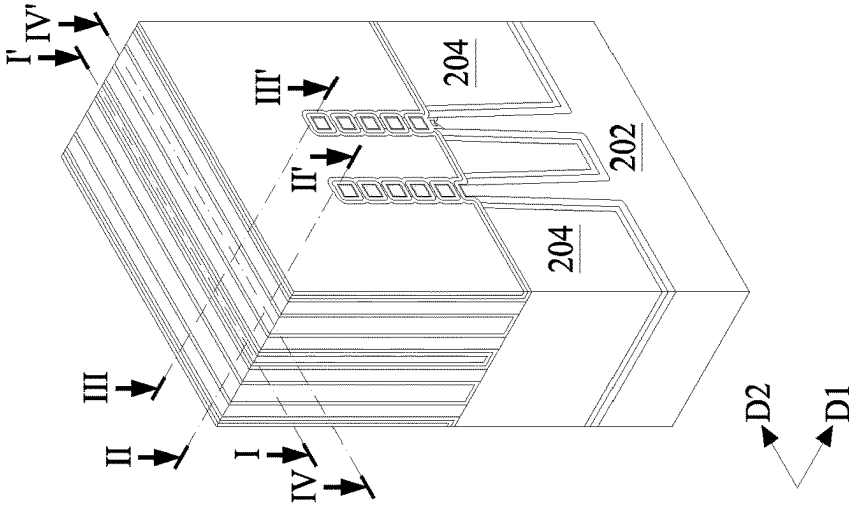


FIG. 11A

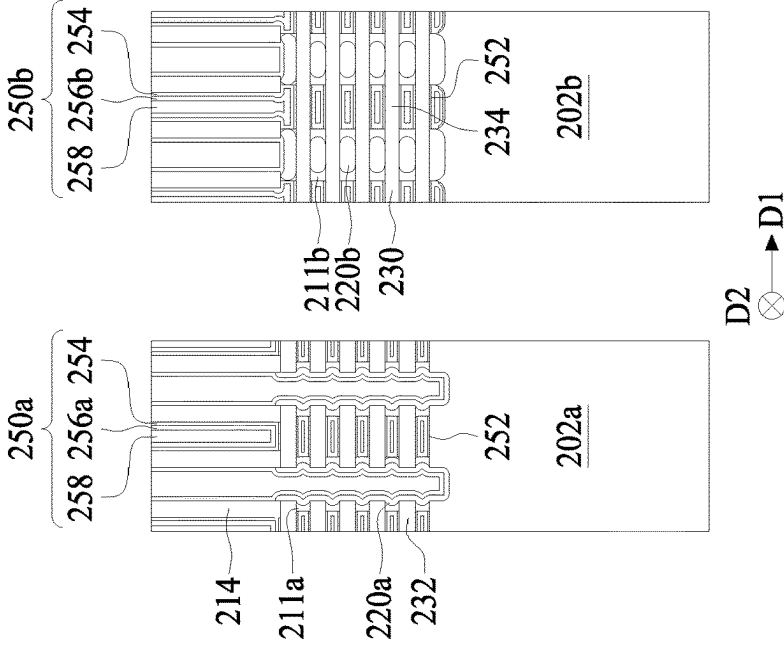
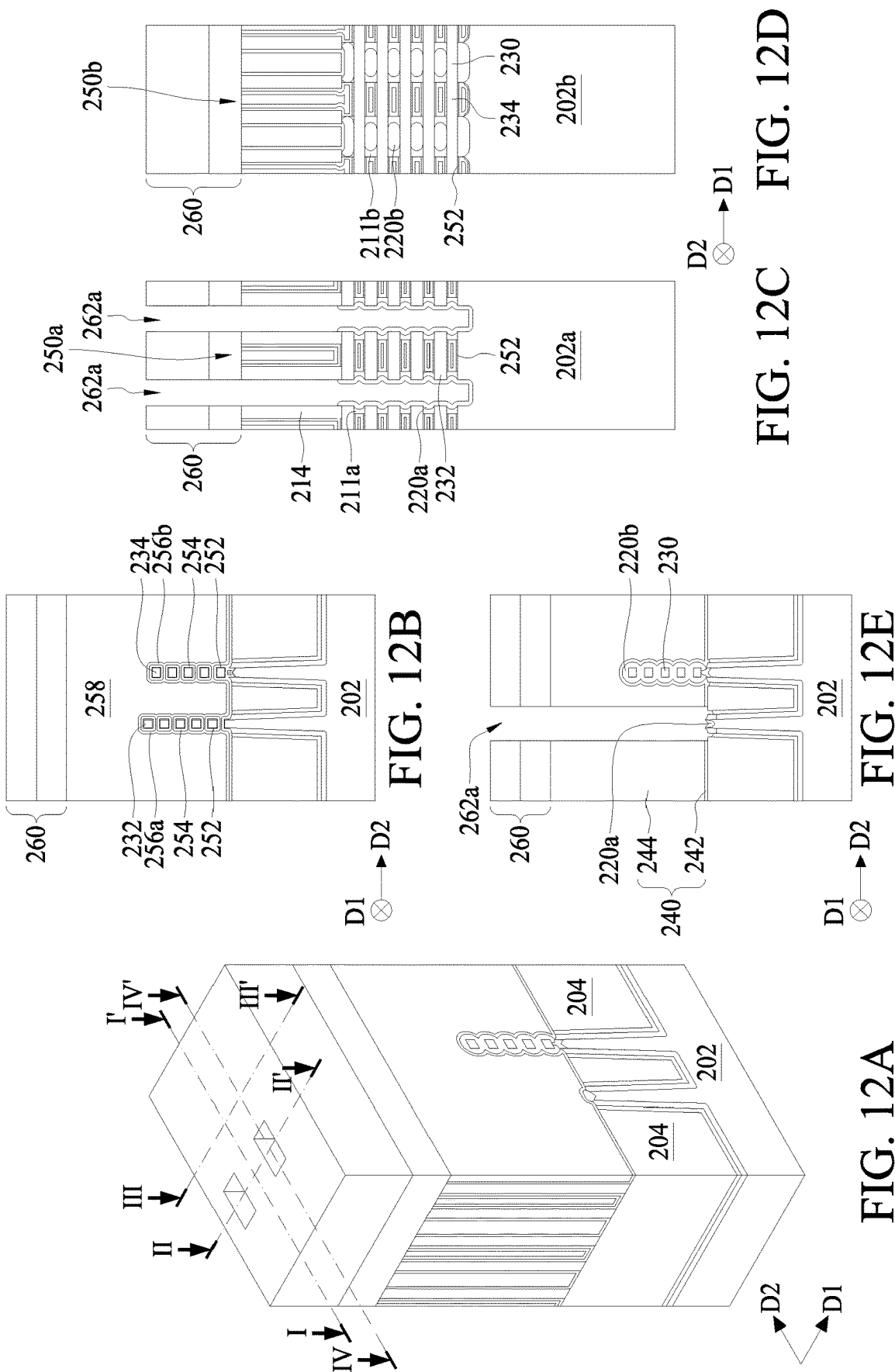


FIG. 11C

FIG. 11D



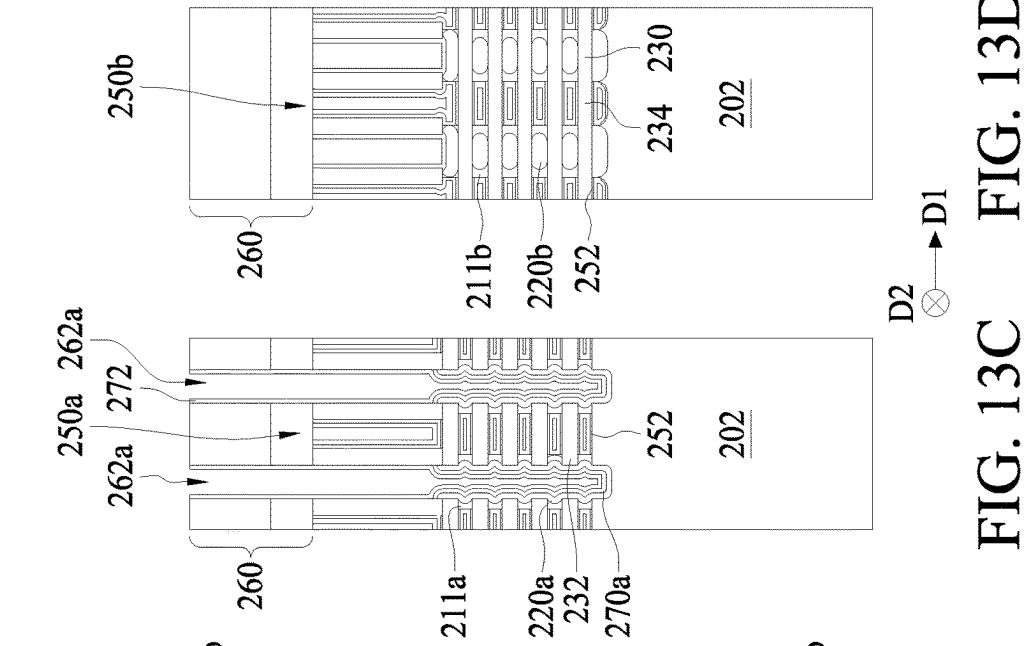
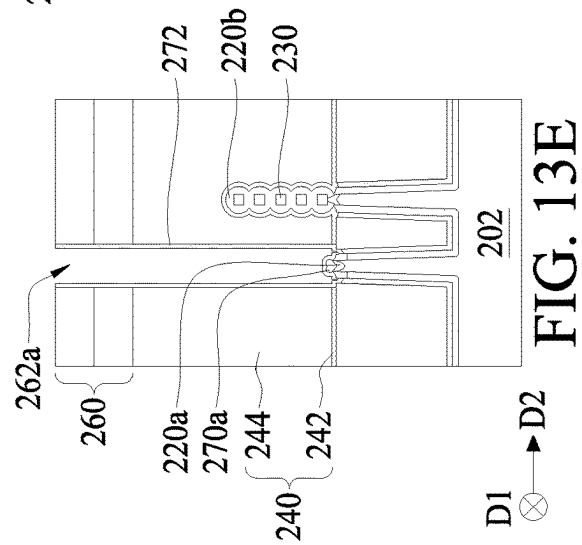
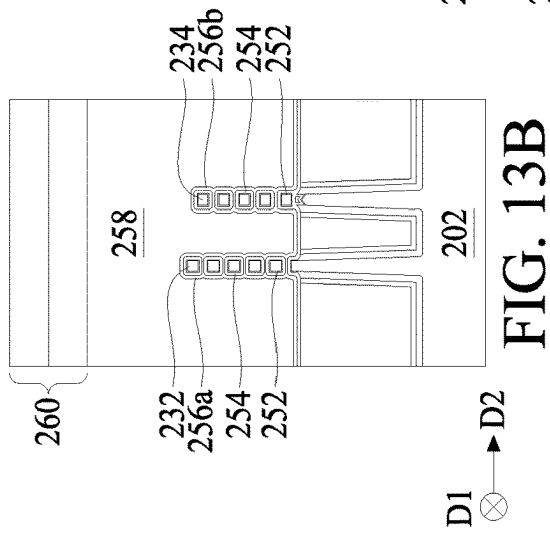
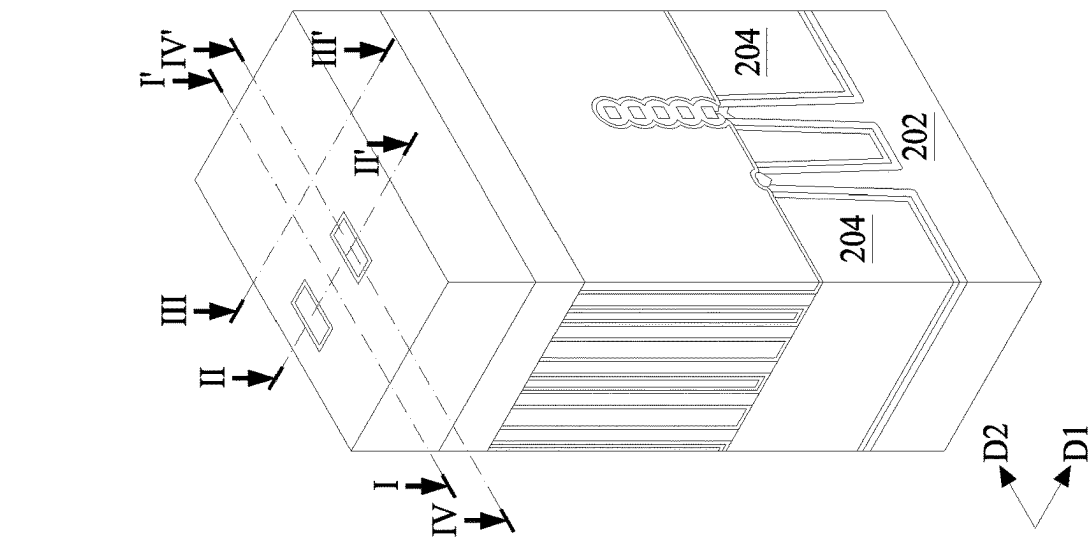
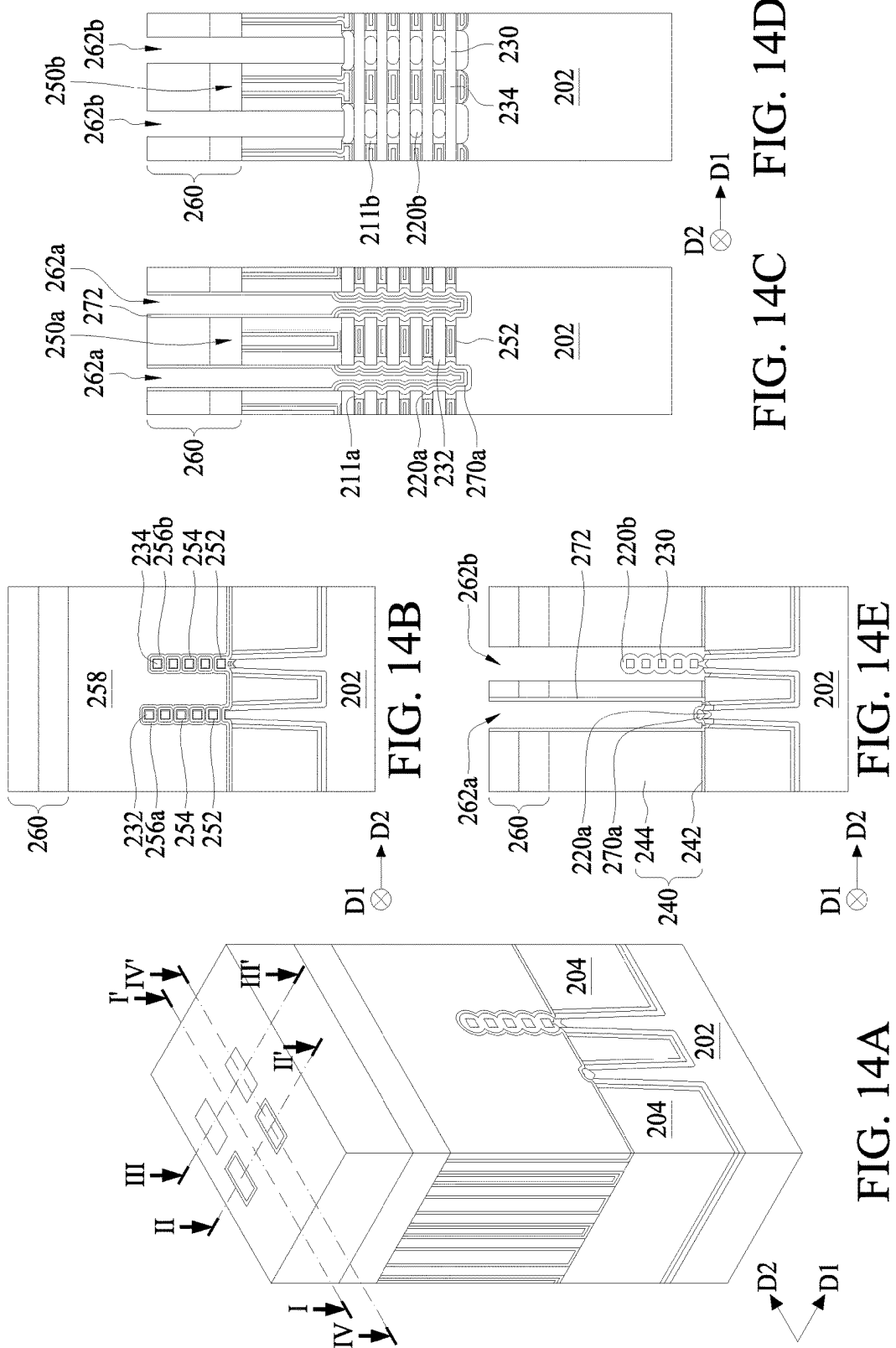


FIG. 13A

FIG. 13B

FIG. 13E

FIG. 13C FIG. 13D



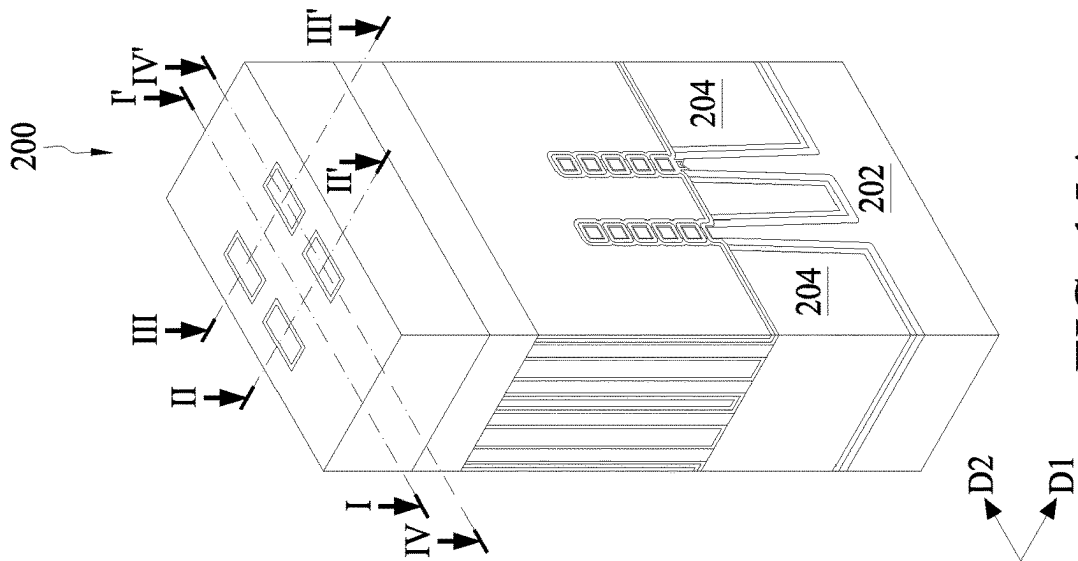


FIG. 15A

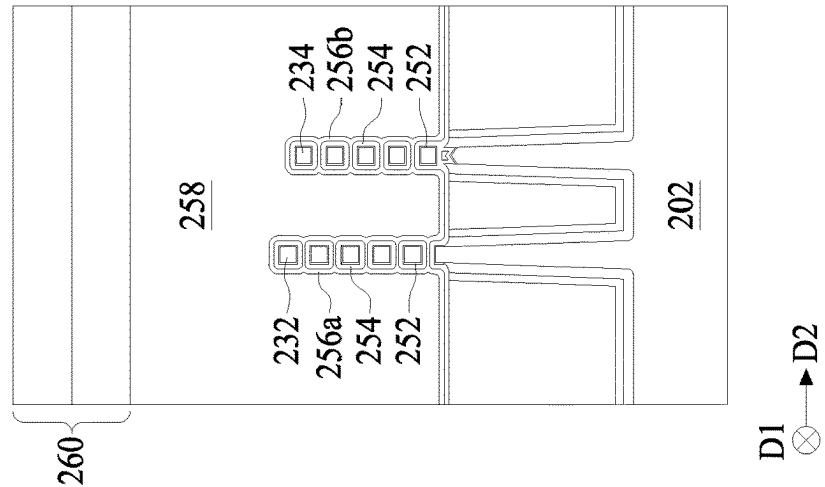


FIG. 15B

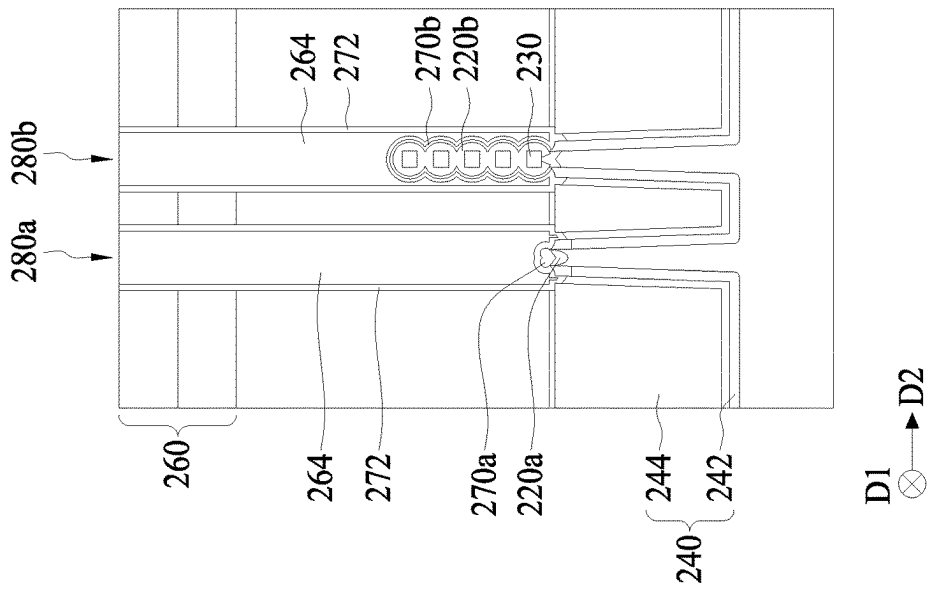


FIG. 15E

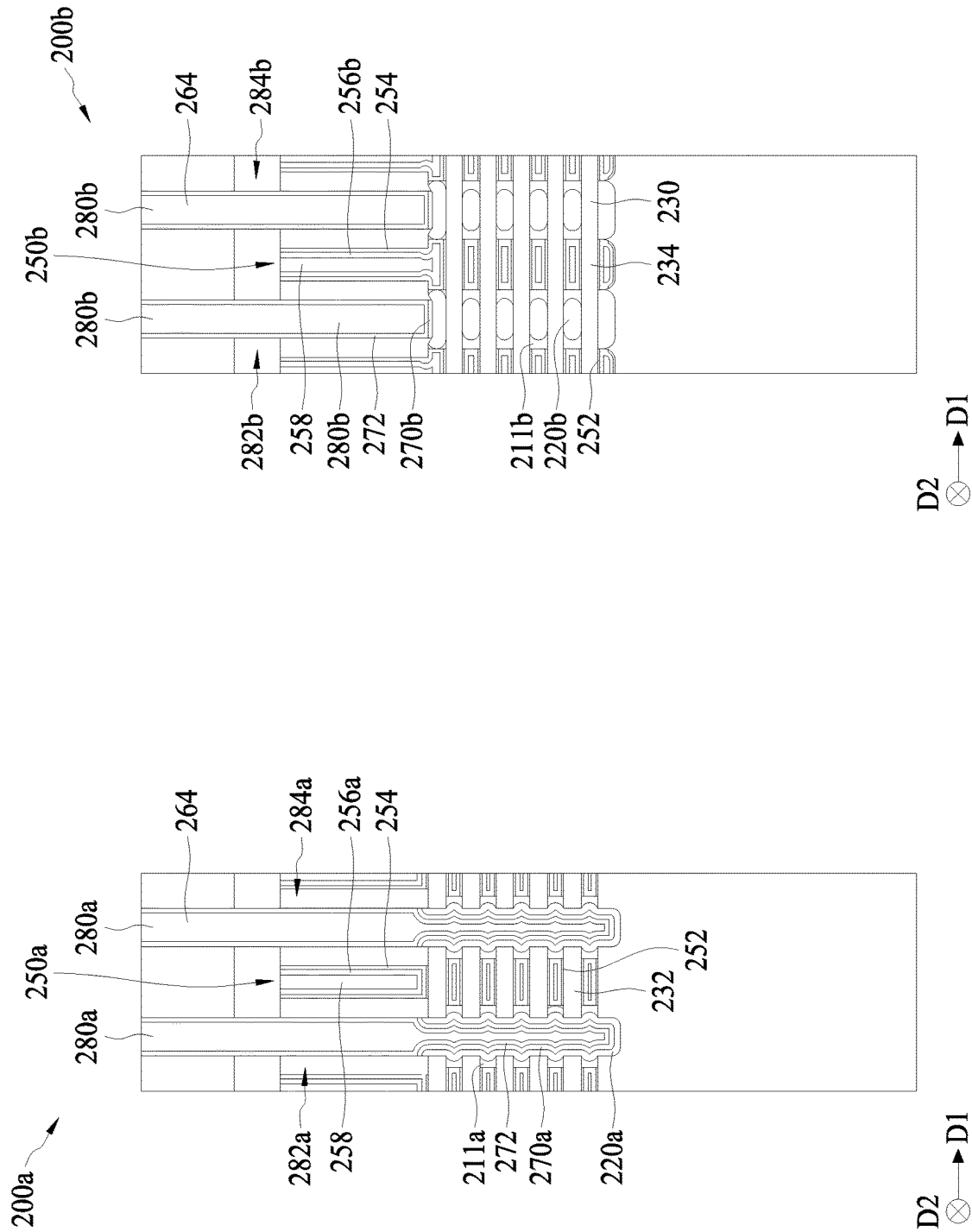
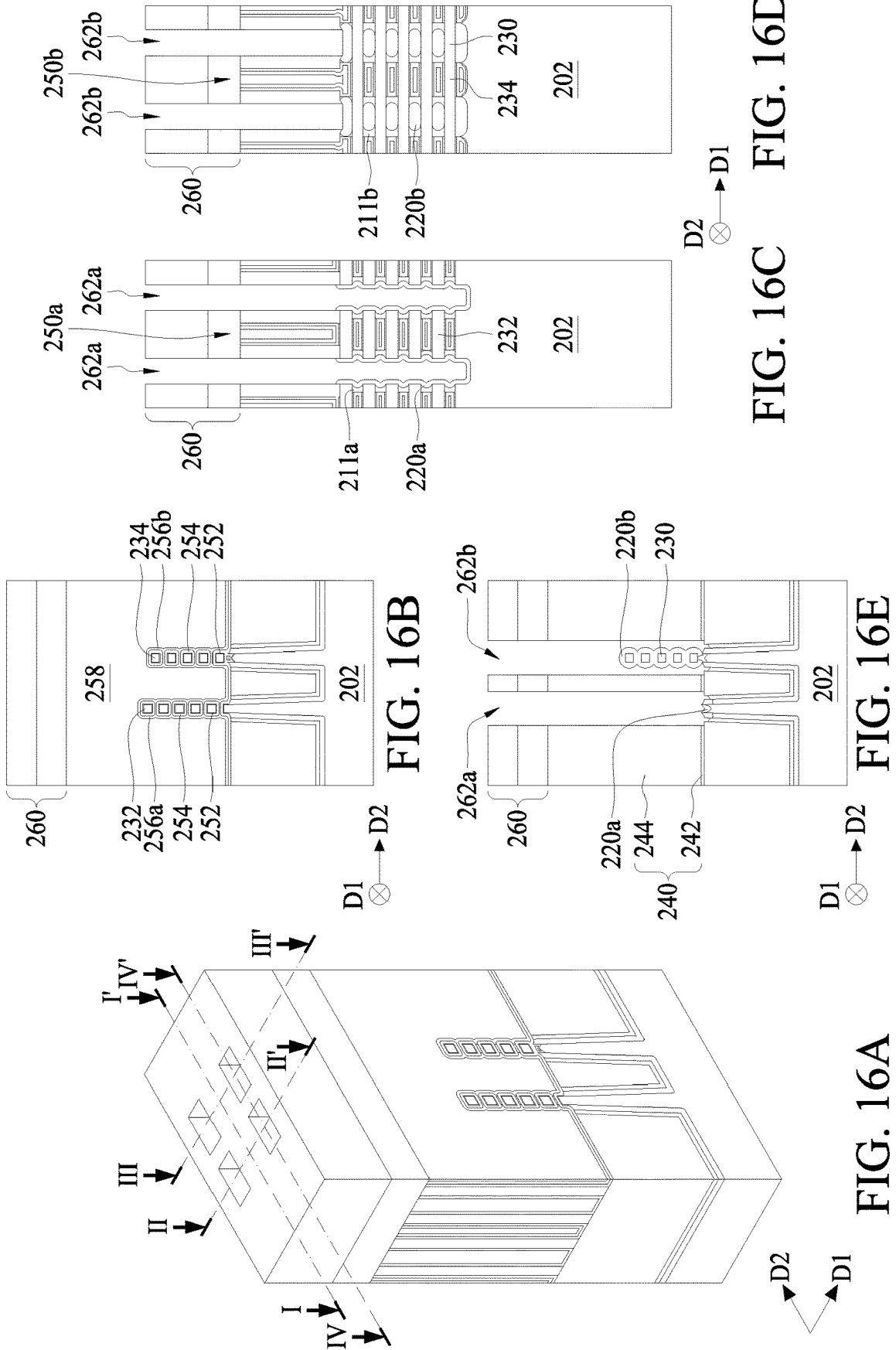


FIG. 15D



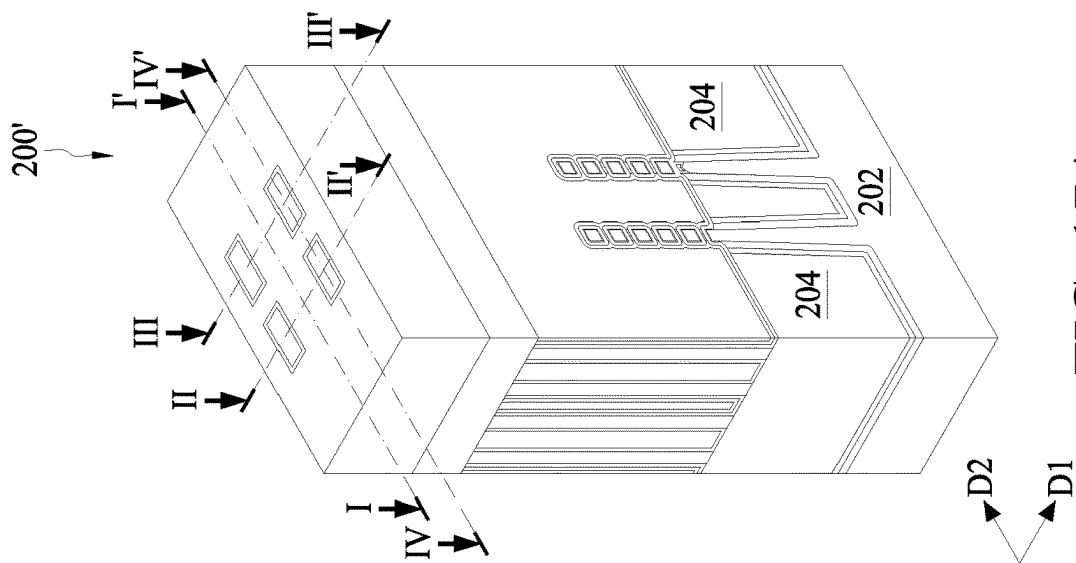


FIG. 17A

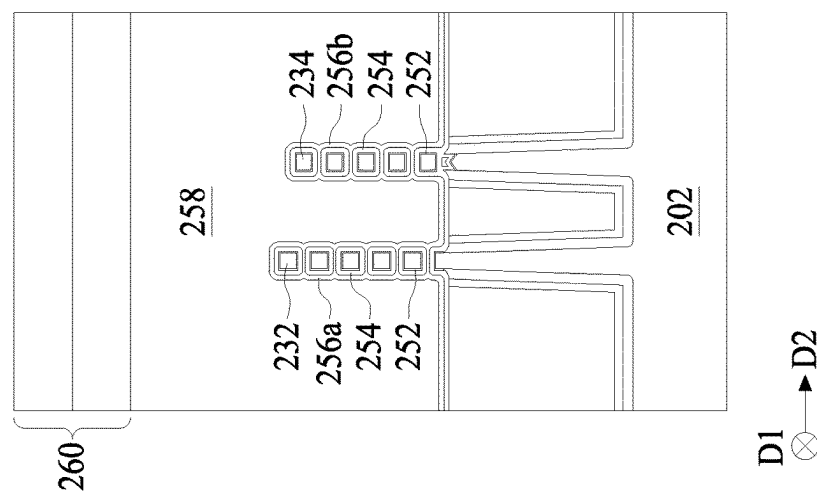


FIG. 17B

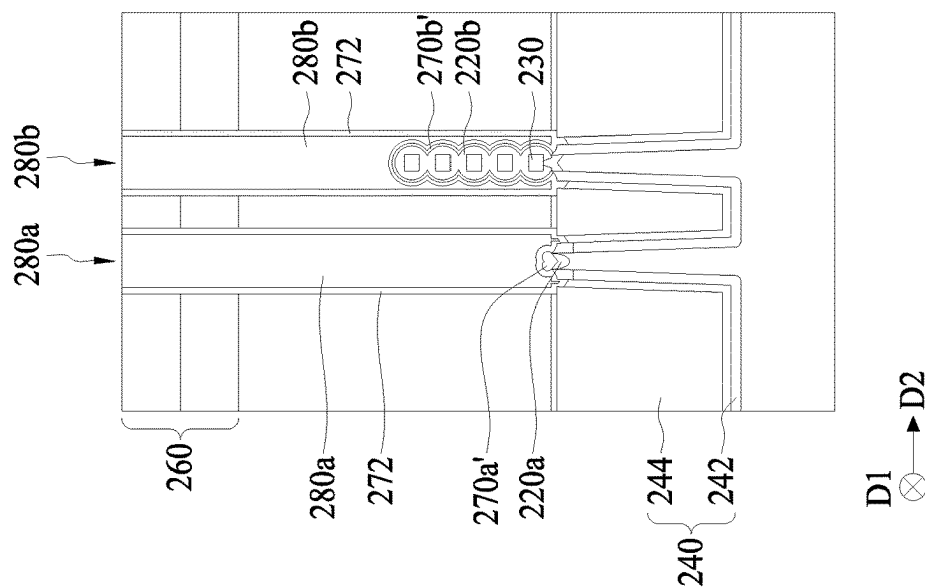


FIG. 17E

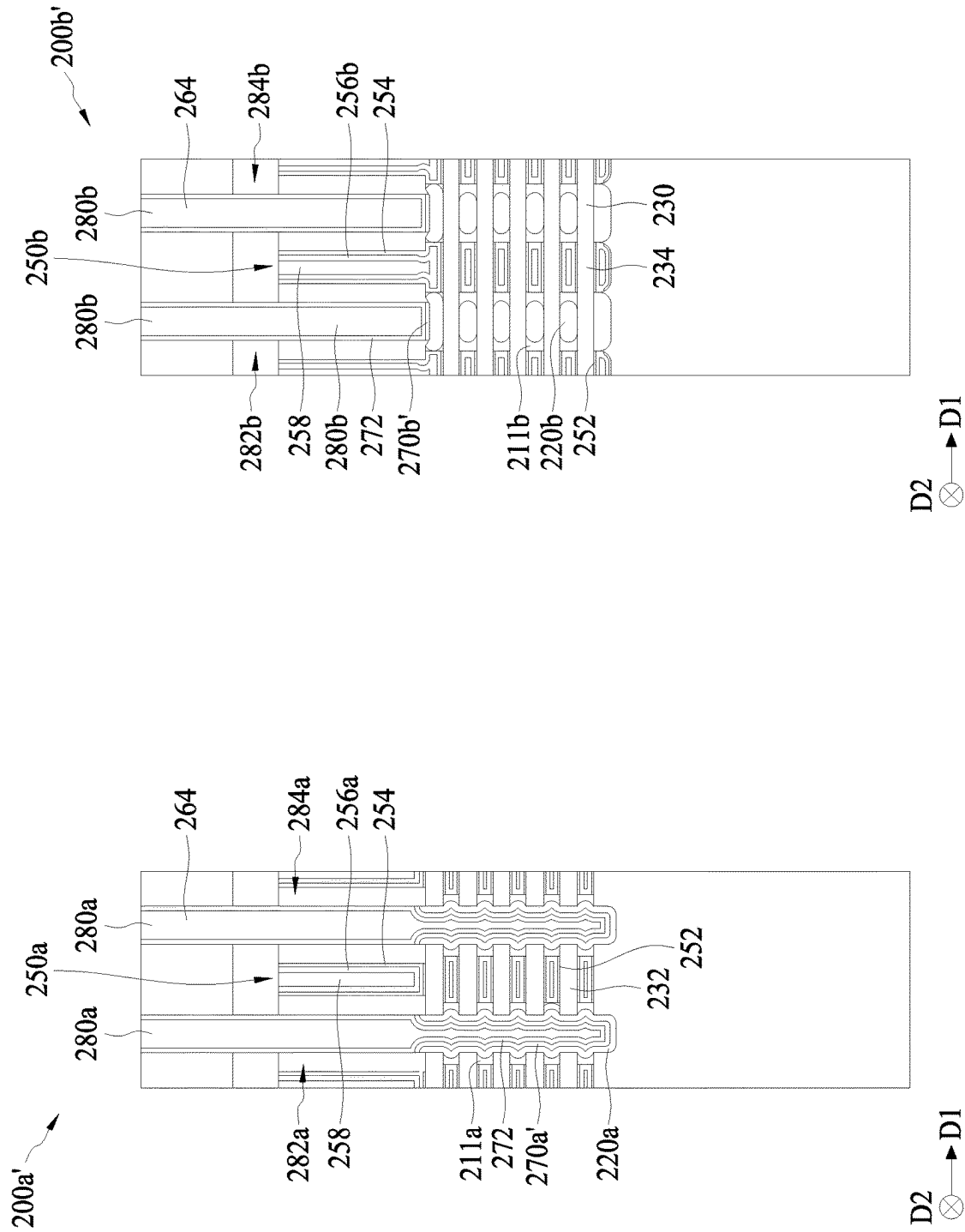


FIG. 17D

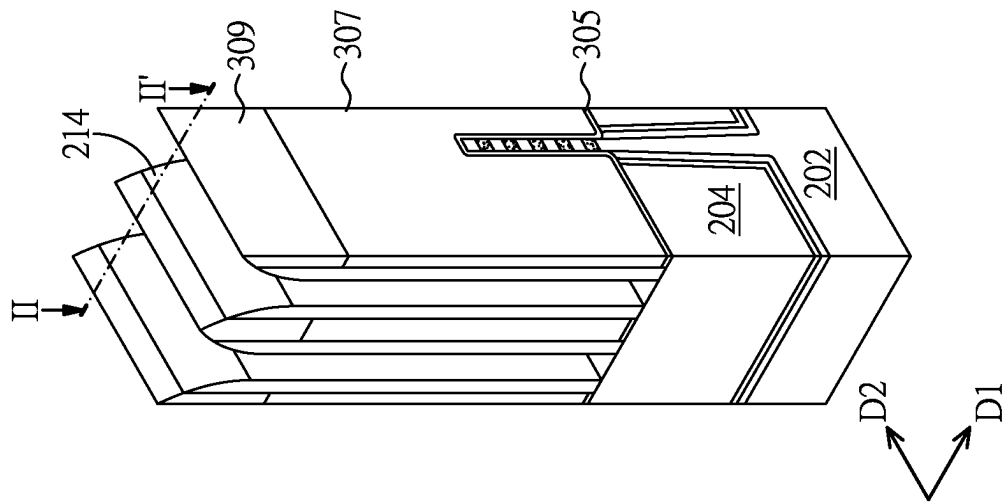


FIG. 18C

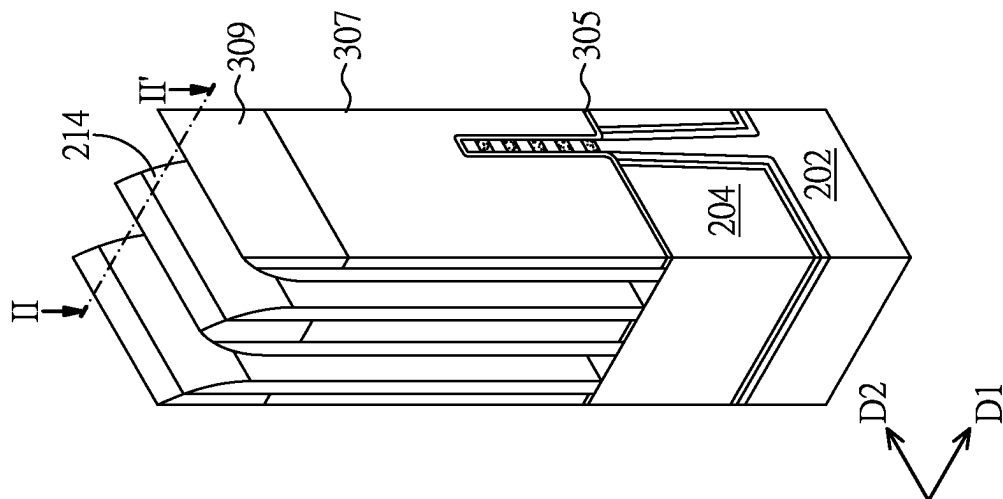


FIG. 18B

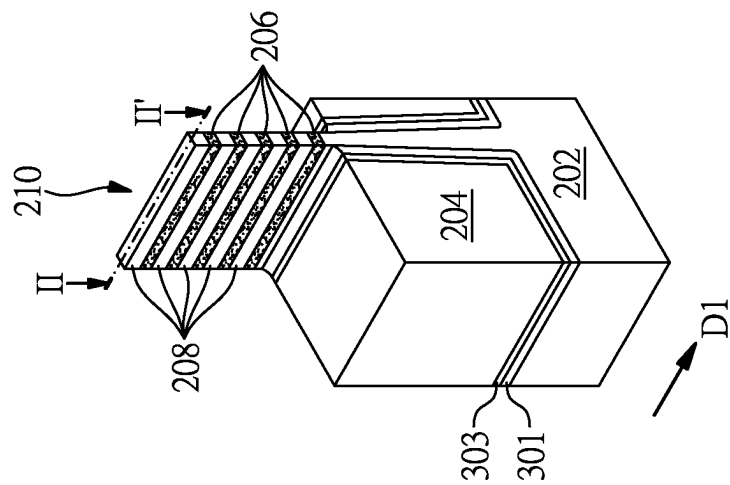


FIG. 18A

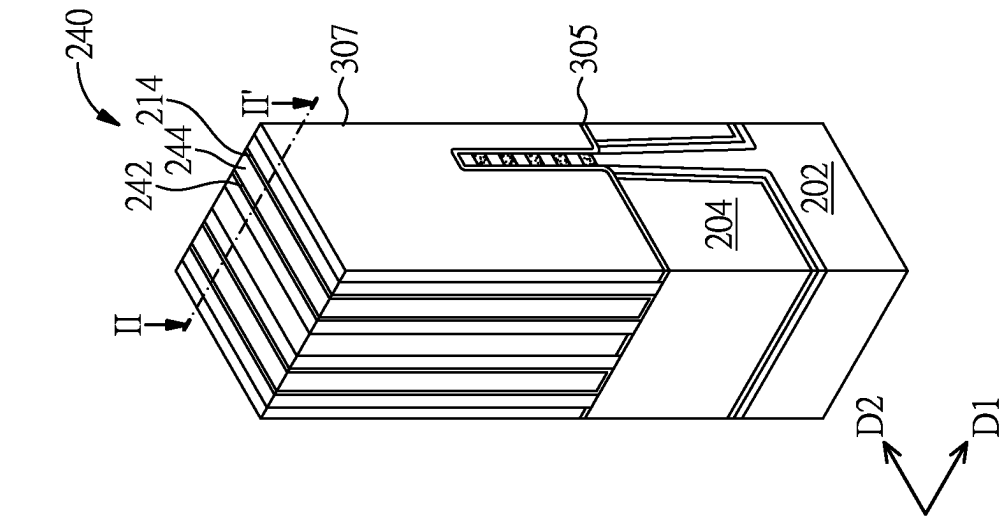


FIG. 18D

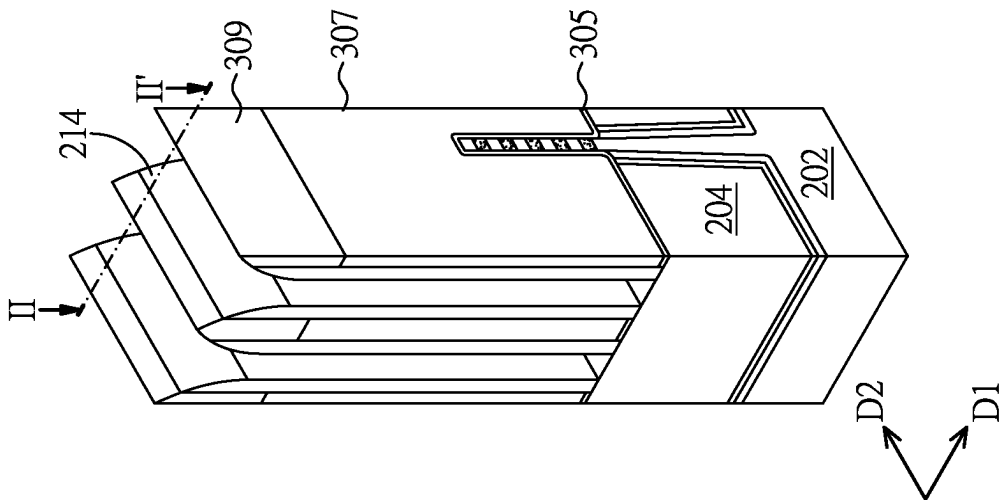


FIG. 18E

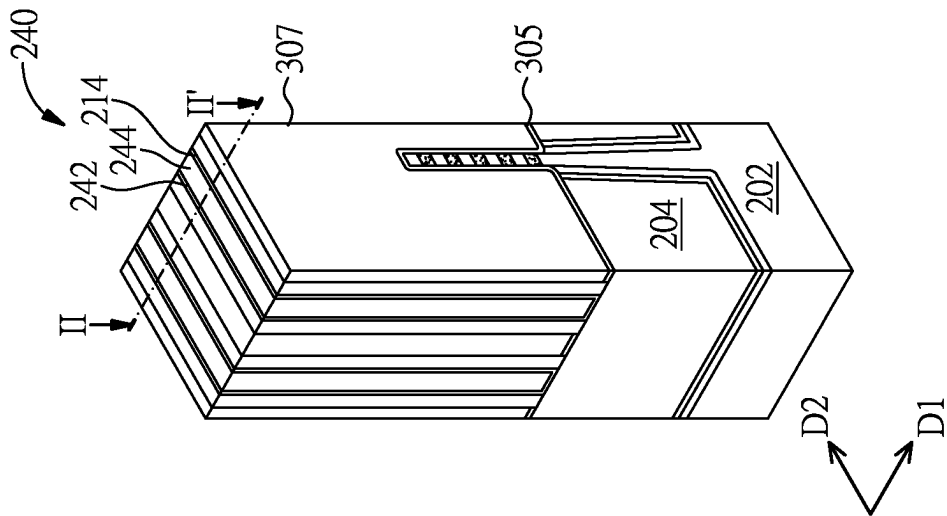


FIG. 18F

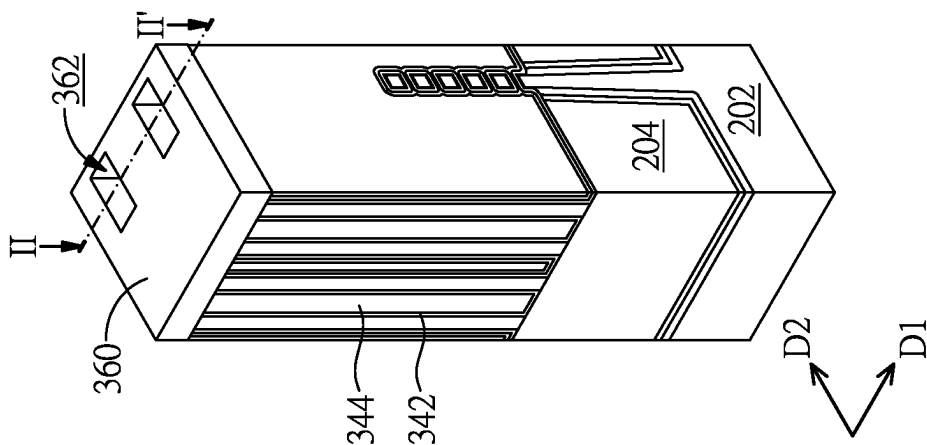


FIG. 18I

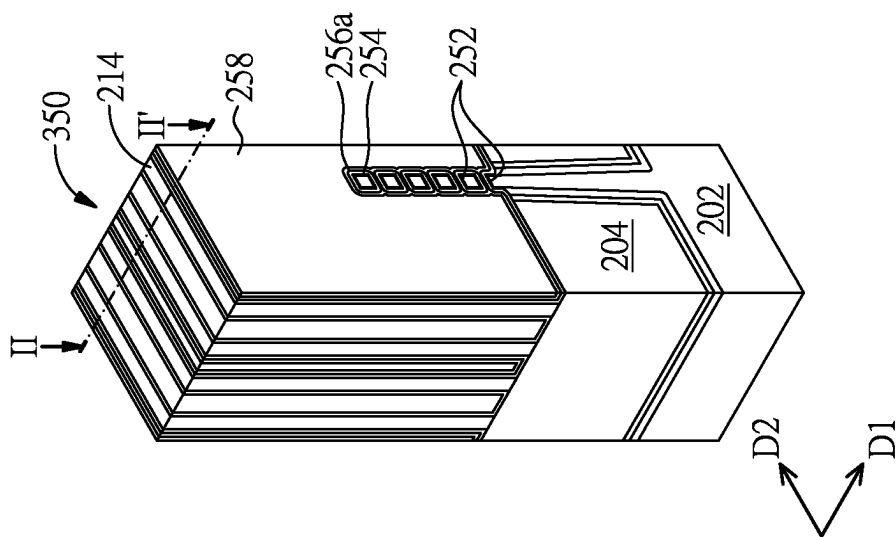


FIG. 18H

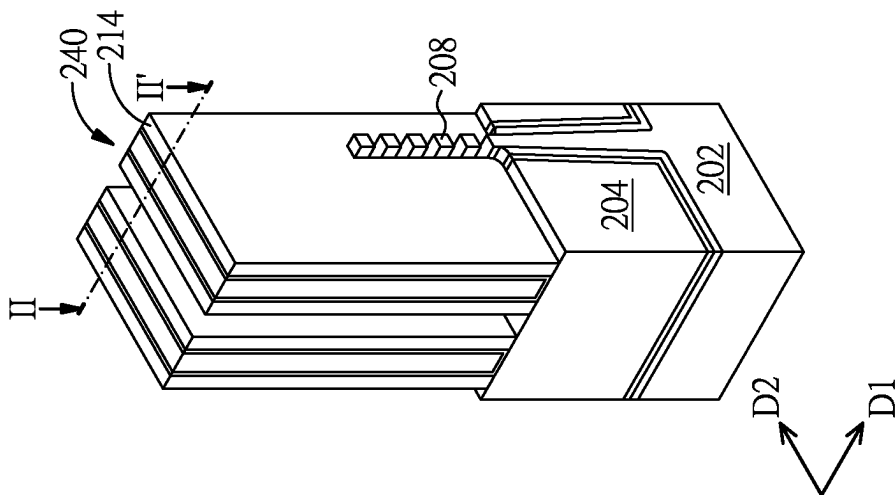


FIG. 18G

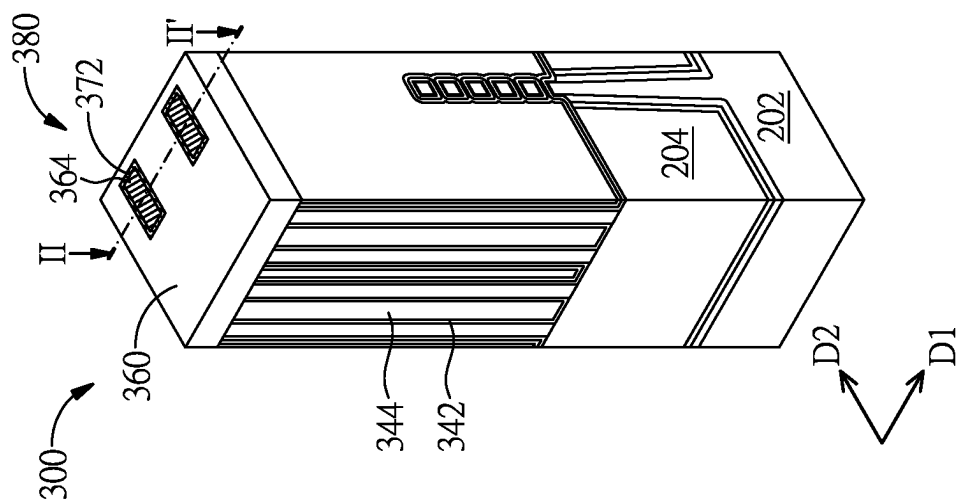


FIG. 18J

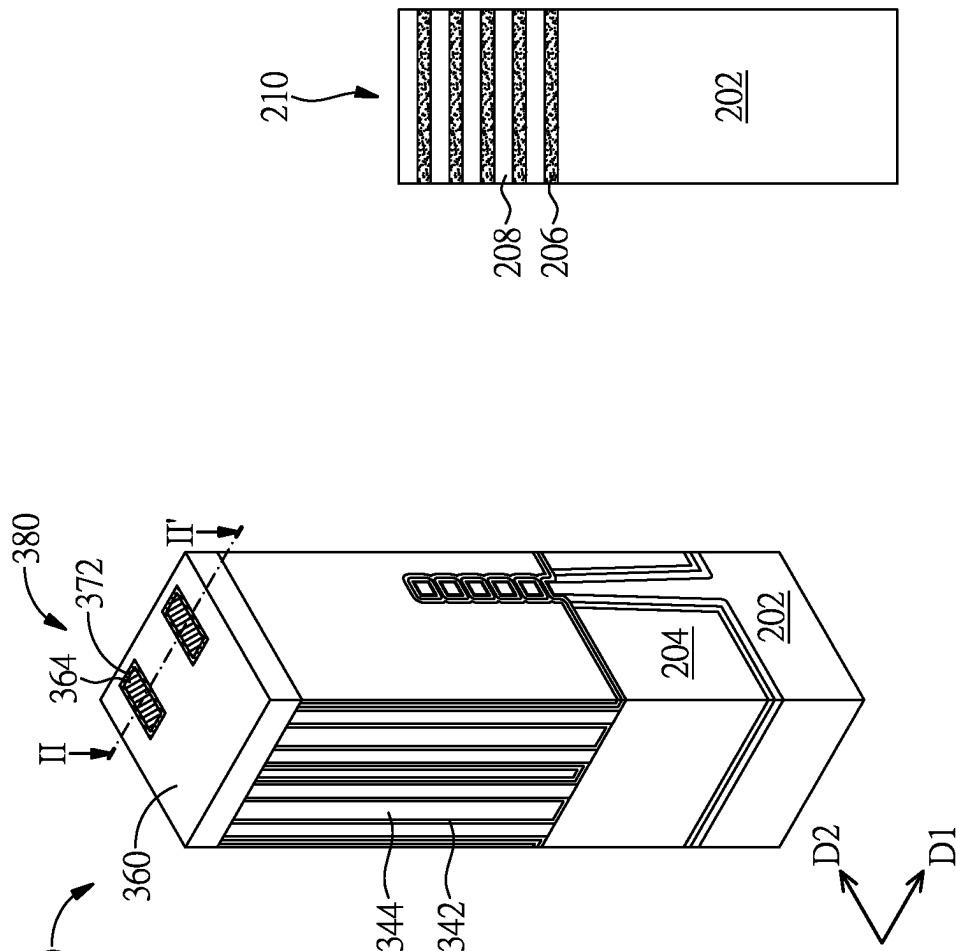


FIG. 18K

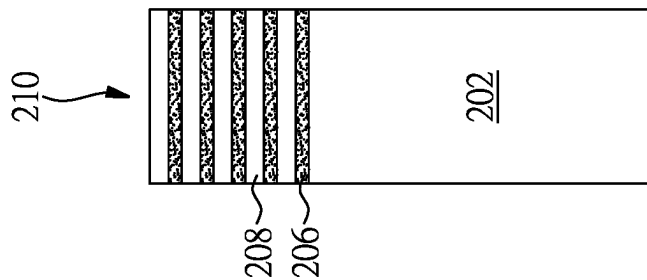


FIG. 19A

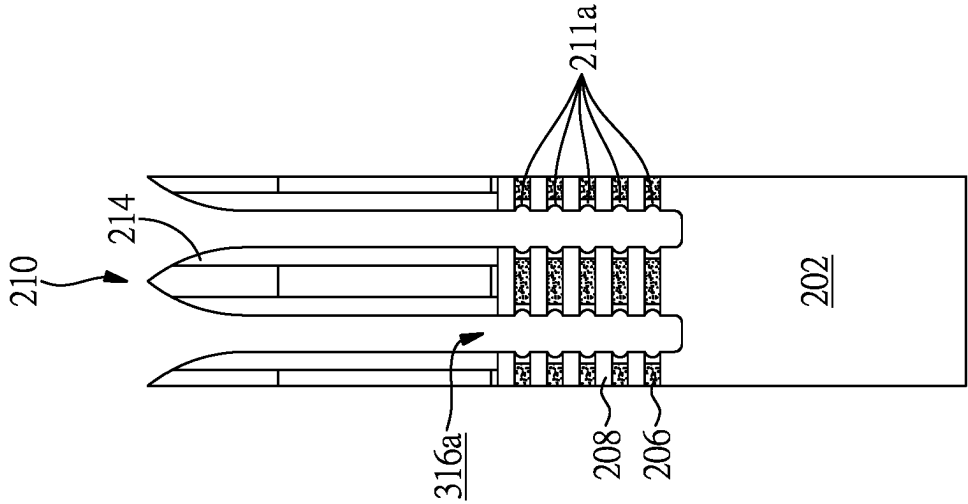


FIG. 19D

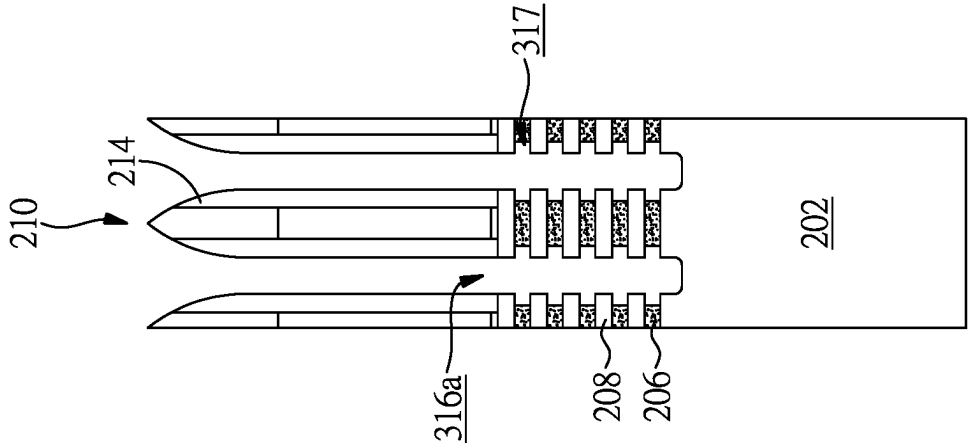


FIG. 19C

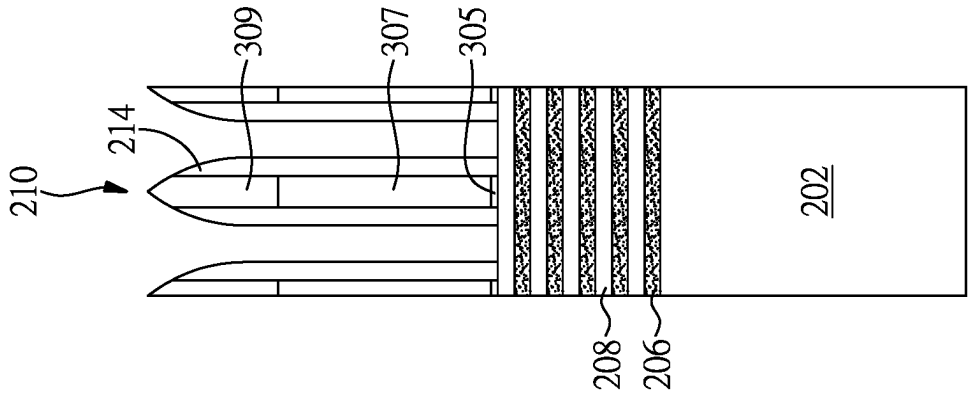


FIG. 19B

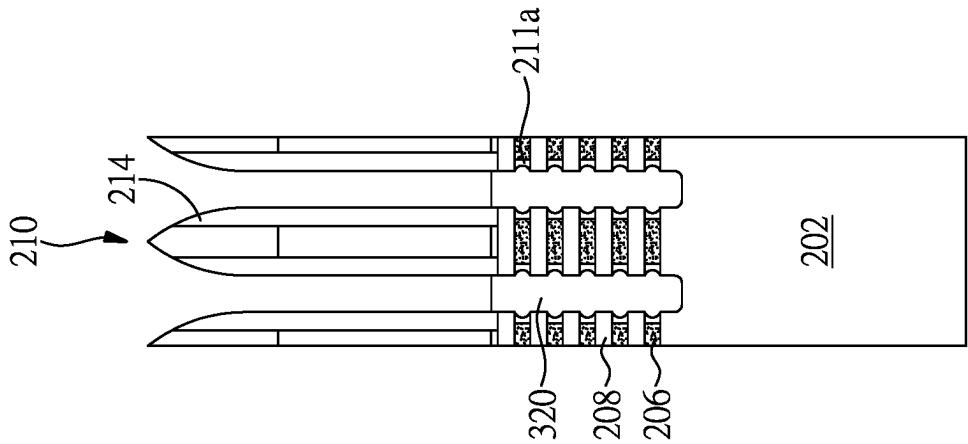


FIG. 19E

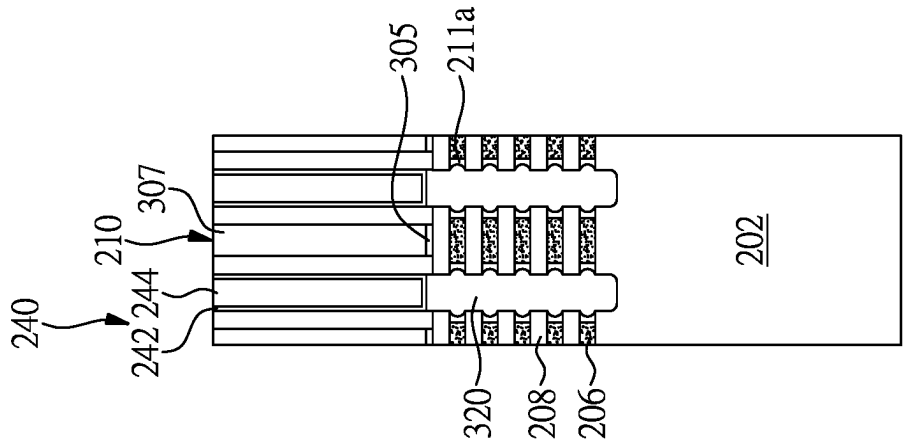


FIG. 19F

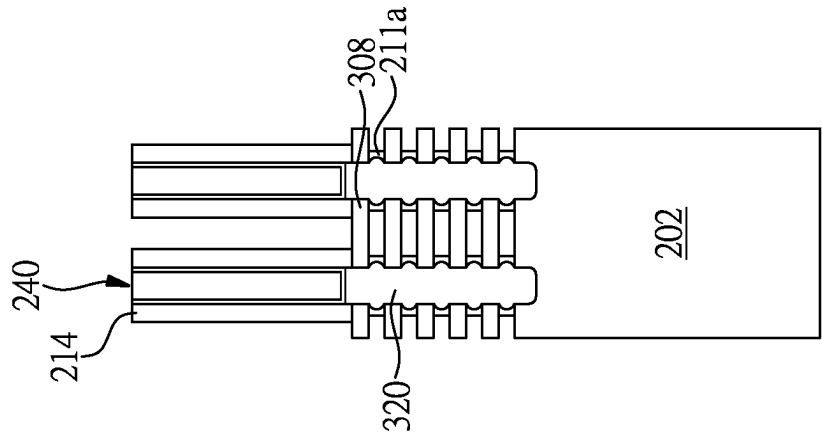


FIG. 19G

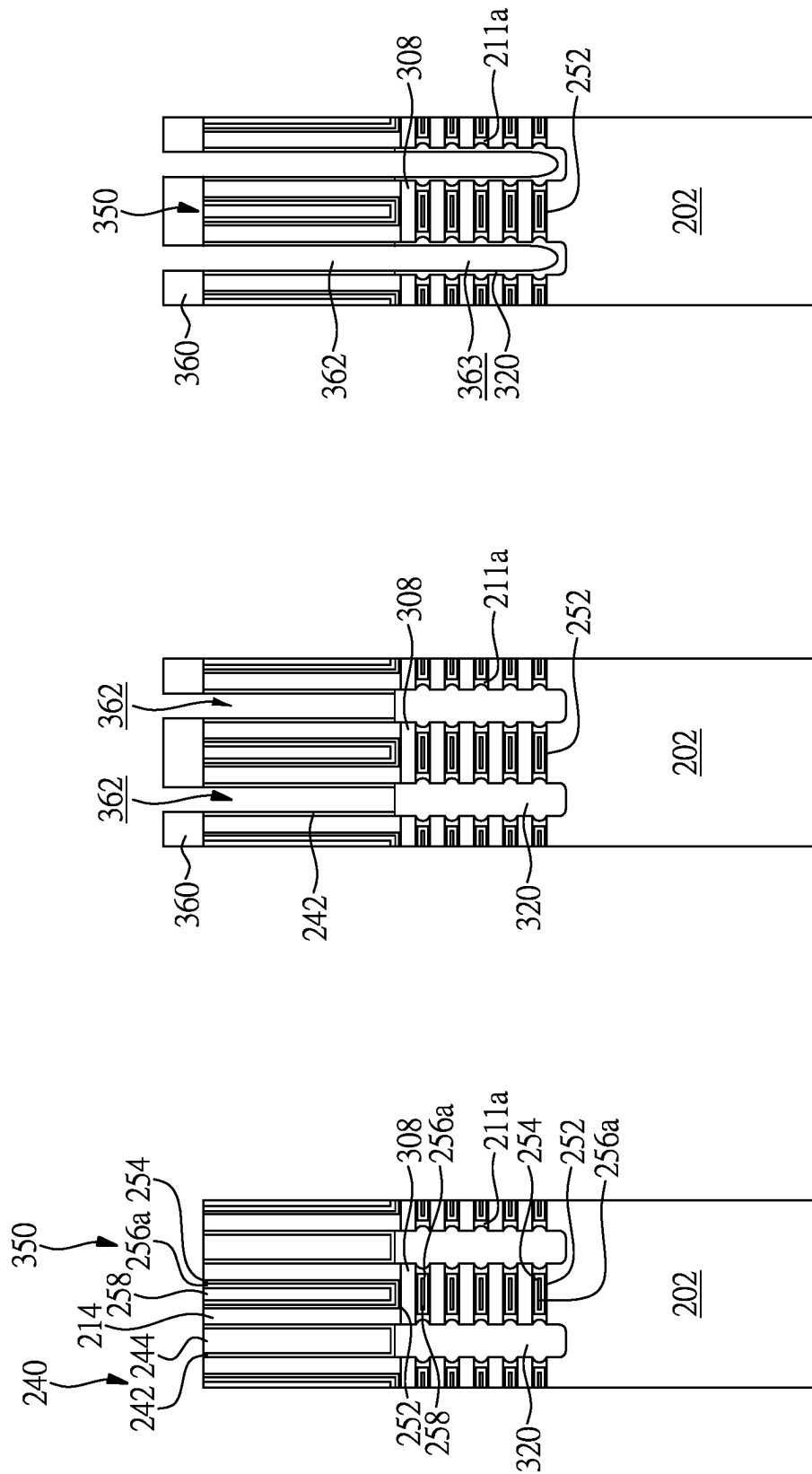


FIG. 19H

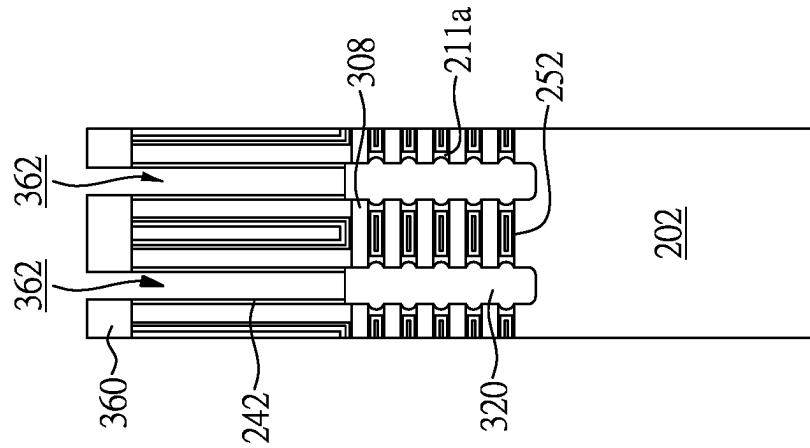


FIG. 19I

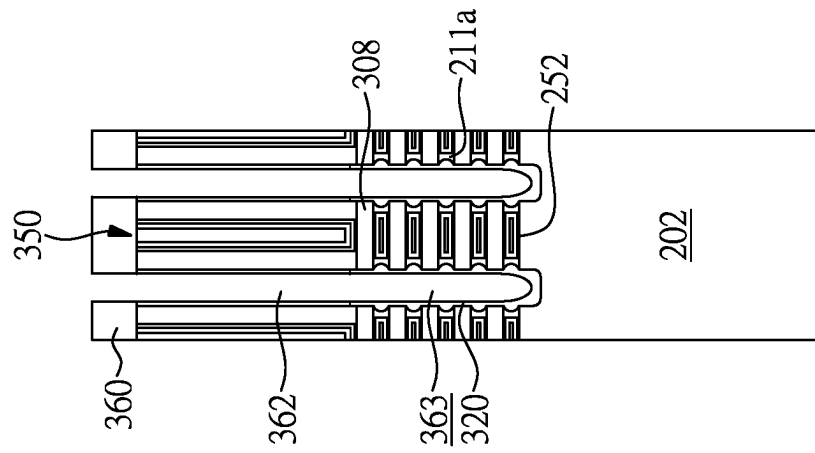


FIG. 19J

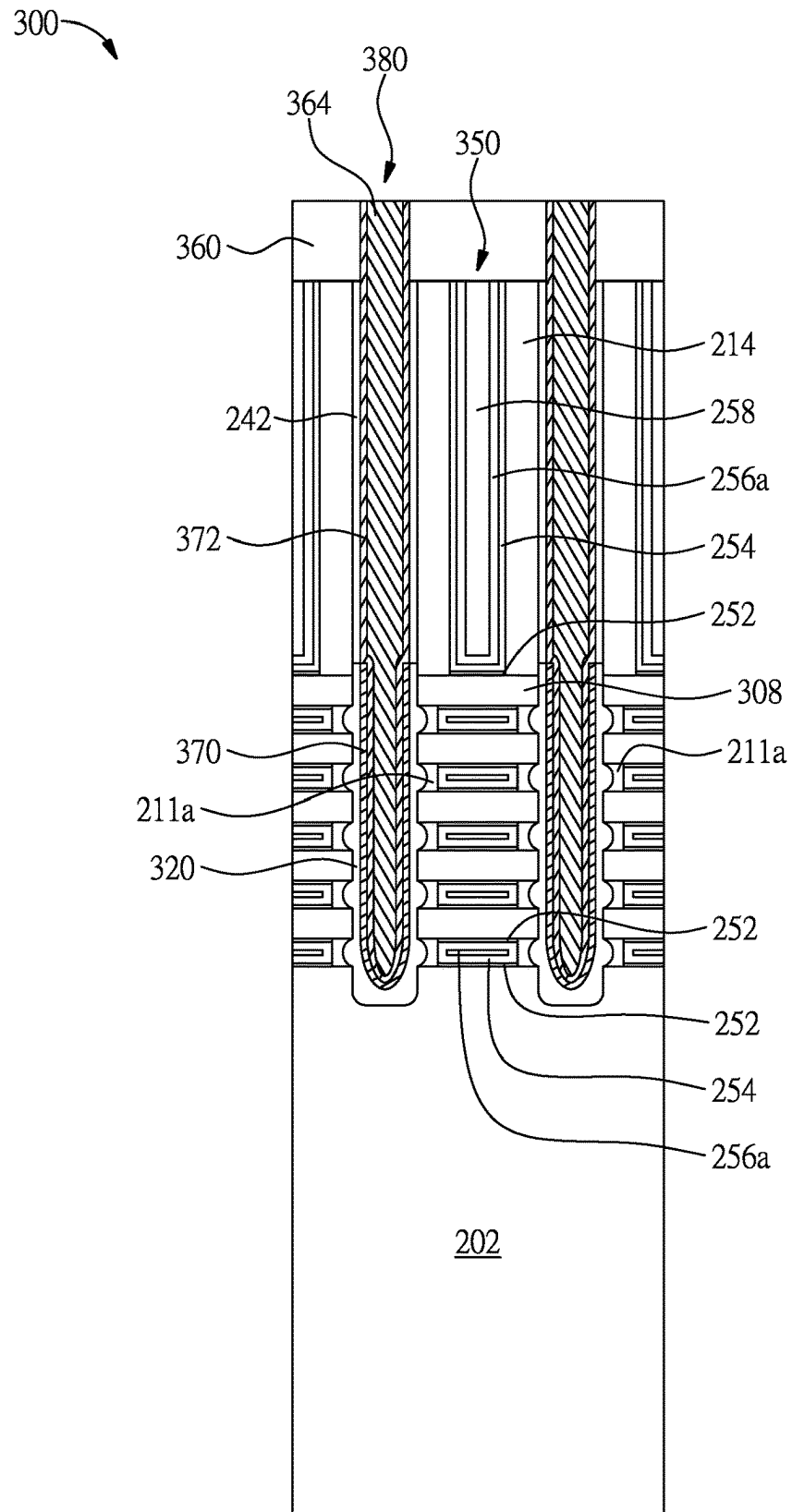


FIG. 19K

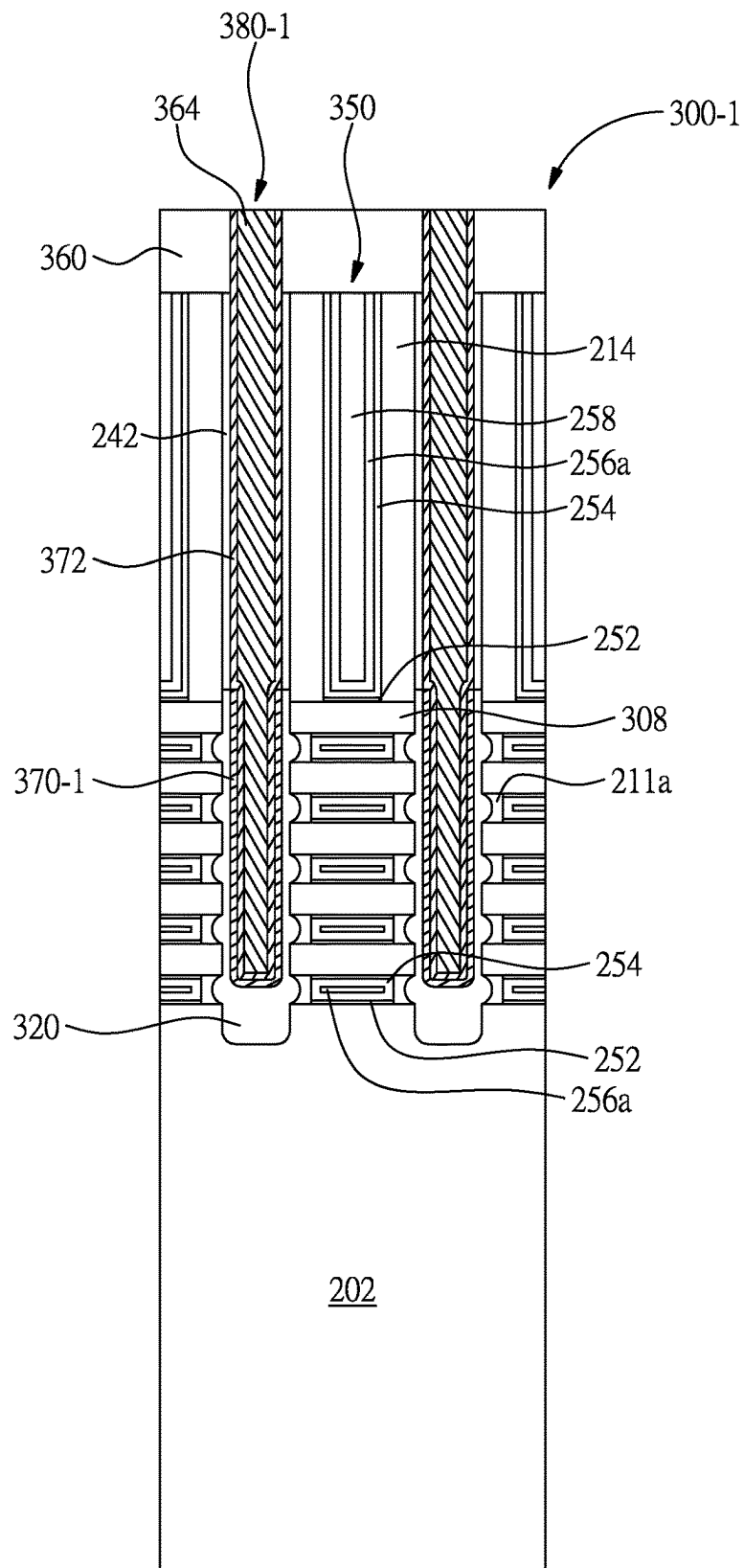


FIG. 20

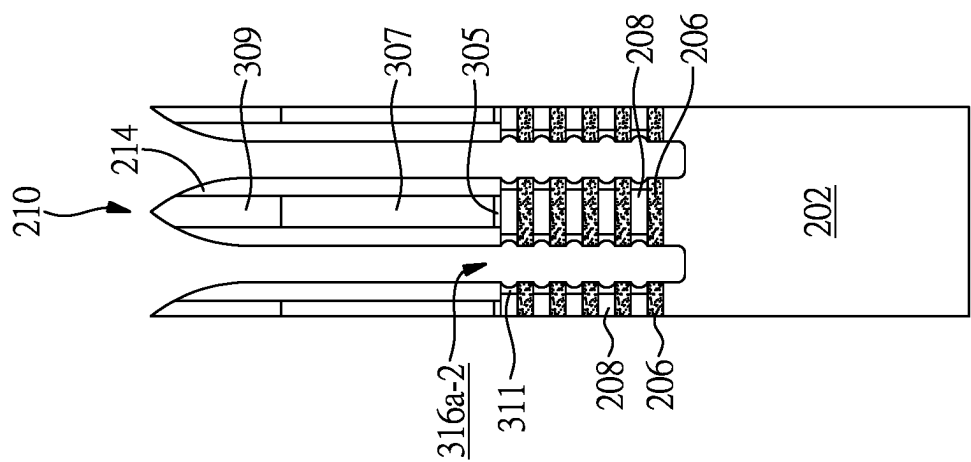


FIG. 21A

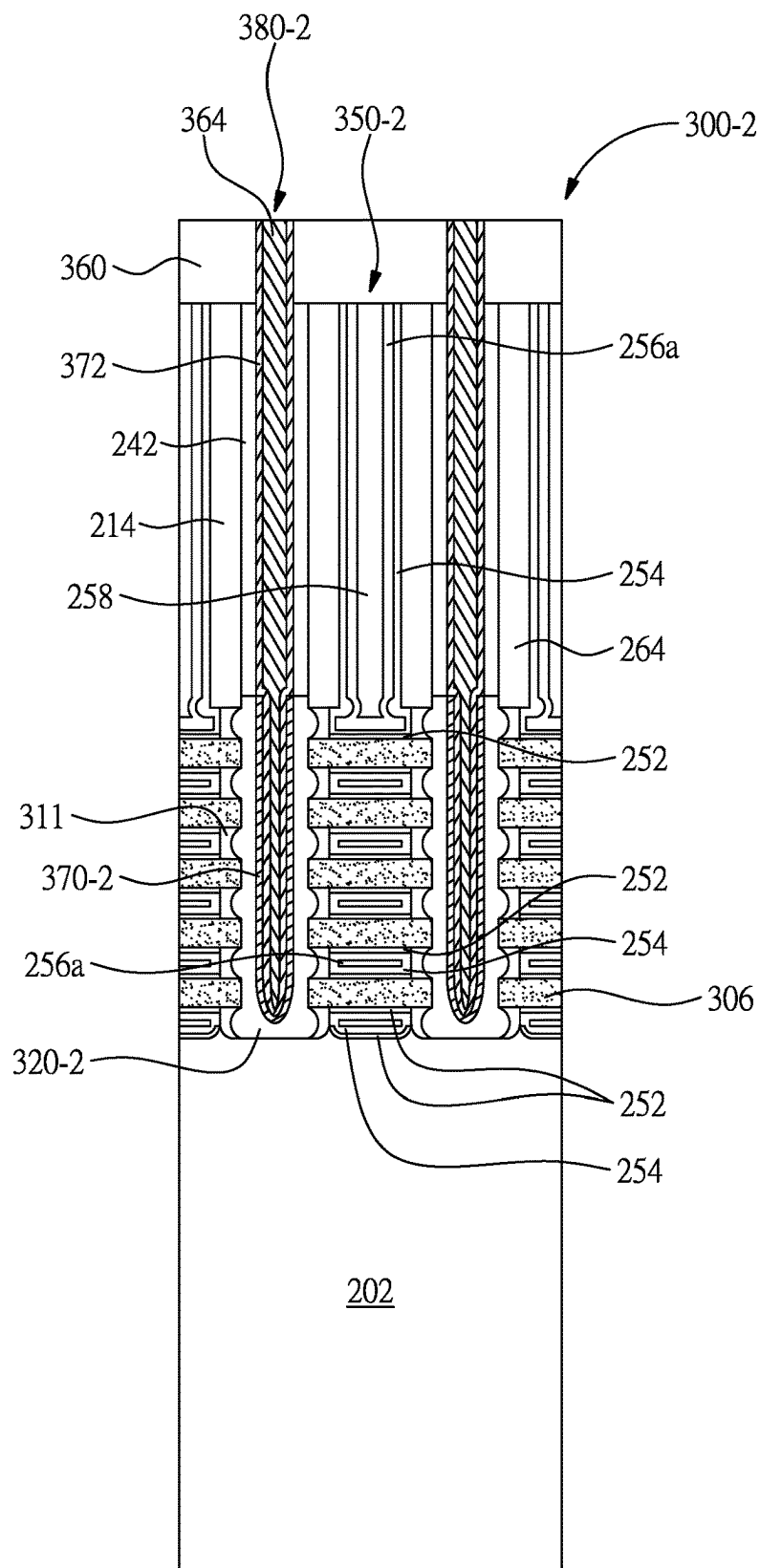


FIG. 21B

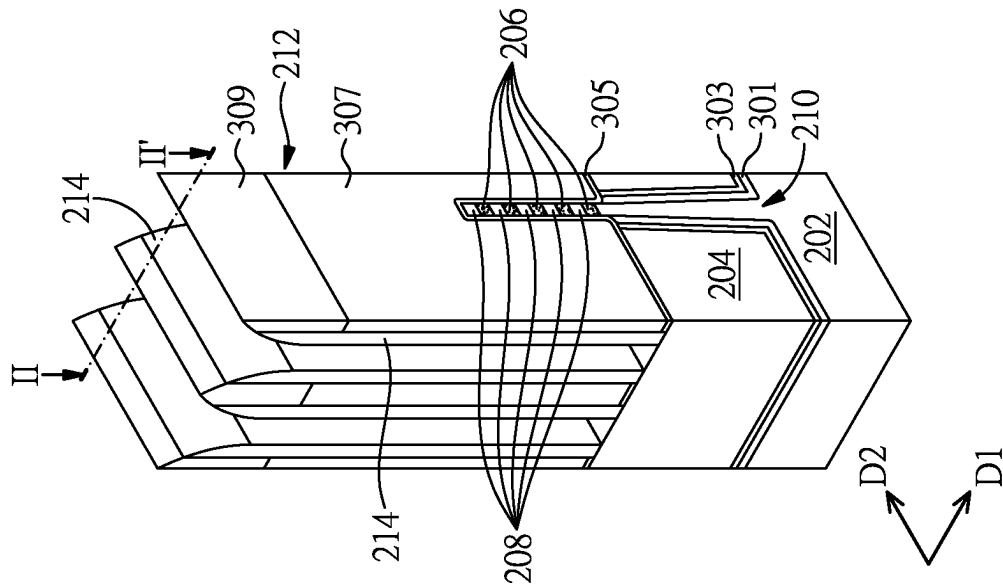


FIG. 22B

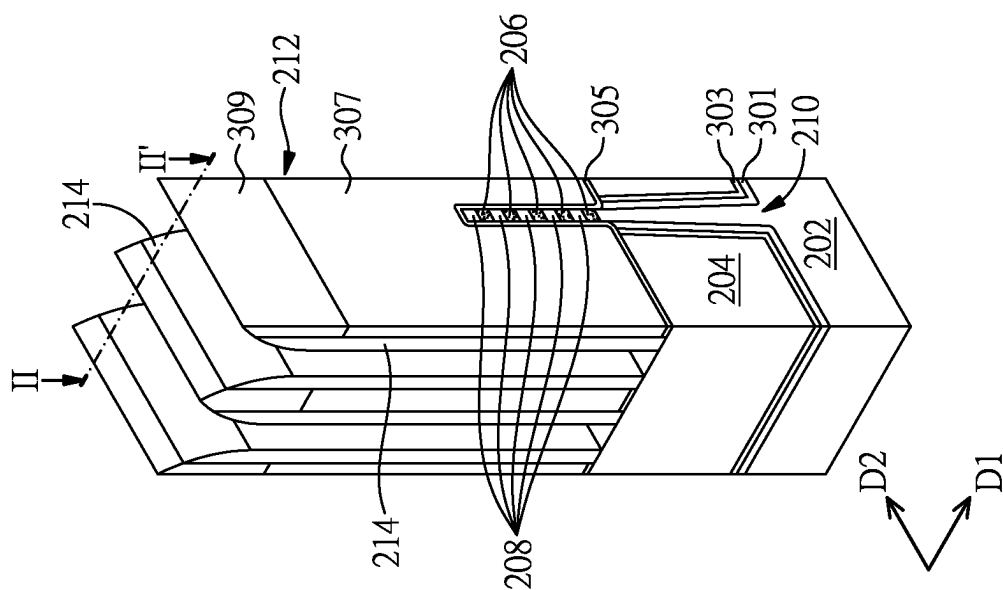


FIG. 22A

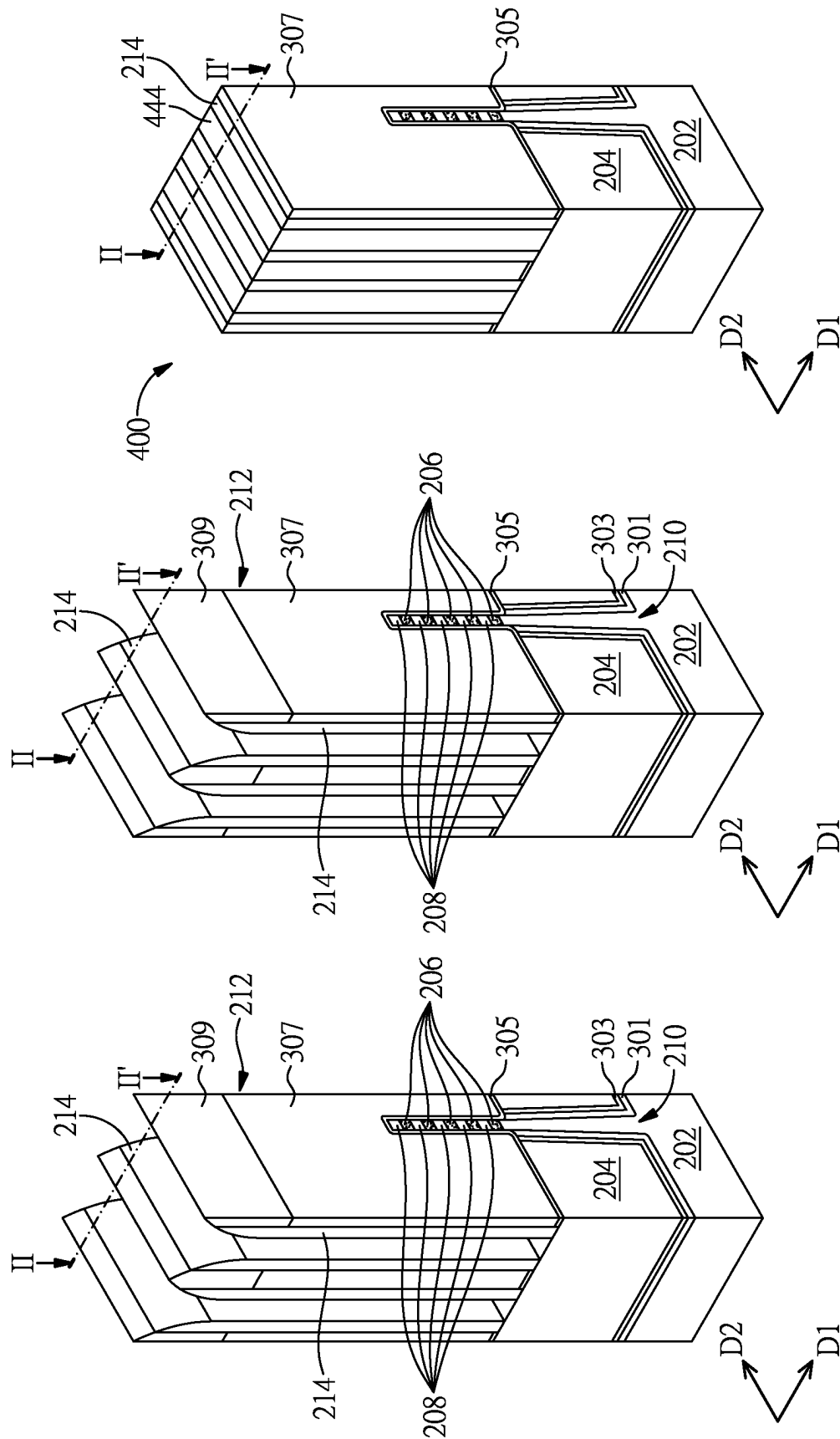


FIG. 22C

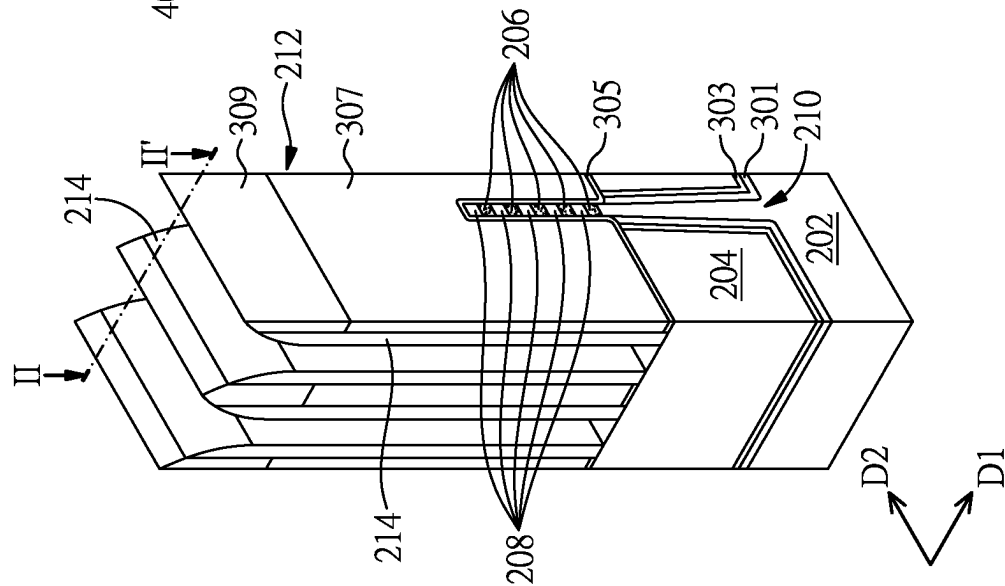


FIG. 22D

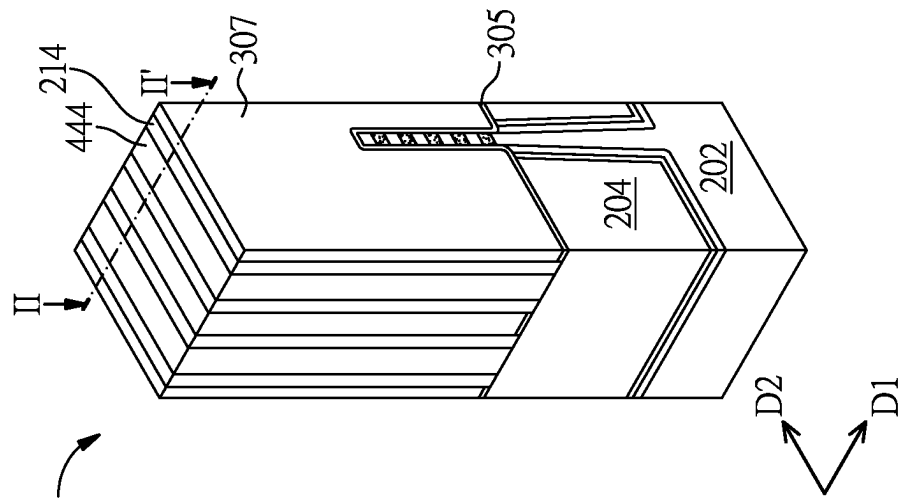


FIG. 22E

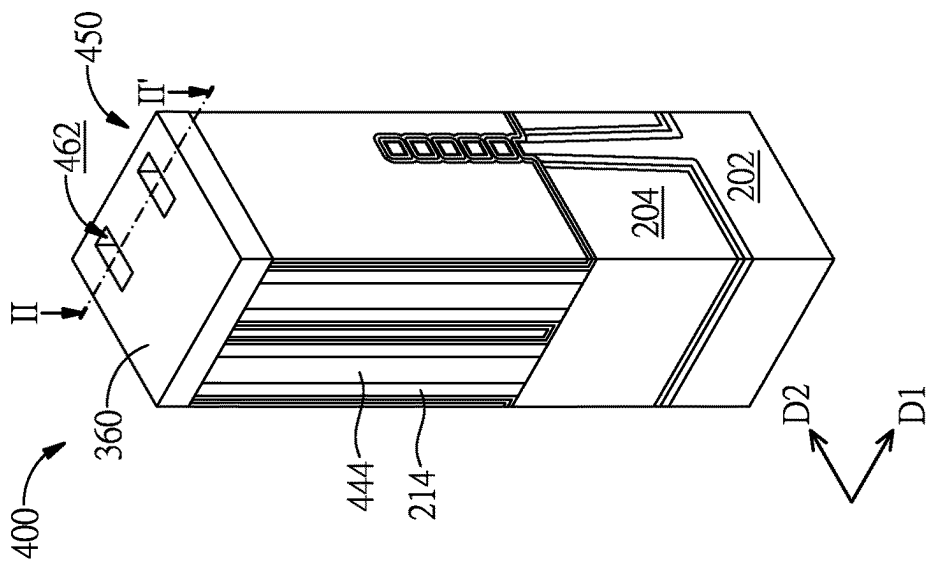


FIG. 22H

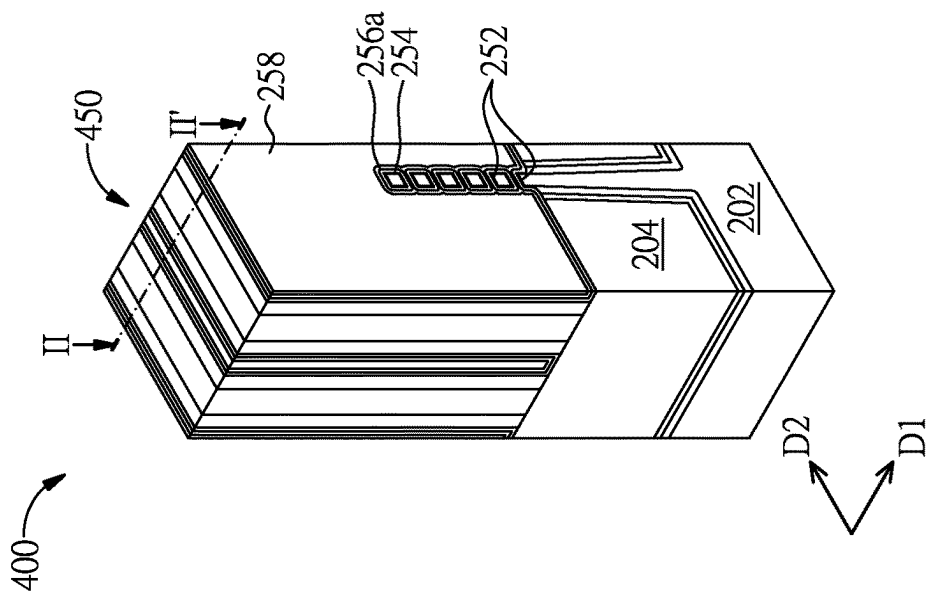


FIG. 22G

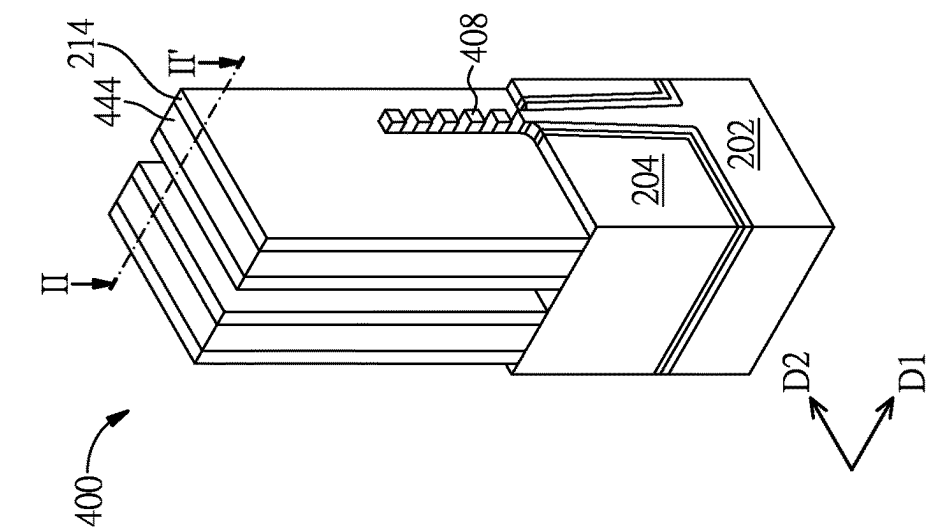


FIG. 22F

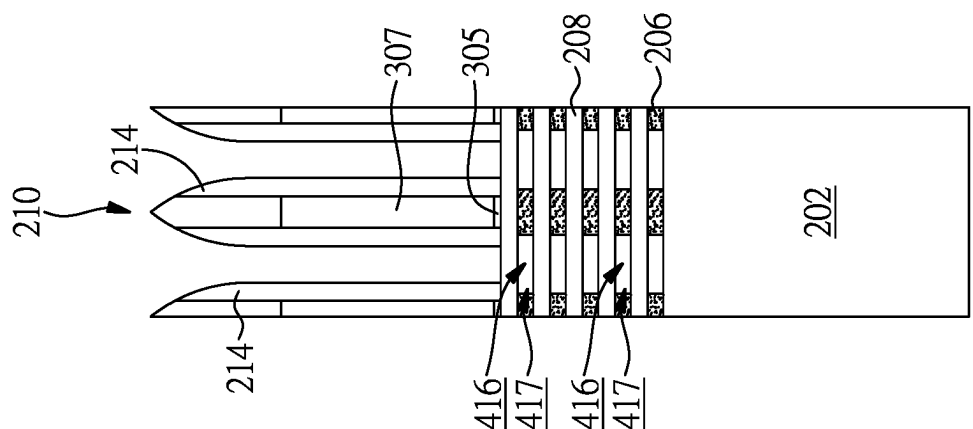


FIG. 23B

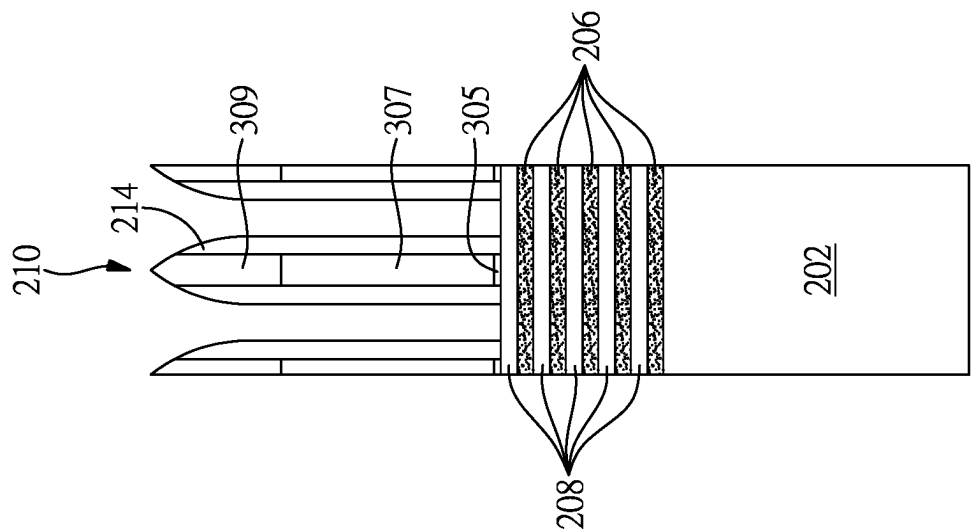


FIG. 23A

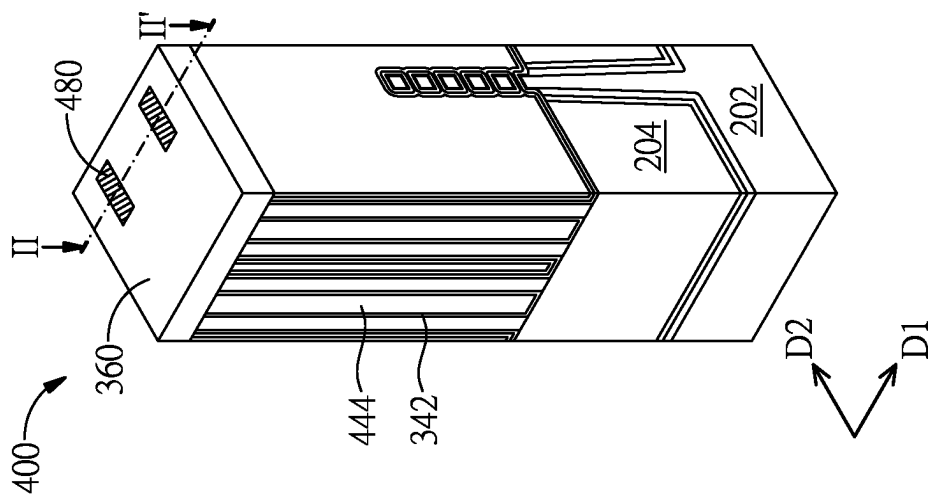


FIG. 22I

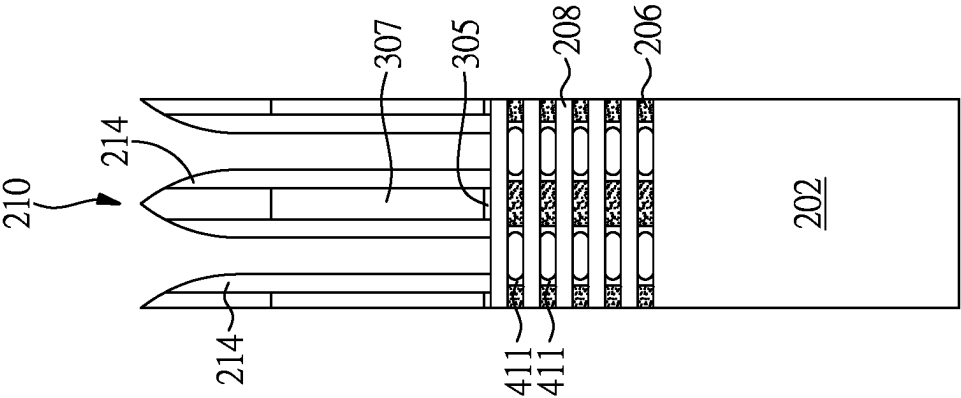


FIG. 23C

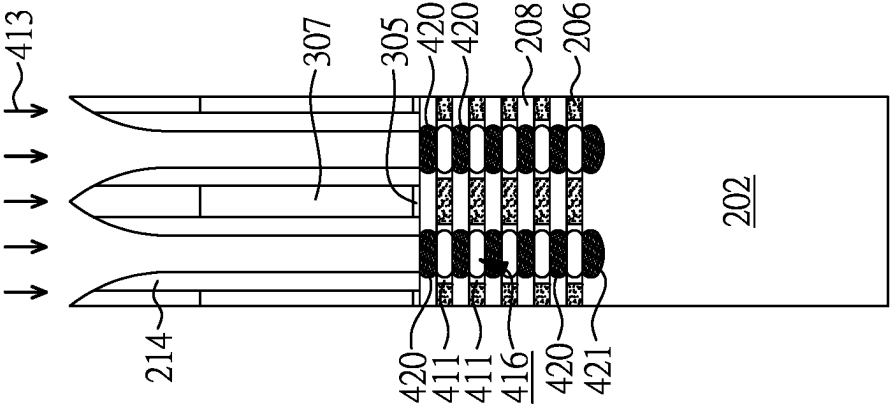


FIG. 23D

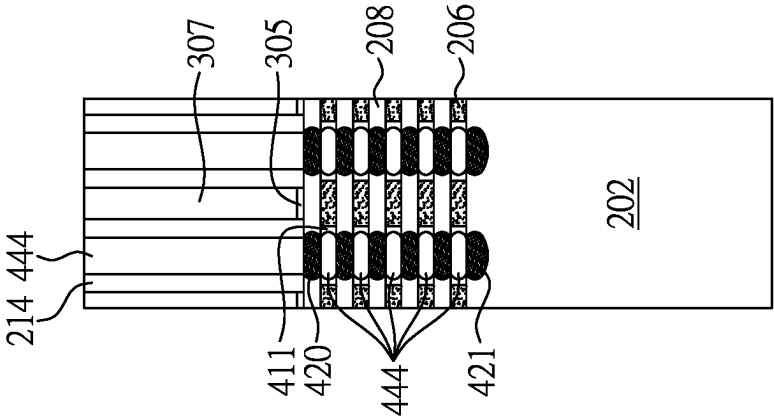


FIG. 23E

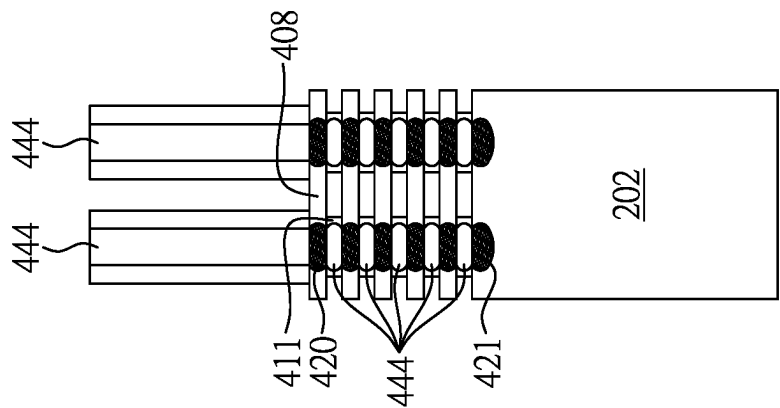


FIG. 23F

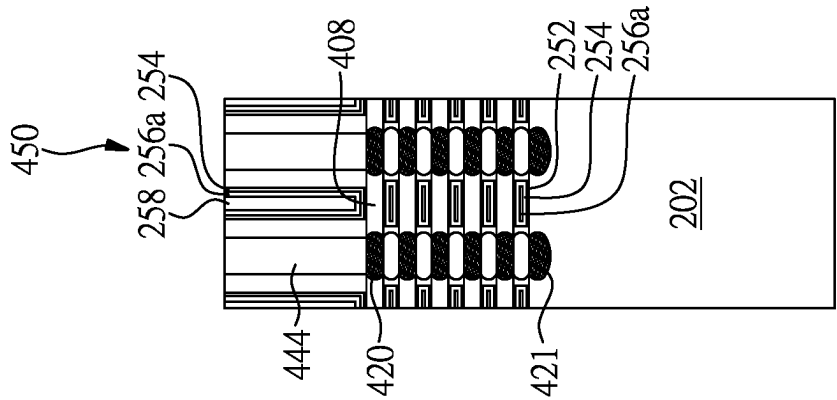


FIG. 23G

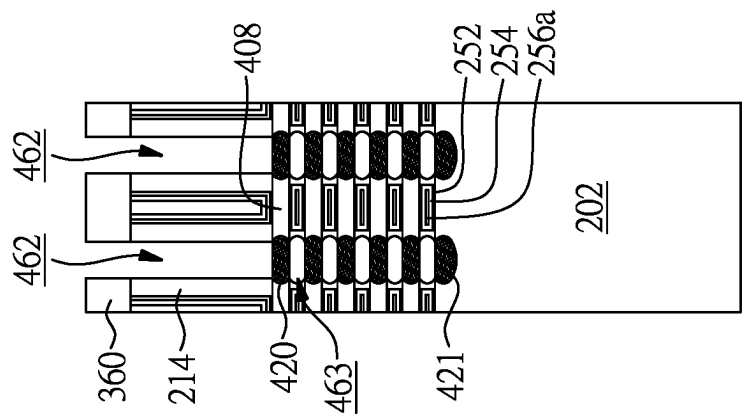


FIG. 23H

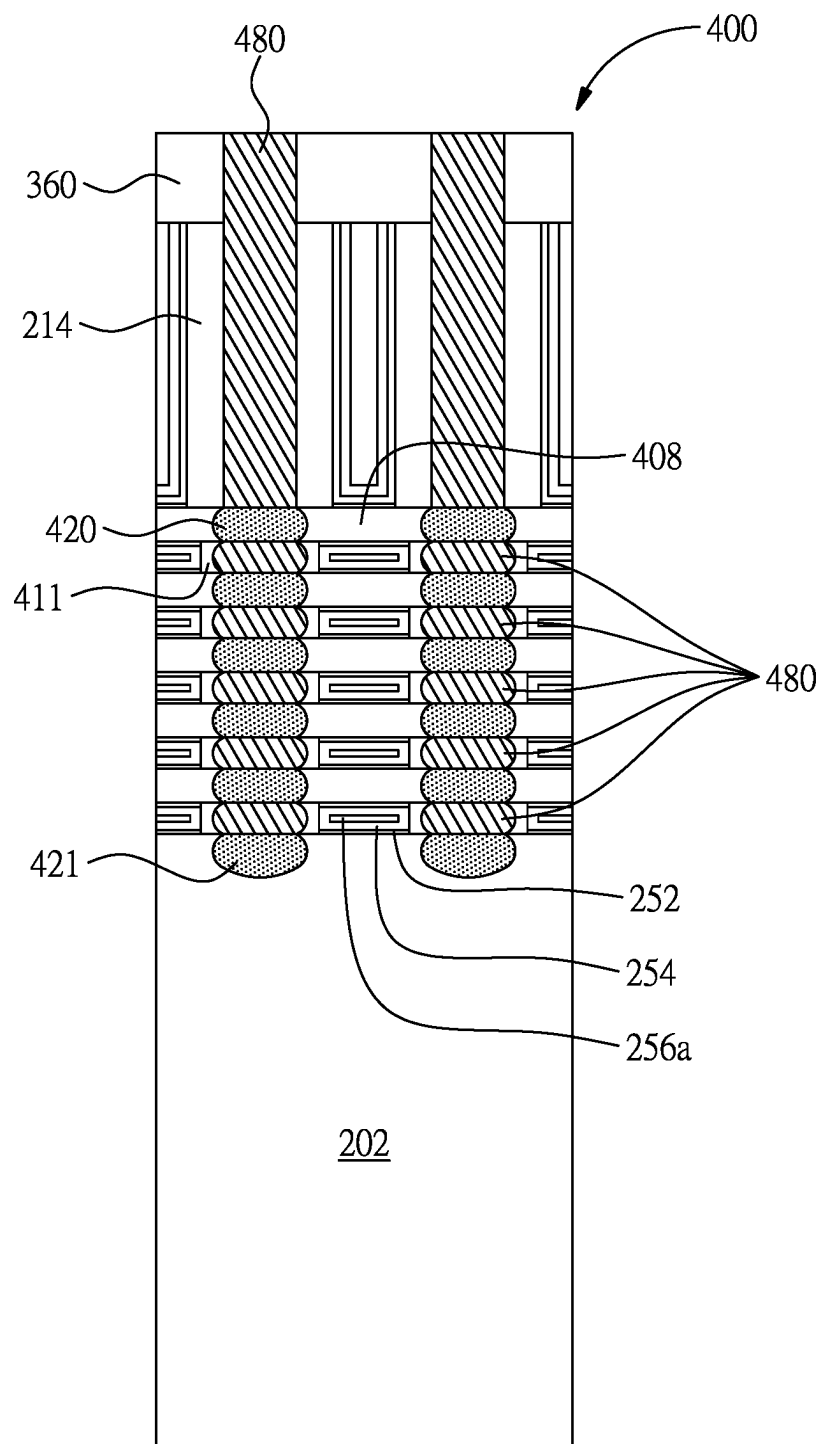


FIG. 23I

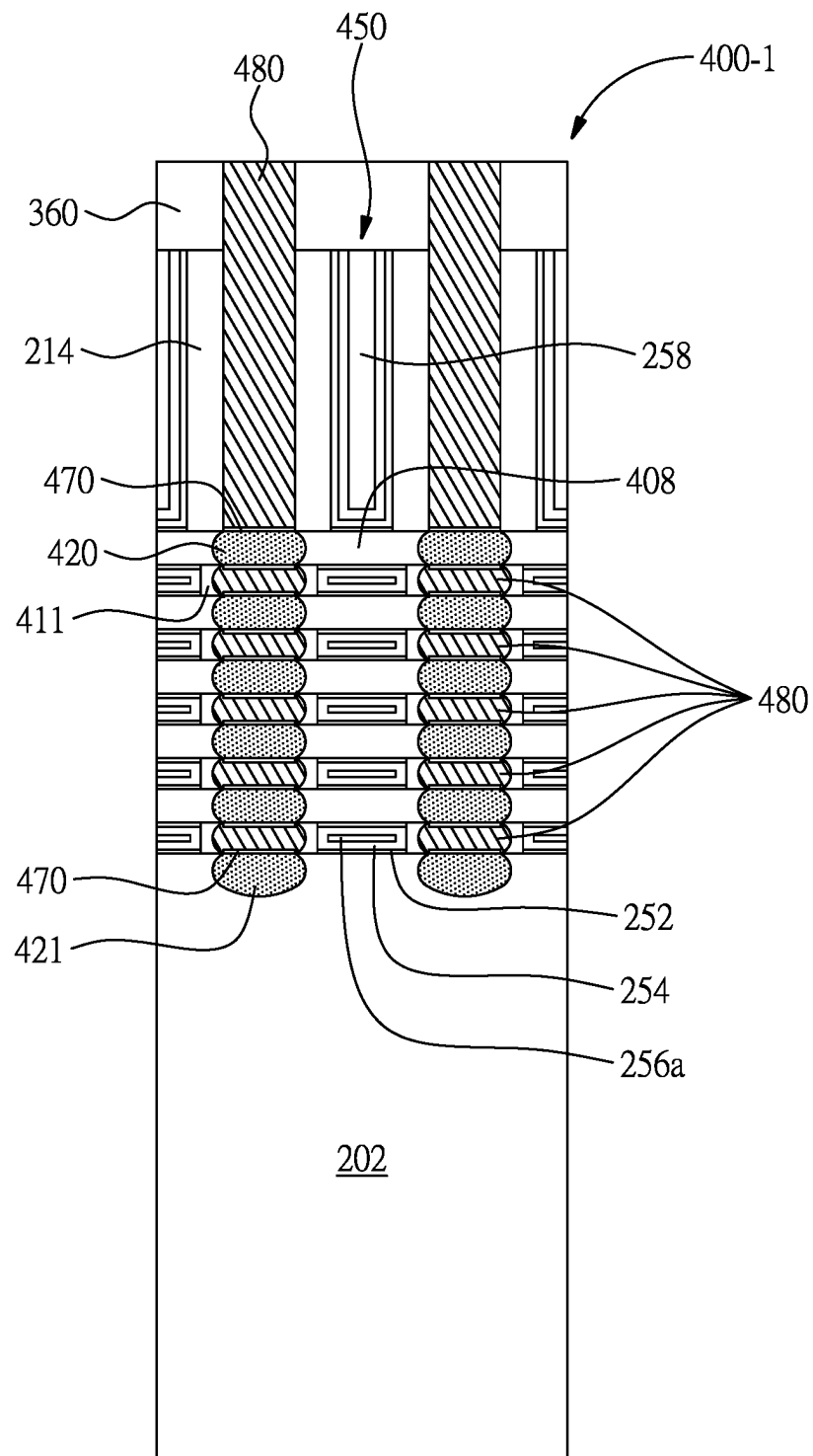


FIG. 24

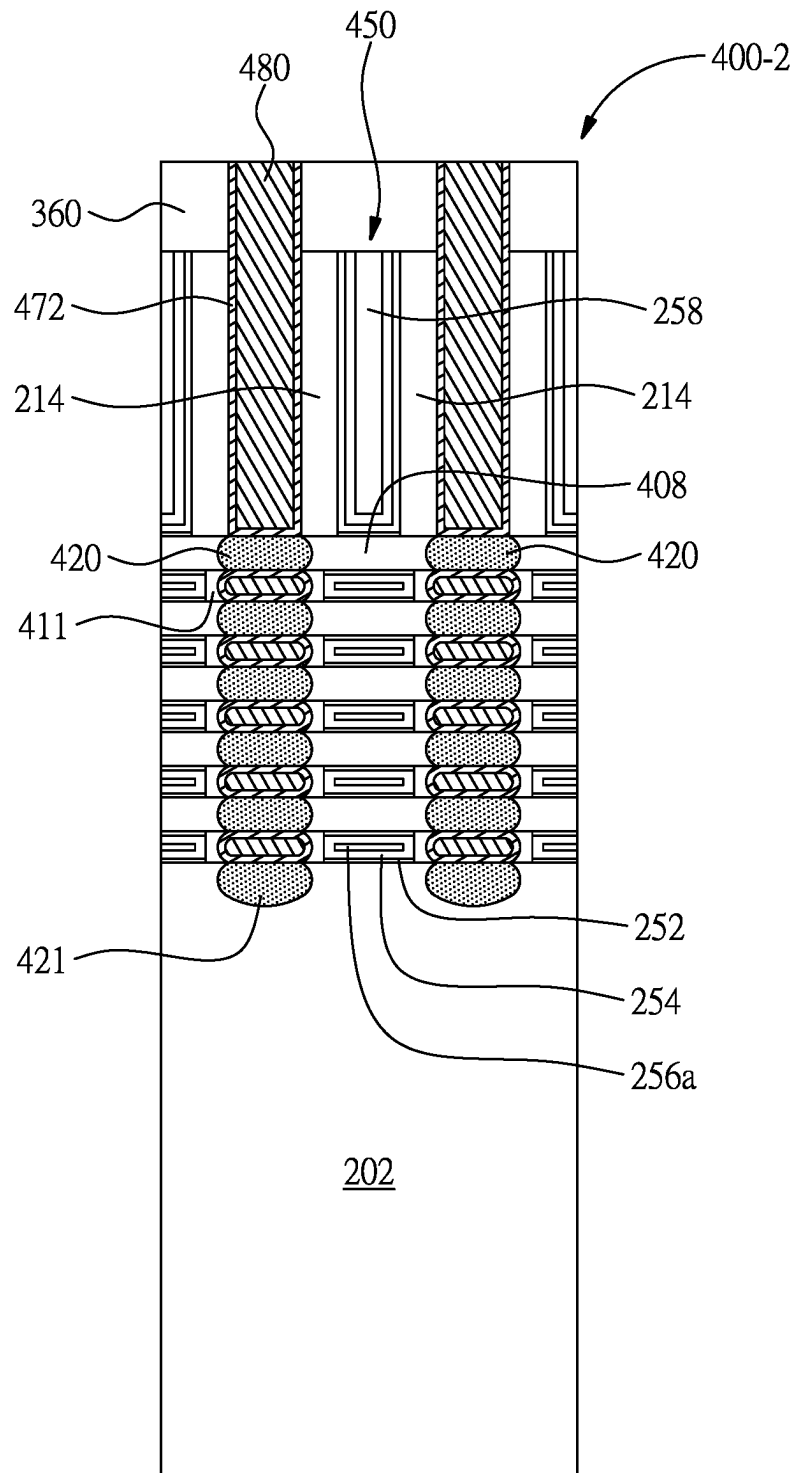


FIG. 25

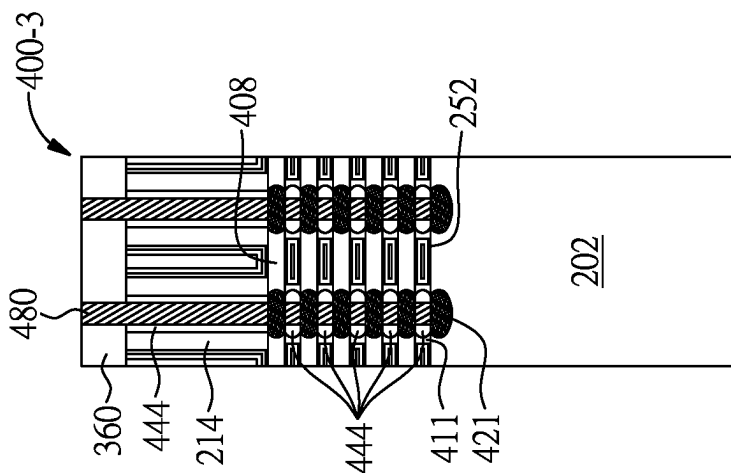


FIG. 26B

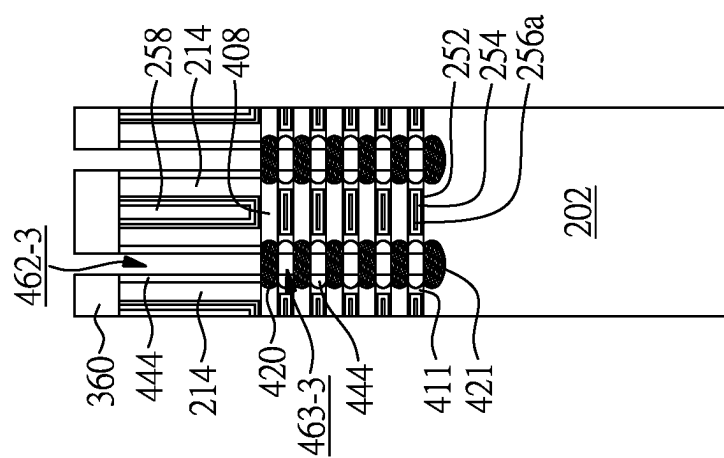


FIG. 26A

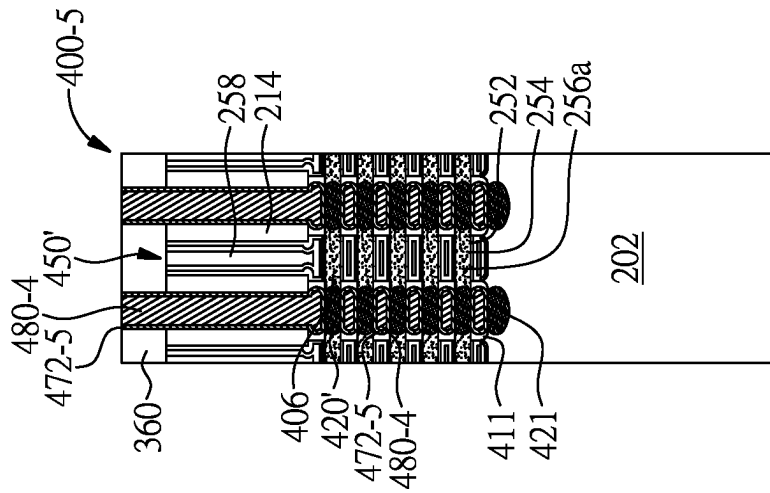


FIG. 28

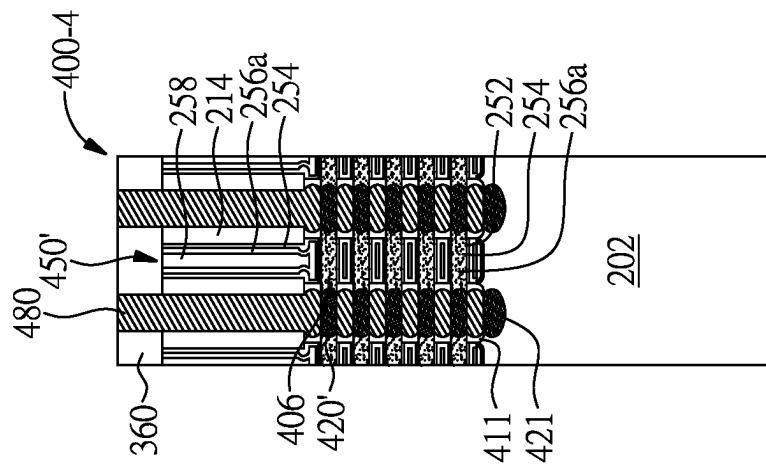


FIG. 27

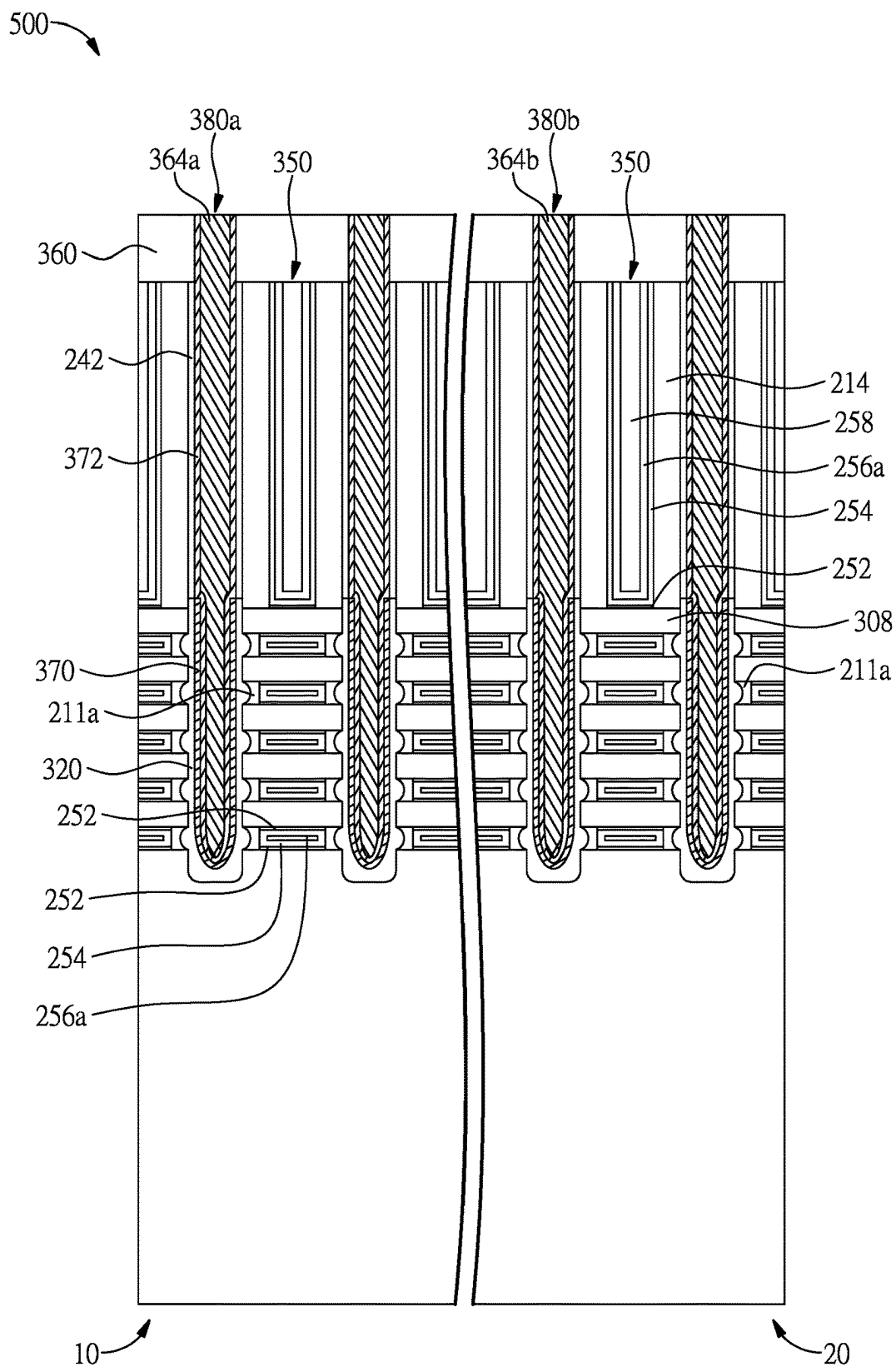


FIG. 29

SEMICONDUCTOR STRUCTURE WITH EXTENDED CONTACT STRUCTURE

PRIORITY CLAIM AND CROSS-REFERENCE

This application is a Divisional application of U.S. patent application Ser. No. 17/666,051, filed on Feb. 7, 2022, which is a Divisional application of U.S. patent application Ser. No. 16/868,625, filed on May 7, 2020, which is a Continuation-in-part application of U.S. patent application Ser. No. 16/681,097, filed on Nov. 12, 2019, which is a Continuation application of U.S. patent application Ser. No. 15/979,123, filed on May 14, 2018, the entirety of which are incorporated by reference herein.

BACKGROUND

As the semiconductor industry develops smaller and smaller nanoscale products and processes in pursuit of higher device density, higher performance, and lower costs, the challenges of downscaling both fabrication and design have led to the development of three-dimensional designs, such as multi-gate field effect transistor (FET) including a fin FET (FinFET) and a gate-all-around (GAA) FET. In a FinFET, a gate electrode is positioned adjacent to three side surfaces of a channel region with a gate dielectric layer interposed therebetween. Because the gate structure surrounds the fin on three sides, the transistor essentially has three gates controlling the current through the fin or channel region. However, the fourth side, the bottom part of the channel region, is positioned far away from the gate electrode and thus is not under close gate control. In contrast to a FinFET, a GAA FET includes an arrangement wherein all side surfaces of the channel region are surrounded by the gate electrode, allowing fuller depletion in the channel region and resulting in fewer short-channel effects due to a steeper sub-threshold current swing (SS) and smaller drain induced barrier lower (DIBL).

Although existing GAA FET devices and methods of fabricating GAA FET devices have been generally adequate for their intended purpose, such devices and methods have not been entirely satisfactory in all aspects.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1 shows a flow chart representing a method for forming a multi-gate semiconductor structure according to aspects of the present disclosure.

FIG. 2 shows a flow chart representing a method for forming a multi-gate semiconductor structure according to aspects of the present disclosure.

FIGS. 3A, 4A, 5A, 6A, 7A, 8A, 9A, 10A, 11A, 12A, 13A, 14A and 15A illustrate a multi-gate semiconductor device at various fabrication stages constructed according to aspects of one or more embodiments of the present disclosure.

FIGS. 3B, 4B, 5B, 6B, 7B, 8B, 9B, 10B, 11B, 12B, 13B, 14B and 15B are cross-sectional views taken along line I-I' of FIGS. 3A, 4A, 5A, 6A, 7A, 8A, 9A, 10A, 11A, 12A, 13A, 14A and 15A, respectively, according to aspects of one or more embodiments of the present disclosure.

FIGS. 3C, 4C, 5C, 6C, 7C, 8C, 9C, 10C, 11C, 12C, 13C, 14C and 15C are cross-sectional views taken along line II-II' of FIGS. 3A, 4A, 5A, 6A, 7A, 8A, 9A, 10A, 11A, 12A, 13A, 14A and 15A, respectively, according to aspects of one or more embodiments of the present disclosure.

FIGS. 3D, 4D, 5D, 6D, 7D, 8D, 9D, 10D, 11D, 12D, 13D, 14D, and 15D are cross-sectional views taken along line III-III' of FIGS. 3A, 4A, 5A, 6A, 7A, 8A, 9A, 10A, 11A, 12A, 13A, 14A and 15A, respectively, according to aspects of one or more embodiments of the present disclosure.

FIGS. 4E, 5E, 6E, 7E, 8E, 9E, 10E, 11E, 12E, 13E, 14E, and 15E are cross-sectional views taken along line IV-IV' of FIGS. 4A, 5A, 6A, 7A, 8A, 9A, 10A, 11A, 12A, 13A, 14A and 15A, respectively, according to aspects of one or more embodiments of the present disclosure.

FIGS. 16A and 17A illustrate a multi-gate semiconductor device at various fabrication stages constructed according to aspects of one or more embodiments of the present disclosure.

FIGS. 16B and 17B are cross-sectional views taken along line I-I' of FIGS. 16A and 17A, respectively, according to aspects of one or more embodiments of the present disclosure.

FIGS. 16C and 17C are cross-sectional views taken along line II-II' of FIGS. 16A and 17A, respectively, according to aspects of one or more embodiments of the present disclosure.

FIGS. 16D and 17D are cross-sectional views taken along line III-III' of FIGS. 16A and 17A, respectively, according to aspects of one or more embodiments of the present disclosure.

FIGS. 16E and 17E are cross-sectional view taken along line IV-IV' of FIGS. 16A and 17A, respectively, according to aspects of one or more embodiments of the present disclosure.

FIGS. 18A to 18K illustrate perspective views of intermediate stages of manufacturing a semiconductor structure in accordance with some embodiments.

FIGS. 19A to 19K are cross-sectional views taken along line II-II' of FIGS. 18A to 18K respectively in accordance with some embodiments.

FIG. 20 illustrates a cross-sectional representation of a semiconductor structure in accordance with some embodiments.

FIGS. 21A and 21B illustrate cross-sectional representation of various stages for forming a semiconductor structure in accordance with some embodiments.

FIGS. 22A to 22I illustrate perspective views of intermediate stages of manufacturing a semiconductor structure in accordance with some embodiments.

FIGS. 23A to 23I are cross-sectional views taken along line II-II' of FIGS. 22A to 22I respectively in accordance with some embodiments.

FIG. 24 illustrates a cross-sectional representation of a semiconductor structure in accordance with some embodiments.

FIG. 25 illustrates a cross-sectional representation of a semiconductor structure in accordance with some embodiments.

FIGS. 26A and 26B illustrate cross-sectional representations of intermediate stages of manufacturing a semiconductor structure in accordance with some embodiments.

FIG. 27 illustrates a cross-sectional representation of semiconductor structures in accordance with some embodiments.

FIG. 28 illustrates a cross-sectional representation of a semiconductor structure in accordance with some embodiments.

FIG. 29 illustrates a cross-sectional representation of a semiconductor structure in accordance with some embodiments.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Furthermore, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. It should be understood that additional operations can be provided before, during, and after the method, and some of the operations described can be replaced or eliminated for other embodiments of the method.

As used herein, the terms such as “first,” “second” and “third” describe various elements, components, regions, layers and/or sections, but these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another. The terms such as “first,” “second” and “third” when used herein do not imply a sequence or order unless clearly indicated by the context.

As used herein, the terms “approximately,” “substantially,” “substantial” and “about” are used to describe and account for small variations. When used in conjunction with an event or circumstance, the terms can refer to instances in which the event or circumstance occurs precisely as well as instances in which the event or circumstance occurs to a close approximation. For example, when used in conjunction with a numerical value, the terms can refer to a range of variation of less than or equal to $\pm 10\%$ of that numerical value, such as less than or equal to $\pm 5\%$, less than or equal to $\pm 4\%$, less than or equal to $\pm 3\%$, less than or equal to $\pm 2\%$, less than or equal to $\pm 1\%$, less than or equal to $\pm 0.5\%$, less than or equal to $\pm 0.1\%$, or less than or equal to $\pm 0.05\%$. For example, two numerical values can be deemed to be “substantially” the same or equal if a difference between the values is less than or equal to $\pm 10\%$ of an average of the values, such as less than or equal to $\pm 5\%$, less than or equal to $\pm 4\%$, less than or equal to $\pm 3\%$, less than or equal to $\pm 2\%$,

less than or equal to $\pm 1\%$, less than or equal to $\pm 0.5\%$, less than or equal to $\pm 0.1\%$, or less than or equal to $\pm 0.05\%$. For example, “substantially” parallel can refer to a range of angular variation relative to 0° that is less than or equal to $\pm 10^\circ$, such as less than or equal to $\pm 5^\circ$, less than or equal to $\pm 4^\circ$, less than or equal to $\pm 3^\circ$, less than or equal to $\pm 2^\circ$, less than or equal to $\pm 1^\circ$, less than or equal to $\pm 0.5^\circ$, less than or equal to $\pm 0.1^\circ$, or less than or equal to $\pm 0.05^\circ$. For example, “substantially” perpendicular can refer to a range of angular variation relative to 90° that is less than or equal to $\pm 10^\circ$, such as less than or equal to $\pm 5^\circ$, less than or equal to $\pm 4^\circ$, less than or equal to $\pm 3^\circ$, less than or equal to $\pm 2^\circ$, less than or equal to $\pm 1^\circ$, less than or equal to $\pm 0.5^\circ$, less than or equal to $\pm 0.1^\circ$, or less than or equal to $\pm 0.05^\circ$.

The gate-all-around (GAA) transistor structures may be patterned by any suitable method. For example, the structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the GAA structure.

GAA transistor structures may include nanowire structures, which are a promising candidate for logic device applications in future technologies. While downscaling device pitch, external resistance of source/drain and metal contact becomes a dominant factor in determining the device performance, more of a factor than channel resistance. As circuit density and device density increase, metal contact dimensions have to be decreased accordingly in order to minimize the ratio of contact area to the total chip area. Contact resistance is normally inverse to contact area. That is, a smaller contact area will correspond to a greater contact resistance. Further, within a limited contact area, resistance of a metal contact will be increased not only due to a smaller metal volume in the limited contact area, but also due to the inferior current spreading in metal. This makes contact resistance a significant and sometimes dominant factor in very large scale integration (VLSI) metal system performance.

It is therefore concluded that electrical contacts and associated contact resistance, which are required to conduct both power and signals throughout the integrated circuitry, are important in the manufacturing and subsequent operation of integrated circuit devices.

It should be noted that the present disclosure presents embodiments in the form of multi-gate transistors or fin-type multi-gate transistors referred to herein as FinFET devices. The FinFET devices may be GAA devices, Omega-gate (a-gate) devices, Pi-gate (H-gate) devices, dual-gate devices, tri-gate devices, bulk devices, silicon-on-insulator (SOI) devices, and/or other configurations. One of ordinary skill may recognize other examples of semiconductor devices that may benefit from aspects of the present disclosure.

FIG. 1 is a flow chart representing a method for forming a multi-gate semiconductor structure **10a** according to aspects of the present disclosure. The method **10a** includes an operation **102**, receiving a substrate including at least a first fin structure and a second fin structure. The method **10a** further includes an operation **104**, disposing a dummy gate

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structure over a portion of the first fin structure and a portion of the second fin structure. The method **10a** further includes an operation **106a**, removing portions of the first fin structure exposed through the dummy gate structure to form at least a first recess in the substrate. The method **10a** further includes an operation **108a**, forming a first semiconductor layer in the first recess. The method **10a** further includes an operation **110**, disposing a dielectric structure over the substrate. The method **10a** further includes an operation **112a**, removing a portion of the dummy gate structure to form a first gate trench in the dielectric structure. The method **10a** further includes an operation **114a**, forming a plurality of first nanowires and a first gate structure in the first gate trench. The method **10a** further includes an operation **116a**, removing a portion of the dielectric structure to form a first opening in the dielectric structure. The method **10a** further includes an operation **118a**, forming a first metal silicide layer over the first semiconductor layer in the first opening. The method **10a** further includes an operation **120a**, disposing a metal layer to fill the first opening. The method **10a** will be further described according to one or more embodiments. It should be noted that the operations of the method for forming the multi-gate semiconductor device **10a** may be rearranged or otherwise modified within the scope of the various aspects. It should be further noted that additional processes may be provided before, during, and after the method **10a**, and that some other processes may be only briefly described herein. Thus other implementations are possible within the scope of the various aspects described herein.

FIG. 2 is a flow chart representing a method for forming a multi-gate semiconductor structure **10b** according to aspects of the present disclosure. In some embodiments, the method **10b** and the method **10a** share similar operations, but the disclosure is not limited thereto. The method **10b** includes the operation **102**, receiving a substrate including at least a first fin structure and a second fin structure. The method **10b** further includes the operation **104**, disposing a dummy gate structure over a portion of the first fin structure and a portion of the second fin structure. The method **10b** further includes an operation **106b**, removing portions of the second fin structure exposed through the dummy gate structure to form at least a second recess in the substrate and a plurality of nanowires suspended in the second recess. The method **10b** further includes an operation **108b**, forming a second semiconductor layer surrounding each of the plurality of second nanowires. The method **10b** further includes the operation **110**, disposing a dielectric structure over the substrate. The method **10b** further includes an operation **112b**, removing a portion of the dummy gate structure to form a second gate trench in the dielectric structure. The method **10b** further includes an operation **114b**, forming a plurality of third nanowires and a second gate structure in the second gate trench. The method **10b** further includes an operation **116b**, removing a portion of the dielectric structure to form a second opening in the dielectric structure. The method **10b** further includes an operation **118b**, forming a second metal silicide layer over the second semiconductor layer. The method **10b** further includes an operation **120b**, disposing a metal layer to fill the second opening. The method **10b** will be further described according to one or more embodiments. It should be noted that the operations of the method for forming the multi-gate semiconductor device **10b** may be rearranged or otherwise modified within the scope of the various aspects. Further, the method **10a** and the method **10b** can be integrated, and thus similar operations can be performed simultaneously. In some embodiments,

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operations **116b** and **118b** of the method **10b** are performed after operations **116a** and **118a** of the method **10a**. In other embodiments, operations **116a** and **118a** of the method **10a** and operations **116b** and **118b** of the method **10b** are simultaneously performed. It should be further noted that additional processes may be provided before, during, and after the method **10b**, and that some other processes may be only briefly described herein. Thus other implementations are possible within the scope of the various aspects described herein.

FIGS. 3A, 4A, 5A, 6A, 7A, 8A, 9A, 10A, 11A, 12A, 13A, 14A, and 15A are drawings illustrating a multi-gate semiconductor device **200** at various fabrication stages constructed according to aspects of one or more embodiments of the present disclosure. FIGS. 3B, 4B, 5B, 6B, 7B, 8B, 9B, 10B, 11B, 12B, 13B, 14B, and 15B are cross-sectional views taken along line I-I' of FIGS. 3A, 4A, 5A, 6A, 7A, 8A, 9A, 10A, 11A, 12A, 13A, 14A, and 15A, respectively, according to aspects of one or more embodiments of the present disclosure. FIGS. 3C, 4C, 5C, 6C, 7C, 8C, 9C, 10C, 11C, 12C, 13C, 14C, and 15C are cross-sectional views taken along line II-II' of FIGS. 3A, 4A, 5A, 6A, 7A, 8A, 9A, 10A, 11A, 12A, 13A, 14A, and 15A, respectively, according to aspects of one or more embodiments of the present disclosure. FIGS. 3D, 4D, 5D, 6D, 7D, 8D, 9D, 10D, 11D, 12D, 13D, 14D, and 15D are cross-sectional views taken along line III-III' of FIGS. 3A, 4A, 5A, 6A, 7A, 8A, 9A, 10A, 11A, 12A, 13A, 14A, and 15A, respectively, according to aspects of one or more embodiments of the present disclosure, and FIGS. 4E, 5E, 6E, 7E, 8E, 9E, 10E, 11E, 12E, 13E, 14E, and 15E are cross-sectional views taken along line IV-IV' of FIGS. 4A, 5A, 6A, 7A, 8A, 9A, 10A, 11A, 12A, 13A, 14A, and 15A, respectively, according to aspects of one or more embodiments of the present disclosure. As shown in FIGS. 3A to 3D, a substrate **202** is provided. In some embodiments, the substrate **202** may be a semiconductor substrate such as a silicon substrate. The substrate **202** may also include other semiconductors such as germanium (Ge), silicon carbide (SiC), silicon germanium (SiGe), or diamond. Alternatively, the substrate **202** may include a compound semiconductor and/or an alloy semiconductor. The substrate **202** may include various layers, including conductive or insulating layers formed on a semiconductor substrate. The substrate **202** may include various doping configurations depending on design requirements as is known in the art. For example, different doping profiles (e.g., n wells, p wells) may be formed on the substrate **202** in regions **202a** and **202b** designed for different device types (e.g., n-type field effect transistors (NFET), or p-type field effect transistors (PFET)), as shown in FIGS. 3C and 3D. The suitable doping may include ion implantation of dopants and/or diffusion processes. The substrate **202** typically has isolation features (e.g., shallow trench isolation (STI) features) **204** interposing the regions **202a** and **202b** providing different device types. Further, the substrate **202** may optionally include an epitaxial layer (epi-layer), may be strained for performance enhancement, may include an SOI structure, and/or may have other suitable enhancement features. A stack including semiconductor layers is formed over the substrate **202**. In some embodiments, a strain relaxed buffer (SRB) layer (not shown) can be formed over the substrate **202**. The SRB layer may be different in composition from the substrate **202** in order to create lattice strain at the interface with the substrate **202**. For example, in some embodiments, the substrate **202** includes silicon and is substantially free of germanium while the SRB layer includes SiGe.

Still referring to FIGS. 3A to 3D, a stack including semiconductor layers is formed over the substrate **202**. In embodiments that include an SRB layer disposed on the substrate **202**, the stack of semiconductor layers may be disposed on the SRB layer. The stack of semiconductor layers may include alternating layers of different compositions. For example, in some embodiments, the stack includes semiconductor layers **206** of a first composition alternating with semiconductor layers **208** of a second composition. By way of example, growth of the layers of the stack may be performed by a molecular beam epitaxy (MBE) process, a metalorganic chemical vapor deposition (MOCVD) process, and/or other suitable epitaxial growth processes. Although five semiconductor layers **206** and five semiconductor layers **208** are shown, it should be understood that the stack may include any number of layers of any suitable composition with various examples including between 2 and 10 semiconductor layers **206** and between 2 and 10 semiconductor layers **208**. As explained below, the different compositions of the layers in the stack (e.g., semiconductor layers **206** and semiconductor layers **208**) may be used to selectively process some of the layers. Accordingly, the compositions may have different oxidation rates, etchant sensitivity, and/or other differing properties. The semiconductor layers **206** and **208** may have thicknesses chosen based on device performance considerations. In some embodiments, the semiconductor layers **206** are substantially uniform in thickness, and the semiconductor layers **208** are substantially uniform in thickness. In some embodiments, the thickness of the semiconductor layers **206** can be less than the thickness of the semiconductor layers **208**, but the disclosure is not limited thereto. For example but not limited thereto, the thickness of the semiconductor layers **206** can be approximately 6 nanometers (nm), and the thickness of the semiconductor layers **208** can be approximately 8 nm.

In some embodiments, the semiconductor layers **208** may include a first semiconductor material such as Si while the semiconductor layers **206** may include the first semiconductor material and a second semiconductor material with a lattice constant greater than a lattice constant of the first semiconductor material. For example, the semiconductor layers **206** may include SiGe, but the disclosure is not limited thereto. Additionally, Ge concentration in the semiconductor layers **206** can be less than or equal to approximately 50%, but the disclosure is not limited thereto. In other embodiments, the semiconductor layers **206** may include other materials such as a compound semiconductor such as SiC, gallium arsenide (GaAs), gallium phosphide (GaP), indium phosphide (InP), indium arsenide (InAs), and/or indium antimonide (InSb), an alloy semiconductor such as SiGe, GaAsP, AlInAs, AlGaAs, InGaAs, GaInP, and/or GaInAsP, or combinations thereof. In some embodiments, the semiconductor layers **206** and **208** may be undoped or substantially dopant-free, where, for example, no doping is performed during the epitaxial growth process. Alternatively, the semiconductor layers **206** and **208** may be doped. For example, the semiconductor layers **206** or **208** may be doped with a p-type dopant such as boron (B), aluminum (Al), In, and Ga for forming a p-type channel, or an n-type dopant such as P, As, Sb, for forming an n-type channel.

Still referring to FIGS. 3A to 3D, at least a first fin structure **210a** and at least a second fin structure **210b** are formed over the substrate **202** from the stack of semiconductor layers **206/208**. The first fin structure **210a** and the second fin structure **210b** may be fabricated using suitable operations including photolithography and etch operations.

In some embodiments, forming the first and second fin structures **210a** and **210b** may further include a trim process to decrease the width and/or the height of the first and second fin structures **210a** and **210b**. The trim process may include wet or dry etching processes. The height and width of the first and second fin structures **210a** and **210b** may be chosen based on device performance considerations. Further, the first and second fin structures **210a** and **210b** can extend along a first direction **D1** as shown in FIGS. 3A to 3D. Accordingly, the substrate **202** including the at least one first fin structure **210a** and at least one second fin structure **210b** is received according to operation **102** of the method **10a** and the method **10b**.

Referring to FIGS. 4A to 4E, in some embodiments, a liner **209** can be formed over the first fin structure **210a**, the second fin structure **210b** and the substrate **202**. Next, a dummy gate structure **212** is disposed over a portion of the first fin structure **210a** and a portion of the second fin structure **210b** according to operation **104** of the method **10a** and the method **10b**. The dummy gate structure **212** may be replaced at a later processing stage by a high-K dielectric layer (HK) and metal gate electrode (MG) as discussed below. In some embodiments, the dummy gate structure **212** is formed over the substrate **202** and extends along a second direction **D2**, which is not parallel with the first direction **D1**. Additionally, the first direction **D1** and the second direction **D2** are in the same plane. As shown in FIGS. 4A to 4D, the portion of the first fin structure **210a** underlying the dummy gate structure **212** may be referred to as the channel region, and the portion of the second fin structure **210b** underlying the dummy gate structure **212** may be referred to as the channel region. The dummy gate structure **212** may also define a source/drain region of the first fin structure **210a**, for example, portions of the first fin structure **210a** adjacent to and on opposing sides of the channel region. Similarly, the dummy gate structure **212** may also define a source/drain region of the second fin structure **210b**, for example, portions of the second fin structure **210b** adjacent to and on opposing sides of the channel region. In some embodiments, the dummy gate structure **212** can include at least a polysilicon layer and a patterned hard mask for defining the dummy gate structure.

Still referring to FIGS. 4A to 4E, a spacer **214** can be disposed over sidewalls of the dummy gate structure **212**, and portions of the first and second fin structures **210a** and **210b** are exposed through the dummy gate structure **212** and the spacer **214**. In some embodiments, the spacer **214** includes insulating materials. As shown in FIGS. 4A, 4C and 4D, the sidewalls of the dummy gate structure **212** are covered by the spacer **214**. In some embodiments, portions of the liner **209** can be removed during or after the forming of the spacer **214**, and thus portions of the first and second fin structures **210a** and **210b** are exposed as shown in FIGS. 4C to 4E.

Referring to FIGS. 5A to 5E, next, the portions of the first fin structure **210a** exposed through the dummy gate structure **212** and the spacer **214** are removed according to operation **106a**. In some embodiments, portions of the semiconductor layers **206** and portions of the semiconductor layers **208** exposed through the dummy gate structure **212** and the spacer **214** are removed, thereby forming at least a first recess **216a** in the substrate **202** as shown in FIG. 5C. In some embodiments, a patterned protecting layer (not shown) is formed over the second fin structure **210b** or deposited over the region **202b**. Thus the second fin structure **210b** is protected and impervious to the formation of the first recess **216a**. The semiconductor layers **206**, and the semiconductor

layers **208** are exposed through sidewalls of the first recess **216a** and the substrate **202** is exposed through a bottom of the first recess **216a**. In some embodiments, a portion of each of the exposed semiconductor layers **206** is removed and thus a plurality of notches (not shown) are formed. In some embodiments, an insulating layer (not shown) is formed over the substrate **202** and a suitable etching operation is then performed. Thus, a plurality of inner spacers **211a** are formed in the notches as shown in FIG. 5C. Consequently, the semiconductor layers **208** and the inner spacers **211a** are exposed. In other words, the semiconductor layers **206** are enclosed by the semiconductor layers **208** and the inner spacers **211a**. In some embodiments, the inner spacers **211a** include one or more insulating materials such as SiN, SiO, SiC, SiOC, SiOCN, other materials, or a combination thereof, but the disclosure is not limited thereto.

Still referring to FIGS. 5A to 5E, a first semiconductor layer **220a** is formed in the first recess **216a** according to operation **108a** of the method **10a**. In some embodiments, the first semiconductor layer **220a** is a doped epitaxial semiconductor layer. In some embodiments, the first semiconductor layer **220a** is a phosphorus-doped silicon (SiP) epitaxial layer, but the disclosure is not limited thereto. Additionally, the first semiconductor layer **220a** covers the semiconductor layers **208**, the inner spacers **211a** and the bottom of the first recess **216a**. Subsequently, the patterned protecting layer is removed from the substrate **202** after the forming of the first semiconductor layer **220a** as shown in FIGS. 5A to 5E. In some embodiments, a thickness of the first semiconductor layer **220a** is between approximately 10 nm and approximately 20 nm, but the disclosure is not limited thereto.

Referring to FIGS. 6A to 6E, next, portions of the second fin structure **210b** exposed through the dummy gate structure **212** and the spacer **214** are removed according to operation **106b** of the method **10b**. In some embodiments, portions of the semiconductor layers **208** are removed, thereby forming at least a second recess **216b** in the substrate **202** according to operation **106b**. Significantly, a plurality of nanowires **230**, which previously comprised the semiconductor layers **206**, are formed in the second recess **216b** according to operation **106b**, as shown in FIGS. 6D and 6E. In some embodiments, a patterned protecting layer (not shown) is formed to fill the first recess **216a** or deposited over the region **202a**, and thus the first semiconductor layer **220a** is protected and impervious to the formation of the second recess **216b** and the plurality of nanowires **230**. As shown in FIGS. 6D and 6E, the plurality of nanowires **230** are suspended in and exposed through the second recess **216b**, the semiconductor layers **208** are exposed through sidewalls of the second recess **216b**, and the substrate **202** is exposed through a bottom of the second recess **216b**.

Referring to FIGS. 7A to 7E, a portion of the exposed semiconductor layers **208** is then removed and thus a plurality of notches (not shown) are formed. In some embodiments, an insulating layer (not shown) is formed over the substrate **202** and a suitable etching operation is subsequently performed. Thus, a plurality of inner spacers **211b** are formed in the notches and over the substrate **202**, as shown in FIG. 7D. In other words, the semiconductor layers **208** are enclosed by the semiconductor layers **206** and the inner spacers **211b**. In some embodiments, the inner spacers **211b** include one or more insulating materials such as SiN, SiO, SiC, SiOC, SiOCN, other materials, or a combination thereof, but the disclosure is not limited thereto.

Still referring to FIGS. 7A to 7E, a second semiconductor layer **220b** is formed in the second recess **216b** according to operation **108b** of the method **10b**. The second semiconductor layer **220b** is formed to surround each of the nanowires **230**, as shown in FIG. 7E. In some embodiments, the second semiconductor layer **220b** includes the first semiconductor material and the second semiconductor material. For example but not limited thereto, the second semiconductor layer **220b** can include SiGe, and a Ge concentration of the second semiconductor layer **220b** is greater than the Ge concentration of the plurality of nanowires **230**, which previously comprised the semiconductor layers **206**. In some embodiments, the Ge concentration of the second semiconductor layer **220b** is greater than 50%, but the disclosure is not limited thereto. In some embodiments, the Ge concentration of the second semiconductor layer **220b** is between approximately 50% and approximately 70%, but the disclosure is not limited thereto. In some embodiments, the second semiconductor layer **220b** is a doped epitaxial semiconductor layer. For example but not limited thereto, the second semiconductor layer **220b** can be a boron-doped silicon germanium (SiGeB) epitaxial layer. Further, the patterned protecting layer is removed from the substrate **202** after the forming of the second semiconductor layer **220b**. In some embodiments, the method **10a** and the method **10b** are integrated such that operations **106b** and **108b** of the method **10b** are performed after operations **106a** and **108a** of the method **10a**. However, operations **106b** and **108b** of the method **10b** can be performed before operations **106a** and **108a** of the method **10a** in other embodiments.

Referring to FIGS. 8A to 8E, a dielectric structure **240** is disposed over the substrate **202** according to operation **110** of the method **10a** and the method **10b**. The dielectric structure **240** fills the first recess **216a** and the second recess **216b**. In some embodiments, the dielectric structure **240** can include an etch-stop layer (e.g., a contact etch stop layer (CESL)) **242** and various dielectric layers (e.g., an inter-layer dielectric (ILD) layer) **244** formed on the substrate **202** after the forming of the second semiconductor layer **220b**. In some embodiments, the CESL **242** includes a SiN layer, a SiCN layer, a SiON layer, and/or other materials known in the art. In some embodiments, the ILD layer **244** includes materials such as tetraethylorthosilicate (TEOS) oxide, undoped silicate glass, or doped silicon oxide such as borophosphosilicate glass (BPSG), fused silica glass (FSG), phosphosilicate glass (PSG), boron doped silicon glass (BSG), and/or other suitable dielectric materials. In some embodiments, after the CESL **242** and the ILD layer **244** are deposited, a planarization process, such as a chemical mechanical planarization (CMP) operation, may be performed to form the dielectric structure **240** and to expose a top surface of the dummy gate structure **212** as shown in FIGS. 8A to 8D. In some embodiments, the planarization is performed to expose at least a top surface of the polysilicon layer of the dummy gate structure **212**.

Referring to FIGS. 9A to 9E, a portion of the dummy gate structure **212** is subsequently removed to form a first gate trench **218a** in the dielectric structure **240** according to operation **112a**. In some embodiments, a patterned protecting layer (not shown) is formed over the region **202b**, and thus elements in the region **202b** are protected and impervious to the formation of the first gate trench **218a**. As shown in FIG. 9C, the spacer **214** is exposed through sidewalls of the first gate trench **218a**, and the first fin structure **210a** is exposed through the first gate trench **218a**. Subsequently, the liner layer **209** disposed over the first fin structure **209** is removed, and the semiconductor layers **206** are then

removed. Accordingly, a plurality of nanowires **232**, which previously comprised the semiconductor layers **208**, are formed in the first gate trench **218a** according to operation **114a** of the method **10a**, as shown in FIGS. **9B** and **9C**. Further, the plurality of nanowires **232** serving as channel regions are suspended in the first gate trench **218a**. In some embodiments, the nanowires **232** can be slightly etched to obtain various desirable dimensions and shapes, and the various desired dimensions and shapes may be chosen based on device performance considerations. As shown in FIG. **9C**, the plurality of nanowires **232** and the inner spacers **211a** are therefore exposed through the first gate trench **218a**. The patterned protecting layer is then removed.

Referring to FIGS. **10A** to **10E**, another portion of the dummy gate structure **212** is then removed to form a second gate trench **218b** in the dielectric structure **240** according to operation **112b** of the method **10b**. In some embodiments, another patterned protecting layer (not shown) is formed over the region **202a**, and thus elements in the region **202a** are protected and impervious to the formation of the second gate trench **218b**. As shown in FIG. **10C**, the spacer **214** is exposed through sidewalls of the second gate trench **218b**, and the second fin structure **210b** is exposed through the second gate trench **218b**. Subsequently, the liner layer **209** disposed over the second fin structure **210b** is removed, and the semiconductor layers **208** are removed. Accordingly, a plurality of nanowires **234**, which previously comprised the semiconductor layers **206**, are formed in the second gate trench **218b** according to operation **114b** of the method **10b**, as shown in FIGS. **10B** and **10D**. Further, the plurality of nanowires **234** serving as channel regions are suspended in the second gate trench **218b**. In some embodiments, the nanowires **234** can be slightly etched to obtain various desirable dimensions and shapes, and the various desired dimensions and shapes may be chosen based on device performance considerations. As shown in FIG. **10D**, the plurality of nanowires **234** and the inner spacers **211b** are therefore exposed through the second gate trench **218b**. The patterned protecting layer is then removed. Additionally, the plurality of nanowires **230** and the plurality of nanowires **234**, both of which previously comprised the semiconductor layers **206**, include the same materials. Further, each of the nanowires **234** is coupled to each of the nanowires **230**, as shown in FIG. **10D**. In other words, each of the nanowires **234** is coupled to a corresponding nanowire **230**. In some embodiments, it is referred that the nanowires **230** and nanowires **234** are the same nanowires, as shown in FIG. **10**.

Referring to FIGS. **11A** to **11E**, an interfacial layer (IL) **252** is formed to surround each of the nanowires **232** exposed in the first gate trench **218a** and each of the nanowires **234** exposed in the second gate trench **218b**, as shown in FIG. **11B**. In some embodiments, the IL **252** may include an oxide-containing material such as SiO or SiON. After the forming of the IL **252**, a gate dielectric layer **254** is formed over the IL **252**. As shown in FIG. **11B**, the gate dielectric layer **254** surrounds each of the nanowires **232** and each of the nanowires **234**. In some embodiments, the gate dielectric layer **254** includes a high-k dielectric material having a high dielectric constant, for example, a dielectric constant greater than that of thermal silicon oxide (~3.9). The high-k dielectric material may include hafnium oxide (HfO₂), zirconium oxide (ZrO₂), lanthanum oxide (La₂O₃), aluminum oxide (Al₂O₃), titanium oxide (TiO₂), yttrium oxide (Y₂O₃), strontium titanate (SrTiO₃), hafnium oxynitride (HfO_xN_y), other suitable metal-oxides, or combinations thereof.

Still referring to FIGS. **11A** to **11E**, after the forming of the gate dielectric layer **254**, a first gate conductive layer **256a** is disposed in the first gate trench **218a** according to operation **114a** of the method **10a**, and a second gate conductive layer **256b** is disposed in the second gate trench **218b** according to operation **114b** of the method **10b**. The first and second gate conductive layers **256a** and **256b** are formed on the gate dielectric layer **254**. In some embodiments, the first gate conductive layer **256a** is formed for an n-channel FET, and the second gate conductive layer **256b** is formed for a p-channel FET. In some embodiments, the first gate conductive layer **256a** can include at least a barrier metal layer (not shown) and a first work function layer, and the second gate conductive layer **256b** can include at least a barrier metal layer (not shown) and a second work function metal layer. The barrier metal layer can include, for example but not limited to, TiN. The first work function metal layer, which provides proper work function to the n-channel FET, includes one or more of TaN, TaAlC, TiN, TiC, Co, TiAl, HfTi, TiSi and TaSi, but the disclosure is not limited thereto. The second work function metal layer, which provides proper work function to the p-channel FET, includes one or more of TiAlC, Al, TiAl, TaN, TaAlC, TiN, TiC and Co, but the disclosure is not limited thereto. Next, a gap-filling metal layer **258** is formed to fill the first gate trench **218a** and the second gate trench **218b**. The gap-filling metal layer **258** can include conductive material, such as Al, Cu, AlCu, or W, but is not limited to the above-mentioned materials. Accordingly, a first gate structure **250a** is formed in the first gate trench **218a**, and a second gate structure **250b** is formed in the second gate trench **218b**, as shown in FIGS. **11A** to **11E**.

Referring to FIGS. **12A** to **12E**, a patterned protecting layer **260** is then formed over the dielectric structure **240** and the first and second gate structures **250a** and **250b**. The patterned protecting layer **260** serves as an etching mask for the subsequent operations. Next, a portion of the dielectric layer **240** is removed through the patterned protecting layer **260**, and thus at least a first opening **262a** is formed in the dielectric structure **240** according to operation **116a** of the method **10a**. Further, the first semiconductor layer **220a** is exposed in a lower portion of the first opening **262a** while the dielectric structure **240** and the spacers **214** are exposed in an upper portion of the first opening **262a**, as shown in FIGS. **12C** and **12E**.

Referring to FIGS. **13A** to **13E**, a first metal silicide layer **270a** is then formed over the first semiconductor layer **220a** according to operation **118a** of the method **10a**. The first metal silicide layer **270a** includes the first semiconductor material and a first metal material. In some embodiments, the first metal silicide layer **270a** can be formed by depositing a metallic layer such as a TiN layer over the first semiconductor layer **220a**. Next, a thermal operation is performed. Consequently, a portion of the first semiconductor layer **220a** reacts with the metallic layer, and the first metal silicide layer **270a** is formed. Therefore, the first metal silicide layer **270a** can include TiSix, but the disclosure is not limited thereto. A thickness of the first semiconductor layer **220a** is reduced to between approximately 5 nm and approximately 15 nm, but the disclosure is not limited thereto. Additionally, since the first metal silicide layer **270a** is formed only over the first semiconductor layer **220a**, the first metal silicide layer **270a** is exposed in the lower portion of the first opening **262a**, as shown in FIG. **13C**. In some embodiments, a glue layer **272** including TiN can be formed over the first metal silicide layer **270a** and sidewalls of the upper portion of the first opening **262a**. However, in other embodiments, the glue layer **272** can be omitted.

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Referring to FIGS. 14A to 14E, a portion of the dielectric layer 240 is further removed to form at least a second opening 262b in the dielectric structure 240 according to operation 116b of the method 10b. Further, the second semiconductor layer 220b is exposed in a lower portion of the second opening 262b while the dielectric structure 240 and the spacers 214 are exposed in an upper portion of the second opening 262b, as shown in FIGS. 14D and 14E.

Referring to FIGS. 15A to 15E, a second metal silicide layer 270b is then formed over the second semiconductor layer 220b according to operation 118b of the method 10b. The second metal silicide layer 270b includes the first semiconductor material, the second semiconductor material, and a second metal material. In some embodiments, the second metal material of the second metal silicide layer 270b is different from the first metal material of the first metal silicide layer 270a. In some embodiments, the first metal silicide layer 270a includes TiSix, and the second metal silicide layer 270b includes NiSiGeX, but the disclosure is not limited thereto. In some embodiment, a Ni layer is formed over the second semiconductor layer 220b, which is exposed in the lower portion of the second opening 262b, by suitable operation, such as chemical vapor deposition (CVD). Subsequently, anneal is performed such that Ni and SiGe are reacted and thus NiSiGeX silicide layer 270b is formed. The superfluous Ni layer is then removed. Additionally, since the second metal silicide layer 270b is formed only over the second semiconductor layer 220b, the second metal silicide layer 270b is exposed in the lower portion of the second opening 262b, as shown in FIG. 15E. In some embodiments, a glue layer 272 including TiN can be formed over the second metal silicide layer 270b and sidewalls of the upper portion of the second opening 262b. However, in other embodiments, the glue layer 272 can be omitted. Additionally, because thermal budget of Ni is lower than that of Ti, the first metal silicide layer 270a is formed before forming the second opening 262b and the second metal silicide layer 270b, but the disclosure is not limited thereto.

Still referring to FIGS. 15A to 15E, a metal layer 264 is next disposed to fill the first opening 262a and the second opening 262b according to operation 120a of the method 10a and operation 120b of the method 10b. In some embodiments, the metal layer 264 includes low-resistivity metal material, such as tungsten (W), but the disclosure is not limited thereto. Accordingly, at least a first conductor, such as a first metal portion 280a, is formed in the first opening 262a and a second conductor, such as a second metal portion 280b, is formed in the second opening 262b. As shown in FIGS. 15C and 15D, a bottom and sidewalls of a lower portion of the first metal portion 280a in the first opening 262a are surrounded by the first silicide layer 270a while sidewalls of an upper portion of the first metal portion 280a in the first openings 262a are surrounded by the spacer 214 and patterned protecting layer 260. Further, the bottom of the second metal portion 280b in the first opening 262a is lower than the plurality of nanowires 232, as shown in FIG. 15C. In contrast to the first metal portion 280a, a lower portion of the second metal portion 280b in the second openings 262b surrounds the second metal silicide layer 270b, as shown in FIGS. 15D and 15E.

Accordingly, a multi-gate semiconductor device 200a is obtained. As shown in FIG. 15C, the multi-gate semiconductor device 200a includes the plurality of nanowires 232, the first gate structure 250a over the plurality of nanowires 232, and source/drain structures 282a and 284a at two ends of each nanowire 232. The source/drain structures 282a and 284a include the first semiconductor layer 220a, the first

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metal portion 280a, and the first metal silicide layer 270a sandwiched between the first semiconductor layer 220a and the lower portion of the first metal portion 280a. Notably, a bottom surface of the first metal portion 280a is lower than the plurality of nanowires 232, as shown in FIG. 15C. Further, each of the first semiconductor layer 220a and the first metal silicide layer 270a substantially includes a U shape. Additionally, the first gate structure 250a can include a metal gate structure, but the disclosure is not limited thereto.

According to the multi-gate semiconductor device 200a, after the forming of the first semiconductor layer 220a and the first metal silicide layer 270a, there is still a space for forming the metal layer 264 in the first opening 262a, and thus the first metal portion 280a is obtained. Accordingly, the lower region of the first metal portion 280a can serve as a portion of the source/drain structures 282a and 284a while an upper region of the first metal portion 280a can serve as a contact plug for providing electrical connection between the source/drain structures 282a and 284a and other devices or circuits. More importantly, the first metal portion 280a can include low-resistivity metal material such as the aforementioned W, thereby reducing contact resistance.

In another embodiment, a multi-gate semiconductor device 200 is provided. The multi-gate semiconductor device 200 includes the multi-gate semiconductor structure 200a and a multi-gate semiconductor structure 200b. In some embodiments, the multi-gate semiconductor device 200 is a CMOS device, and the multi-gate semiconductor structure 200a is an n-channel FET and the multi-gate semiconductor structure 200b is a p-channel FET. As shown in FIGS. 15A to 15E, the multi-gate semiconductor device 200 includes the plurality of nanowires 232 serving as channel regions for the n-channel multi-gate semiconductor structure 200a and the plurality of nanowires 234 serving as channel regions for the p-channel multi-gate semiconductor structure 200b. The multi-gate semiconductor device 200 further includes the first gate structure 250a disposed over the plurality of nanowires 232, the second gate structure 250b disposed over the plurality of nanowires 234, the first source/drain structures 282a and 284a disposed at two ends of each nanowire 232, and second source/drain structures 282b and 284b disposed at two ends of each nanowire 234. It should be noted that the first source/drain structures 282a and 284a include a conductor such as the first metal portion 280a, the first semiconductor layer 220a disposed around sidewalls and a bottom of the lower portion of the first metal portion 280a, and the first metal silicide layer 270a disposed between the lower portion of the first metal portion 280a and the first semiconductor layer 220a. The second source/drain structures 282b and 284b include the plurality of nanowires 230, the second metal silicide layer 270b disposed over the plurality of nanowires 230, and the second semiconductor layer 220b disposed between the second metal silicide layer 270b and the plurality of nanowires 230. Further, the nanowires 230 and the nanowires 234 are the same nanowires. It is referred that a portion of each nanowire surrounded by the second gate structure 250b serve as channel regions and are referred to as a first portion 234, while another portion of each nanowire adjacent to and on opposite sides of the channel region form a part of the second source/drain structures 282b and 284b and are referred to as a second portion 230.

As mentioned above, the first metal silicide layer 270a and the second metal silicide layer 270b can include different semiconductor materials and different metal materials. In some embodiments, the first metal silicide layer 270a

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includes TiSi while the second metal silicide layer **270b** includes NiSiGe. It should be noted that for the n-channel multi-gate semiconductor structure **200a**, the contact resistance is reduced by forming the low-resistivity first metal portion **280a** having the bottom surface lower than the plurality of nanowires **232**. For the p-channel multi-gate semiconductor structure **200b**, the contact resistance is reduced by forming the Ni-silicide layer, because Ni resistance is lower than Ti resistance. Accordingly, the contact resistance of the multi-gate semiconductor device **200** is reduced by the dual contact formation.

FIGS. **16A** and **17A** illustrate a multi-gate semiconductor device **200'** at various fabrication stages constructed according to aspects of one or more embodiments of the present disclosure. FIG. **16B** and FIG. **17B** are cross-sectional views taken along line I-I' of FIGS. **16A** and **17A**, respectively, according to aspects of one or more embodiments of the present disclosure, FIGS. **16C** and **17C** are cross-sectional views taken along line II-II' of FIGS. **16A** and **17A**, respectively, according to aspects of one or more embodiments of the present disclosure, FIGS. **16D** and **17D** are cross-sectional views taken along line III-III' of FIGS. **16A** and **17A**, respectively, according to aspects of one or more embodiments of the present disclosure, and FIGS. **16E** and **17E** are cross-sectional views taken along line IV-IV' of FIGS. **16A** and **17A**, respectively, according to aspects of one or more embodiments of the present disclosure. It should be noted that similar elements in FIGS. **3A** to **15E** and FIGS. **16A** to **17E** are designated by the same numerals. Further, similar elements in FIGS. **3A** to **15E** and FIGS. **16A** to **17E** can include similar materials and can be formed by similar steps; therefore such redundant details are omitted in the interest of brevity.

Please refer to FIGS. **16A** to **16E**. In some embodiments, operations **102**, **104**, **106a** and **108a**, **106b** and **108b**, **110**, **112a** and **114a**, and **112b** and **114b** are performed, and operations **116a** and **116b** are simultaneously performed after the forming of the first and second gate structures **250a** and **250b**. Consequently, a first opening **262a** and a second opening **262b** are simultaneously formed in the dielectric layer **240**. As shown in FIGS. **16A** to **16E**, the first semiconductor layer **220a** is exposed in the first opening **262a** and the second semiconductor layer **220b** is exposed in the second opening **262b**. In some embodiments, the first semiconductor layer **220a** forms a bottom and sidewalls of a lower portion of the first opening **262a** while the second semiconductor layer **220b** protrudes from a bottom of the second opening **262b**, as shown in FIGS. **16C** and **16E**.

Referring to FIGS. **17A** to **17E**, a first metal silicide layer **270a'** is formed over the first semiconductor layer **220a** and a second metal silicide layer **270b'** is formed over the second semiconductor layer **220b** according to operations **118a** and **118b**. Notably, the operations **118a** and **118b** are performed at the same time, and thus the first metal silicide layer **270a'** and the second metal silicide layer **270b'** are simultaneously formed. The first metal silicide layer **270a'** includes the first semiconductor material and a first metal material, and the second metal silicide layer **270b'** includes the first semiconductor material, the second semiconductor material and a second metal material. Notably, the first metal material and the second metal material are the same. In some embodiments, the first metal silicide layer **270a'** includes TiSi, and the second metal silicide layer **270b'** includes TiSiGe, but the disclosure is not limited thereto.

Still referring to FIGS. **17A** to **17E**, a glue layer such as a TiN layer is then formed over the first metal silicide layer **270a'**, the second metal silicide layer **270b'**, sidewalls of an

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upper portion of the first opening **262a**, and sidewalls of an upper portion of the second opening **262b**. However, in some embodiments, formation of the glue layer can be omitted. Subsequently, a metal layer **264** is formed to fill the first opening **262a** and the second opening **262b** according to operation **120**.

Accordingly, a multi-gate semiconductor device **200'** is provided. The multi-gate semiconductor device **200'** includes the multi-gate semiconductor structure **200a'** and a multi-gate semiconductor structure **200b'**. In some embodiments, the multi-gate semiconductor device **200'** is a CMOS device, the multi-gate semiconductor structure **200a'** is an n-channel FET, and the multi-gate semiconductor structure **200b'** is a p-channel FET. As shown in FIGS. **17A** to **17E**, the multi-gate semiconductor device **200'** includes the plurality of nanowires **232** serving as channel regions for the n-channel multi-gate semiconductor structure **200a'** and the plurality of nanowires **234** serving as channel regions for the p-channel multi-gate semiconductor structure **200b'**. The multi-gate semiconductor device **200'** further includes the first gate structure **250a** disposed over the plurality of nanowires **232**, the second gate structure **250b** disposed over the plurality of nanowires **234**, the first source/drain structures **282a** and **284a** disposed at two ends of each nanowire **232**, and the second source/drain structures **282b** and **284b** disposed at two ends of each nanowire **234**. It should be noted that the first source/drain structures **282a** and **284a** include the first conductor such as the first metal portion **280a**, the first semiconductor layer **220a** disposed around sidewalls and a bottom of the lower portion of the first metal portion **280a**, and the first metal silicide layer **270a'** disposed between the lower portion of the first metal portion **280a** and the first semiconductor layer **220a**. The second source/drain structures **282b** and **284b** include the plurality of nanowires **230**, the second metal silicide layer **270b'** disposed over the plurality of nanowires **230**, and the second semiconductor layer **220b** disposed between the second metal silicide layer **270b'** and the plurality of nanowires **230**. Further, the nanowires **230** and the nanowires **234** are the same nanowires. It is referred that a portion of each nanowire surrounded by the second gate structure **250b** serve as channel regions and are referred to as a first portion **234**, while another portion of each nanowire adjacent to and on opposite sides of the channel region form a part of the second source/drain structures **282b** and **284b** and are referred to as a second portion **230**.

As mentioned above, the first metal silicide layer **270a'** and the second metal silicide layer **270b'** can include different semiconductor materials but the same metal materials. In some embodiments, the first metal silicide layer **270a'** includes TiSi while the second metal silicide layer **270b'** includes TiSiGe. It should be noted that for the n-channel multi-gate semiconductor structure **200a'**, the contact resistance is reduced by forming the low-resistivity first metal portion **280a**. However, by simultaneously forming the first opening **262a** and the second opening **262b** and simultaneously forming the first metal silicide layer **270a'** and the second metal silicide layer **270b'**, the methods **10a** and **10b** are integrated and simplified while contact resistance of the multi-gate semiconductor device **200'** is reduced.

According to one embodiment of the present disclosure, a multi-gate semiconductor structure is provided. The multi-gate semiconductor structure includes a plurality of nanowires, a gate structure disposed over the plurality of nanowires, and source/drain structures at two ends of each of the plurality of nanowires. The source/drain structures include a semiconductor layer, a metal portion, and a metal silicide

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layer. Further, a bottom surface of the metal portion is lower than the plurality of nanowires.

According to another embodiment, a multi-gate semiconductor device is provided. The multi-gate semiconductor device includes a plurality of first nanowires and a plurality of second nanowires, a first gate structure disposed over the plurality of first nanowires and a second gate structure disposed over a first portion of the plurality of second nanowires, first source/drain structures disposed at two ends of each of the plurality of first nanowires, and second source/drain structures disposed at two ends of each of the first portions of the second nanowires. The first source/drain structures further include a conductor, a first semiconductor disposed around a bottom and sidewalls of a portion of the conductor, and a first metal silicide layer disposed between the conductor and the first semiconductor layer. The second source/drain structures further include a second portion of the second nanowires, a second metal silicide layer disposed over the second portions of the second nanowires, and a second semiconductor layer disposed between the second metal silicide layer and the second portions of the second nanowires.

According to one embodiment of the present disclosure, a method for forming a multi-gate semiconductor device is provided. The method includes the following operations. A substrate including at least a first fin structure and a second fin structure is received. A dummy gate structure is disposed over a portion of the first fin structure and a portion of the second fin structure. Portions of the first fin structure exposed through the dummy gate structure are removed to form at least a first recess in the substrate. A first semiconductor layer is formed in the first recess. A dielectric structure is disposed over the substrate. A portion of the dummy gate structure is removed to form a first gate trench in the dielectric structure. A plurality of first nanowires and a first gate structure are formed in the first gate trench. A portion of the dielectric structure is removed to form a first opening exposing the first semiconductor layer. A first metal silicide layer is formed over the first semiconductor layer. A metal layer is formed to fill the first opening, where a bottom of the metal layer in the first opening is lower than the plurality of nanowires.

FIGS. 18A to 18K illustrate perspective views of intermediate stages of manufacturing a semiconductor structure 300 in accordance with some embodiments. FIGS. 19A to 19K are cross-sectional views taken along line II-II' of FIGS. 18A to 18K respectively in accordance with some embodiments. Some processes and materials for forming the semiconductor structure 300 may be similar to, or the same as, those for forming the multi-gate semiconductor devices 200 and 200' described above and are not repeated herein.

Similar to FIGS. 3A to 3D, a fin structure 210 similar to the first fin structure 210a described above is formed over the substrate 202 from the alternately stacked semiconductor layers 206/208, and the isolation feature 204 is formed around the fin structure 210, as shown in FIGS. 18A and 19A in accordance with some embodiments. In some embodiments, a first liner layer 301 and a second liner layer 303 are formed before the isolation feature 204 is formed.

In some embodiments, the first liner layer 301 and the second liner layer 303 are made of different materials. In some embodiments, the first liner layer 301 and the second liner layer 303 are made of materials such as silicon oxide, silicon nitride, silicon oxynitride (SiON), other applicable insulating materials, or a combination thereof. In some embodiments, the first liner layer 301 is made of an oxide and the second liner layer 303 is made of a nitride. The first

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liner layer 301, the second liner layer 303, and isolation features 204 may be formed by using LPCVD, PECVD, high-density plasma CVD (HDP-CVD), high aspect ratio process (HARP), flowable CVD (FCVD), ALD, another suitable method, or a combination thereof.

After the isolation feature 204 is formed, dummy gate structures 212 are formed across the fin structure 210, and the spacers 214 are formed on the sidewalls of the dummy gate structures 212, as shown in FIGS. 18B and 19B in accordance with some embodiments. The portion of the fin structure 210 underlying the dummy gate structure 212 may be referred to as the channel region, and the portions of the fin structure 210 adjacent to and on opposite sides of the channel region may be defined as the source/drain regions.

In some embodiments, the dummy gate structure 212 includes a gate dielectric layer 305, a gate electrode layer 307, and a mask layer 309. In some embodiments, the gate dielectric layer 305 is made of one or more dielectric materials, such as silicon oxide, silicon nitride, silicon oxynitride (SiON), HfO₂, HfZrO, HfSiO, HfTiO, HfAlO, or a combination thereof. In some embodiments, the dielectric material is formed using thermal oxidation, CVD, ALD, physical vapor deposition (PVD), another suitable method, or a combination thereof.

In some embodiments, the gate electrode layer 307 is made of a conductive material. In some embodiments, the conductive material includes polycrystalline-silicon (poly-Si), poly-crystalline silicon-germanium (poly-SiGe), metallic nitrides, metallic silicides, metals, or a combination thereof. In some embodiments, the conductive material is formed using CVD, PVD, or a combination thereof.

In some embodiments, the mask layer 309 is a bi-layers structure including an oxide layer and a nitride layer. In some embodiments, the oxide layer is silicon oxide, and the nitride layer is silicon nitride.

The formation of the dummy gate structure 212 may include conformally forming a dielectric material as the gate dielectric layer 305 along the substrate 102, the fin structure 210, and the isolation feature 204. After the dielectric material is formed, a conductive material may be formed over the dielectric material as the gate electrode layer 307 and the mask layer 309 may be formed over the conductive material. After the mask layer 309 is formed, the dielectric material and the conductive material may be patterned through the mask layer 309 to form the dummy gate structure 212. Similar to FIGS. 4A to 4E, after the dummy gate structures 212 are formed, the spacers 214 are formed on the sidewalls of the dummy gate structures 212 in accordance with some embodiments.

Next, portions of the semiconductor layers 206 and 208 of the fin structure 210 not covered by the dummy gate structures 212 and the spacers 214 are recessed to form recesses 316a, as shown in FIGS. 18C and 19C in accordance with some embodiments. In some embodiments, the fin structure 210 is recessed by performing an etching process. After the recesses 316a are formed, the semiconductor layers 206 are laterally etched from the recesses 316a to form notches 317 between the semiconductor layers 208 in accordance with some embodiments.

After the notches 317 are formed, the inner spacers 211a are formed in the notches 317, as shown in FIGS. 18D and 19D in accordance with some embodiments. Next, source/drain structures 320 are formed in the recesses 316a, as shown in FIGS. 18E and 19E in accordance with some embodiments. In some embodiments, the source/drain structures 320 are formed using a process similar to that used to form the first semiconductor layer 220a. In some embodi-

ments, the source/drain structures **320** are formed using an epitaxial growth process, such as MBE, MOCVD, VPE, other applicable epitaxial growth processes, or a combination thereof. In some embodiments, the source/drain structures **320** are made of any applicable material for an n-type semiconductor device and a p-type semiconductor device, such as Ge, Si, GaAs, AlGaAs, SiGe, GaAsP, SiP, SiC, SiCP, or a combination thereof.

In some embodiments, the source/drain structures **320** are in-situ doped during the epitaxial growth process. For example, the source/drain structures **320** may be the epitaxially grown SiGe doped with boron (B). For example, the source/drain structures **320** may be the epitaxially grown Si doped with carbon to form silicon:carbon (Si:C) source/drain features, phosphorous to form silicon:phosphor (Si:P) source/drain features, or both carbon and phosphorous to form silicon carbon phosphor (SiCP) source/drain features. In some embodiments, the source/drain structures **320** are doped in one or more implantation processes after the epitaxial growth process.

After the source/drain structures **320** are formed, the dielectric structure **240** including the contact etch stop layer (CESL) **242** and the inter-layer dielectric (ILD) layer **244** is formed, and a polishing process is performed until a top surface of the gate electrode layer **307** is exposed, as shown in FIGS. **18F** and **19F** in accordance with some embodiments.

Next, the dummy gate structures **212** are removed to expose the semiconductor layers **206** and **208** of the fin structure **210**, and the semiconductor layers **206** of the fin structure **210** in the channel region are then removed to form nanostructures **308** from the semiconductor layers **208**, as shown in FIGS. **18G** and **19G** in accordance with some embodiments. As shown in FIG. **19G**, gaps are formed between the nanostructures **308** in accordance with some embodiments.

Afterwards, gate structures **350** are formed over the nanostructures **308** and wrapped around the nanostructures **308**, as shown in FIGS. **18H** and **19H** in accordance with some embodiments. Similar to the first gate structure **250a**, the gate structure **350** includes an interfacial layer (IL) **252**, a gate dielectric layer **254**, a first gate conductive layer **256a**, and a gap-filling metal layer **258** in accordance with some embodiments. In some embodiments, the nanostructures **308** are wrapped by the interfacial layer **252**, the gate dielectric layer **254**, and the first gate conductive layer **256a**. In some embodiments, the nanostructures **308** are also wrapped by the gap-filling metal layer **258** (not shown).

After the gate structures **350** are formed, a dielectric layer **360** is formed over the inter-layer dielectric layer **244**, and contact openings **362** are formed through the dielectric layer **360** and the inter-layer dielectric layer **244**, as shown in FIGS. **18I** and **19I** in accordance with some embodiments. More specifically, the dielectric layer **360** and the inter-layer dielectric layer **244** may be etched through openings of a mask (not shown) to form the contact openings **362**, and the source/drain structures **320** are exposed by the contact openings **362** in accordance with some embodiments.

In some embodiments, the dielectric layer **360** is made of silicon oxide, silicon nitride, silicon oxynitride, phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), and/or other applicable low-k dielectric materials. The dielectric layer **360** may be formed by chemical vapor deposition (CVD), physical vapor deposition, (PVD), atomic layer deposition (ALD), or other applicable processes. In some embodiments, an etch stop layer (not shown) is formed before the dielectric layer **360** is formed.

After the contact openings **362** are formed, trenches **363** are formed in the source/drain structures **320**, as shown in FIGS. **18J** and **19J** in accordance with some embodiments. More specifically, the source/drain structures **320** are etched through the contact opening **362** to form the trenches **363** in accordance with some embodiments. In some embodiments, a bottom portion of the trenches **363** is lower than the bottommost nanostructure **308**. In some embodiments, the bottom portion of the trenches **363** is lower than the bottom surface of the gate structures **350**.

Next, metal silicide layers **370** are formed on the trenches **363**, as shown in FIGS. **18K** and **19K** in accordance with some embodiments. The materials and processes for forming the metal silicide layer **370** may be similar to, or the same as, those for forming the first metal silicide layer **270a** described above and are not repeated herein.

In some embodiments, the metal silicide layers **370** are formed on the surface of the source/drain structures **320** exposed by the trenches **363**, and a bottom portion of the metal silicide layer **370** is lower than the bottommost nanostructures **308**, and a top portion of the metal silicide layer **370** is higher than the topmost nanostructures **308**. In some embodiments, the bottom portion of the metal silicide layer **370** is lower than the bottom surface of the gate structures **350**, and a top portion of the metal silicide layer **370** is lower than a top surface of the gate structures **350**.

After the metal silicide layers **370** are formed, contacts **380** are formed in the contact openings **362** and in the trenches **363** over the metal silicide layers **370**, as shown in FIGS. **18K** and **19K** in accordance with some embodiments. In some embodiments, the contact **380** includes a glue layer **372** and a metal layer **364**. The materials and processes for forming the glue layer **372** and the metal layer **364** may be similar to, or the same as, those for forming the glue layer **272** and the metal layer **264** described above and are not repeated herein.

Since the trench **363** is formed in the source/drain structure **320** before the contact **380** is formed, the contact **380** can extend into the source/drain structure **320** and be laterally next to, instead of much higher than, the nanostructures **308** in accordance with some embodiments. Accordingly, the parasitic resistance of the channels (i.e. the nanostructures **308**), especially the lower channels close to the substrate **202**, may be reduced, and the performance of the semiconductor structure **300** may therefore be improved.

In some embodiments, a bottom portion of the contact **380** is lower than the bottommost nanostructures **308**, and a top portion of the contact **380** is higher than the topmost nanostructures **308**. In addition, the bottom portion of the contact **380** is lower than the bottom surface of the gate structures **350**, and a top portion of the contact **380** is higher than a top surface of the gate structures **350** in accordance with some embodiments.

Furthermore, the inner spacers **211a** are formed between the neighboring nanostructures **308**, so that the gate structures **350** are separated from the source/drain structures **320** by the inner spacers **211a**, as shown in FIG. **19K** in accordance with some embodiments. In some embodiments, some portions of the source/drain structures **320** are sandwiched between the contacts **380** and the nanostructures **308** and some portions of the source/drain structures **320** are sandwiched between the contacts **380** and inner spacers **211a**.

FIG. **20** illustrates a cross-sectional representation of a semiconductor structure **300-1** in accordance with some embodiments. The semiconductor structure **300-1** is similar to the semiconductor structure **300** described above, except a bottom portion of the contact in the semiconductor struc-

ture **300-1** does not extend into the substrate **202** in accordance with some embodiments. Processes and materials for forming the semiconductor structure **300-1** are substantially similar to, or the same as, those for forming the semiconductor structure **300** and are not repeated herein.

More specifically, after the processes shown in FIGS. **18A-18I** and **19A-19I** are performed, a trench is formed in the source/drain structure **320** and a metal silicide layer **370-1** and a contact **380-1** are formed in the trench, as shown in FIG. **20** in accordance with some embodiments. In addition, the trench formed in the source/drain structure **320** does not extend into the substrate **202**, and therefore the bottommost portion of a metal silicide layer **370-1** and the bottommost portion of the contact **380-1** are both higher than the top surface of the substrate **202** in accordance with some embodiments.

In some embodiments, the bottommost portion of the metal silicide layer **370-1** is lower than the bottommost nanostructures **308** but is higher than the bottommost portion of the gate structure **350**. In some embodiments, the topmost portion of the metal silicide layer **370-1** is higher than the topmost nanostructures **308** but is lower than the topmost portion of the gate structure **350**. In some embodiments, the topmost portion of the contact **380-1** is higher than both the topmost nanostructures **308** and the top surface of the gate structure **350**.

Although the contact **380-1** does not extend into the substrate **202**, the contact **380-1** still extends into the source/drain structure **320** and therefore is also laterally next to the nanostructures **308** in accordance with some embodiments. Accordingly, the parasitic resistance of the channels may be reduced, and the performance of the semiconductor structure **300-1** may be improved.

FIGS. **21A** and **21B** illustrate cross-sectional representation of various stages for forming a semiconductor structure **300-2** in accordance with some embodiments. The semiconductor structure **300-2** is similar to the semiconductor structure **300** described above, except nanostructures are formed from the semiconductor layers **206** instead of the semiconductor layers **208** in accordance with some embodiments. Processes and materials for forming the semiconductor structure **300-2** are substantially similar to, or the same as, those for forming the semiconductor structure **300** and are not repeated herein.

More specifically, the processes shown in FIGS. **18A, 18B, 19A, and 19B** are performed, and recesses **316-2** are formed through the fin structure and inner spacers **311** are formed between the semiconductor layers **206**, as shown in FIG. **21A** in accordance with some embodiments. Similar to the inner spacers **211a**, the inner spacers **311** may be formed by recessing the semiconductor layers **208** to form notches (similar to the notches **317**) and forming the inner spacers **311** in the notches. The materials for forming the inner spacers **311** are similar to, or the same as, those for forming the inner spacers **211a** described above and are not repeated herein.

Afterwards, the processes shown in FIGS. **18E to 18G and 19E to 19G** may be performed. More specifically, source/drain structures **320-2** are formed and the semiconductor layers **208** in the channel region are removed to form nanostructures **306** from the semiconductor layers **206**, as shown in FIG. **21B** in accordance with some embodiments. Next, gate structures **350-2** are formed over the nanostructures **306** and wrapped around the nanostructures **306** in accordance with some embodiments. Similar to the gate structure **350**, the gate structure **350-2** includes the interfacial layer **252**, the gate dielectric layer **254**, the first gate

conductive layer **256a**, and the gap-filling metal layer **258** in accordance with some embodiments. In some embodiments, some portions of the gate structures **350-2** extend under the spacers **214** and therefore are overlapped with the spacers **214**. In some embodiments, the width of the portion of the gate structure **350-2** wrapped around the nanostructure **306** is greater than the width of the portion of the gate structure **350-2** between the spacers **214**. In some embodiments, the bottom surface of the gate structure **350-2** is substantially level with the bottommost portion of the source/drain structure **320-2**.

After the gate structures **350-2** are formed, processes shown in FIGS. **18I-18K** and **19I-19K** may be performed to form the semiconductor structure **300-2**. More specifically, the dielectric layer **360** is formed over the gate structures **350-2**, and metal silicide layers **370-2** and contacts **380-2** are formed extending into source/drain structures **320-2**, as shown in FIG. **21B** in accordance with some embodiments.

The contacts **380-2** are similar to the contact **380** and include the glue layer **372** and the metal layer **364** in accordance with some embodiments. In some embodiments, the bottommost portion of the contact **380-2** is higher than a bottom surface of the gate structure **350-2**. In some embodiments, the bottommost portion of the contact **380-2** is higher than the bottommost nanostructures **306**.

As described previously, since the contact **380-2** extends into the source/drain structure **320** and is also laterally next to the nanostructures **306** in accordance with some embodiments, the parasitic resistance of the channels may be reduced, and the performance of the semiconductor structure **300-2** may be improved.

FIGS. **22A to 22I** illustrate perspective views of intermediate stages of manufacturing a semiconductor structure **400** in accordance with some embodiments. FIGS. **23A to 23I** are cross-sectional views taken along line II-II' of FIGS. **22A to 22I** respectively in accordance with some embodiments. Processes and materials for forming the semiconductor structure **400** may be similar to, or the same as, those for forming the multi-gate semiconductor devices **200** and **200'** and the semiconductor structures **300, 300-1, and 300-2** described above and are not repeated herein.

Similar to FIGS. **19A** and **19B**, the fin structure **210** is formed over the substrate **202** from the alternately stacked semiconductor layers **206** and **208**, the isolation feature **204** is formed around the fin structure **210**, and the dummy gate structures **212** are formed across the fin structure **210**, as shown in FIGS. **22A** and **23A** in accordance with some embodiments. After the dummy gate structures **212** are formed, portions of the semiconductor layers **206** of the fin structure **210** not covered by the dummy gate structures **212** and the spacers **214** are removed to form gaps **416** between nanostructures **408** formed from the semiconductor layers **208**, as shown in FIGS. **22B** and **23B** in accordance with some embodiments. After the gaps **416** are formed, the semiconductor layers **206** are laterally etched from the gaps **416** to form notches **417** under the spacers **214**, as shown in FIGS. **22B** and **23B** in accordance with some embodiments.

Afterwards, the inner spacers **411** are formed in the notches **417** between the nanostructures **408**, as shown in FIGS. **22C** and **23C** in accordance with some embodiments. Processes and materials for forming the inner spacers **411** may be similar to, or the same as, those for forming the inner spacers **211a** described previously and are not repeated herein.

Next, an implantation process **413** is performed to form doped source/drain regions **420** in the nanostructures **308**, as shown in FIGS. **22D** and **23D** in accordance with some

embodiments. In some embodiments, dopants are implanted into the nanostructures **408** not covered by the dummy gate structures **212** and the spacers **214** and are driven under the spacers **214** by performing a diffusion process, so that the doped source/drain regions **420** are partially overlapped the spacers **214** and the inner spacers **411**. In some embodiments, the dopants are p-type dopants, such as boron or BF_2 . In some embodiments, the dopants are n-type dopants, such as phosphorus or arsenic. In some embodiments, the dopants are further implanted in the substrate **202** during the implantation process **413**, so that additional doped regions **421** are formed in the substrate **202**, as shown in FIG. 23D. Since the source/drain regions **420** and the doped regions **421** in the substrate **202** are both formed by the same implantation process **413**, the dopants in the source/drain regions **420** and in the doped regions **421** are the same in accordance with some embodiments.

After the source/drain regions **420** are formed, an inter-layer dielectric (ILD) layer **444** is formed, as shown in FIGS. 22E and 23E in accordance with some embodiments. Processes and materials for forming the inter-layer dielectric layer **444** may be similar to, or the same as, those for forming the inter-layer dielectric layer **244** described previously and are not repeated herein. More specifically, the gaps **416** between the source/drain regions **421** of the nanostructures **308** are fully filled with the inter-layer dielectric layer **444**, so that the source/drain regions **421** of the nanostructures **308** are surrounded by the inter-layer dielectric layer **444**, as shown in FIG. 23E in accordance with some embodiments. That is, some portions of the ILD layers **444** extend under the source/drain regions **421** and are in direct contact with the inner spacers **411** and the doped regions **421** in accordance with some embodiments.

Next, the dummy gate structures **212** are removed to expose the semiconductor layers **206** and **208** of the channel region of the fin structure **210**, and the semiconductor layers **206** of the fin structure **210** in the channel region are then removed to form the nanostructures **408** from the semiconductor layers **208**, as shown in FIGS. 22F and 23F in accordance with some embodiments.

Afterwards, gate structures **450** are formed over the nanostructures **408** and wrapped around the nanostructures **408**, as shown in FIGS. 22G and 23G in accordance with some embodiments. Similar to the first gate structure **350**, the gate structure **450** includes an interfacial layer **252**, a gate dielectric layer **254**, a first gate conductive layer **256a**, and a gap-filling metal layer **258** in accordance with some embodiments.

After the gate structures **450** are formed, the dielectric layer **360** is formed over the inter-layer dielectric layer **444**, and contact openings **462** are formed through the dielectric layer **360** and the inter-layer dielectric layer **444**, as shown in FIGS. 22H and 23H in accordance with some embodiments. More specifically, a mask (not shown) with openings may be formed over the dielectric layer **360**, and the dielectric layer **360** and the inter-layer dielectric layer **444** under the openings may be etched to form the contact openings **462** through the dielectric layer **360** and the inter-layer dielectric layer **244**. Furthermore, the portions of the inter-layer dielectric layer **444** sandwiched between the source/drain regions **420** of the nanostructures **408** under the opening are also removed to form gaps **463** between the source/drain regions **420**, as shown in FIG. 23H in accordance with some embodiments. Accordingly, the source/drain regions **420** of nanostructures **408** are exposed by the contact openings **462** in accordance with some embodiments.

Next, contacts **480** are formed in the contact openings **462** and in the gaps **463**, as shown in FIGS. 22I and 23I in accordance with some embodiments. The materials and processes for forming the contacts **480** may be similar to, or the same as, those for forming the metal layer **264** described above and are not repeated herein.

Since the contact **480** in the semiconductor structure **400** is formed in the contact opening **462** and in the gaps **463**, a bottom portion of the contact **480** is lower than the bottommost nanostructures **408**, and a top portion of the contact **480** is higher than the topmost nanostructures **408** in accordance with some embodiments. In addition, the bottom portion of the contact **480** is substantially level with the bottom surface of the gate structures **450**, and a top portion of the contact **480** is higher than a top surface of the gate structures **450** in accordance with some embodiments.

In addition, the inner spacers **411** are formed between the neighboring the source/drain regions **420** of the nanostructures **408**, so that the gate structures **450** are separated from the contacts **480** by the inner spacers **411**, as shown in FIG. 23I in accordance with some embodiments. In some embodiments, the contacts **480** are in direct contact with the inner spacers **411**. Furthermore, some portions of the contacts **480** are sandwiched between the source/drain regions **420** of the nanostructures **408** and the doped regions **421** of the substrate **202**.

Similarly, since the contact **480** extends between the nanostructures **408**, the distances between the nanostructures **408** and the contact **480** are reduced, such as compared to the contact which is formed above the source/drain structure, which is higher than the top portion of the entire nanostructure, in accordance with some embodiments. Accordingly, the parasitic resistance of the channels may be reduced, and the performance of the semiconductor structure **400** may be improved.

FIG. 24 illustrates a cross-sectional representation of a semiconductor structure **400-1** in accordance with some embodiments. The semiconductor structure **400-1** is similar to the semiconductor structure **400** described above, except metal silicide layers **470** are formed before the formation of the contacts **480** in accordance with some embodiments. Processes and materials for forming the semiconductor structure **400-1** are substantially similar to, or the same as, those for forming the semiconductor structure **400** and are not repeated herein.

More specifically, after the processes shown in FIGS. 22A-22H and 23A-23H are performed, the metal silicide layers **470** are formed around the source/drain regions **470** of the nanostructures **408**, as shown in FIG. 24 in accordance with some embodiments. In addition, the metal silicide layers **470** cover the top surface of the doped region **421** of the substrate **202** in accordance with some embodiments. Processes and materials for forming the metal silicide layers **470** are substantially similar to, or the same as, those for forming the metal silicide layers **370** and are not repeated herein.

In some embodiments, the bottommost portion of the metal silicide layer **470** is lower than the bottommost nanostructures **408** and is substantially level with the bottommost portion of the gate structure **450**. In some embodiments, the topmost portion of the metal silicide layer **470** is higher than the topmost nanostructures **408** but is lower than the topmost portion of the gate structure **450**. In some embodiments, a portion of the contact **480** extends into a space surrounded by the inner spacers **411** and the metal silicide layers **470**.

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FIG. 25 illustrates a cross-sectional representation of a semiconductor structure 400-2 in accordance with some embodiments. The semiconductor structure 400-2 is similar to the semiconductor structure 400 described above, except glue layers 472 are formed before the formation of the contacts 480 in accordance with some embodiments. Processes and materials for forming the semiconductor structure 400-2 are substantially similar to, or the same as, those for forming the semiconductor structure 400 and are not repeated herein.

More specifically, after the processes shown in FIGS. 22A-22H and 23A-23H are performed, the glue layers 472 are formed around the source/drain regions 470 of the nanostructures 408, as shown in FIG. 25 in accordance with some embodiments. In addition, the glue layers 472 cover the top surface of the doped region 421 of the substrate 202 and the sidewalls of the inner spacers 411 in accordance with some embodiments. Processes and materials for forming the glue layers 472 are substantially similar to, or the same as, those for forming the glue layers 372 and are not repeated herein.

In some embodiments, the bottommost portion of the glue layer 472 is lower than the bottommost nanostructures 408 and is substantially level with the bottommost portion of the gate structure 450. In some embodiments, the topmost portion of the glue layer 472 is higher than the topmost nanostructures 408 but is lower than the topmost portion of the gate structure 450. In some embodiments, a portion of the glue layer 472 is sandwiched between the inner spacers 411 and the contacts 480.

FIGS. 26A and 26B illustrate cross-sectional representations of intermediate stages of manufacturing a semiconductor structure 400-3 in accordance with some embodiments. The semiconductor structure 400-3 is similar to the semiconductor structure 400 described above, except some portions of the inter-layer dielectric layer 444 over the source/drain regions 420 are not removed in accordance with some embodiments. Processes and materials for forming the semiconductor structure 400-3 are substantially similar to, or the same as, those for forming the semiconductor structure 400 and are not repeated herein.

More specifically, after the processes shown in FIGS. 22A-22G and 23A-23G are performed, contact openings 462-3 and gaps 463-3 are formed through the dielectric layer 360 and the inter-layer dielectric layer 444, as shown in FIG. 26A in accordance with some embodiments. Similar to those shown in FIG. 23F and described previously, a mask (not shown) with openings may be formed over the dielectric layer 360, and the dielectric layer 360 and the inter-layer dielectric layer 444 under the openings may be etched to form the contact openings 462-3 and the gaps 463-3. However, opening of the mask used for forming the semiconductor structure 400-3 may be relatively narrow and therefore the portions of the inter-layer dielectric layer 444 sandwiched between the source/drain regions 420 are only partially removed, such that some portions of the inter-layer dielectric layer 444 remain on the sidewalls of the spacers 214 and the inner spacers 411, as shown in FIG. 26A in accordance with some embodiments.

After the contact openings 462-3 and the gaps 463-3 are formed, the contacts 480 are formed in the contact openings 462-3 and gaps 463-3 with some portions of the inter-layer dielectric layer 444 sandwiched between the contact 480 and the inner spacers 411, as shown in FIG. 26B in accordance with some embodiments. In some embodiments, the source/drain regions 420 of the nanostructures 408 are partially surrounded by the inter-layer dielectric layer 444.

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FIG. 27 illustrates a cross-sectional representation of a semiconductor structure 400-4 in accordance with some embodiments. The semiconductor structure 400-4 is similar to the semiconductor structure 400 described above, except nanostructures are formed from the semiconductor layers 206 instead of the semiconductor layers 208 in accordance with some embodiments. Processes and materials for forming the semiconductor structure 400-4 are substantially similar to, or the same as, those for forming the semiconductor structure 400 and are not repeated herein.

More specifically, the processes similar to those shown in FIGS. 22A-22I and 23A-23I may be performed, except the semiconductor layers 208 are removed while the semiconductor layers 206 are not removed in the processes shown in FIGS. 23B and 23F. Accordingly, nanostructures 406 are formed from the semiconductor layers 206 and are wrapping by the gate structure 450'. In addition, source/drain regions 420' are formed in the nanostructures 406 and are surrounded by the contacts 480-4 in accordance with some embodiments. Processes and materials for forming the gate structure 450' and the source/drain regions 420' are substantially similar to, or the same as, those for forming the gate structure 350-2 and the source/drain regions 420, respectively, and are not repeated herein.

FIG. 28 illustrates a cross-sectional representation of a semiconductor structure 400-5 in accordance with some embodiments. The semiconductor structure 400-5 is similar to the semiconductor structure 400-4 described above, except glue layers 472-5 are formed before the formation of the contacts 480-5 in accordance with some embodiments. Processes and materials for forming the semiconductor structure 400-5 are substantially similar to, or the same as, those for forming the semiconductor structure 400-4 and are not repeated herein. In addition, processes and materials for forming the glue layers 472-5 are substantially similar to, or the same as, those for forming the glue layers 472 and are not repeated herein.

The semiconductor structures 300, 300-1, 300-2, 400, and 400-1 to 400-5 described above may be used in a PMOS device or in an NMOS device. In some embodiments, a semiconductor device includes at least one PMOS device and at least NMOS device having the structures described above.

FIG. 29 illustrates a cross-sectional representation of a semiconductor structure 500 in accordance with some embodiments. In some embodiments, the semiconductor structure 500 includes an NMOS region 10 and a PMOS region 20, and the structure shown in the NMOS region 10 and the PMOS region 20 are substantially the same as the semiconductor structure 300 shown in FIG. 19K, except the physical properties of the materials in contacts 380a in the NMOS region 10 and contacts 380b in the PMOS region 20 are different.

More specifically, the contact 380a in the NMOS region 10 includes a glue layer 372 and a metal layer 364a, and the contact 380b in the PMOS region 20 includes the glue layer 372 and a metal layer 364b in accordance with some embodiments. The metal layer 364a and the metal layer 364b may exert different stresses on the semiconductor structure 500. In some embodiments, the metal layer 364a is made of a first material having tensile stress and the metal layer 364b is made of a second material having compressive stress. Since the contacts 380a and 380b can provide additional tensile stress and the compressive stress to the channels in the NMOS region 10 and the PMOS region 20, the performance of the semiconductor structure 500 may be improved.

In some embodiments, the metal layer **364a** and the metal layer **364b** are made of the same metal element, such as tungsten, but include different dopants. In some embodiments, the metal layer **364a** and the metal layer **364b** are made of the same metal element but are formed under different deposition conditions (e.g. various temperatures and pressures).

In some embodiments, an additional nucleation and/or adhesive layer is formed before forming the metal layer **364a** and the metal layer **364b**, so that the metal materials deposited thereon have different stresses. In some embodiments, the additional nucleation and/or adhesive layer is made of WN, WC, WCN, WSiN, or the like.

In some embodiments, the metal layer **364a** is made of tungsten with a tensile stress and is formed by performing a chemical vapor deposition (CVD) process including using a tungsten-containing precursor (e.g. WF₆) and reducing gases (e.g. H₂) at temperature in a range from about 250° C. to about 400° C. In some embodiments, the CVD process further comprising using HF.

Although the structures shown in the NMOS region **10** and the PMOS region **20** of the semiconductor structure **500** are substantially the same as the semiconductor structure **300** shown in FIG. 19K, the concept of the disclosure is not intended to be limiting. That is, the NMOS region may include any one of the semiconductor structures described above (e.g. the semiconductor structures **300**, **300-1**, **300-2**, **400**, and **400-1** to **400-5**), and the PMOS region may include any one of the semiconductor structures described above (e.g. the semiconductor structures **300**, **300-1**, **300-2**, **400**, and **400-1** to **400-5**). In addition, other than the stress provided by the contacts are different, the semiconductor structures of the PMOS region and the NMOS region may be different or substantially the same. For example, the semiconductor structure may include a NMOS region having a structure the same as that shown in FIG. **20** and a PMOS region having a structure the same as that shown in FIG. **25**.

In a gate-all-around (GAA) transistor structure, a numbers of nanostructures may be formed over a substrate and are surrounded by a gate structure. The nanostructures may be used as channel regions, and source/drain structures may be formed at opposite sides of the channel regions and contacts may be formed over the source/drain structures. The lower nanostructures closer to the substrate may have relatively larger parasitic resistance and lower drive current due to the longer current paths from the channels to the contacts and current crowding.

In some embodiments of the present application, contacts (e.g. the contacts **280a**, **380**, **380-1**, **380-2**, **480**, **480-4**, **480-5**, **380a**, and **380b**) are formed extending to a lower portion of the semiconductor structure (e.g. closer to, or extending into, the top surface of the substrate **202**), so that the current paths from the lower channels to the contacts can be shortened. Accordingly, the parasitic resistance may be reduced and the drive current may be improved. In addition, since the contacts extends into, or between, the source/drain structures/regions, the contact areas between the contacts and the source/drain structures/regions are therefore increased, and the resistance of the semiconductor structures may be reduced.

Furthermore, the contacts formed in the NMOS region and PMOS region may be made of materials providing different stresses. In some embodiments, the contacts (e.g. the contacts **380a** and **380b**) can provide additional tensile stress and the compressive stress to the channels in the

NMOS region and the PMOS region, and therefore the performance of the semiconductor structure **500** may be improved.

Embodiments for forming semiconductor structures are provided. The semiconductor may include nanostructures formed over a substrate and a gate structure wrapped around the nanostructures. In addition, a source/drain structure may be formed at one side of the nanostructure, and a contact formed over the source/drain structure. The contact may extend below a topmost surface of the source/drain structure, so that the current paths between the nanostructures and the contact may be reduced, and the performance of the semiconductor structure may be improved.

In some embodiments, a semiconductor structure is provided. The semiconductor structure includes a substrate and nanostructures formed over the substrate. In addition, the nanostructures includes channel regions and source/drain regions. The semiconductor structure further includes a gate structure vertically sandwiched the channel regions of the nanostructures and a contact wrapping around and vertically sandwiched between the source/drain regions of the nanostructures.

In some embodiments, a semiconductor structure is provided. The semiconductor structure includes a substrate and nanostructures formed over the substrate and vertically separated from each other. In addition, each of the nanostructures includes a first channel region and a source/drain region adjacent to the first channel region. The semiconductor structure further includes a gate structure wrapping around the first channel regions of the nanostructures and a contact formed over the source/drain regions of the nanostructures. The semiconductor structure further includes first inner spacers vertically sandwiched between the nanostructures and laterally separated the contact from the gate structure.

In some embodiments, a semiconductor structure is provided. The semiconductor structure includes a substrate and nanostructures longitudinally oriented along a first direction and vertically separated from each other. In addition, the nanostructures include channel regions and doped regions at opposite sides of the channel regions and a gate structure wrapping around the channel regions of the nanostructures. The semiconductor structure further includes a contact formed over the doped regions of the nanostructures and a gate spacer formed between the contact and the gate structure. In addition, the doped regions of the nanostructures partially extends under the gate spacer.

In some embodiments, a semiconductor structure is provided. The semiconductor structure includes a substrate and nanostructures formed over the substrate. The semiconductor structure further includes a gate structure surrounding the nanostructures and a source/drain structure attached to the nanostructures. The semiconductor structure further includes a contact formed over the source/drain structure and extending into the source/drain structure.

In some embodiments, a semiconductor structure is provided. The semiconductor structure includes a substrate and nanostructures formed over the substrate. The semiconductor structure further includes inner spacers sandwiched between the nanostructures and a source/drain structure adjacent to the inner spacers. The semiconductor structure further includes a contact formed in the source/drain structure and protruding from a top surface of the source/drain structure.

In some embodiments, a semiconductor structure is provided. The semiconductor structure includes a substrate and nanostructures formed over the substrate. In addition, the

nanostructures include channel regions and source/drain regions. The semiconductor structure further includes a gate structure wrapping around the channel regions of the nanostructures and a contact wrapping around the source/drain regions of the nanostructures.

The gate all around (GAA) transistor structures described above may be patterned by any suitable method. For example, the structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the GAA structure.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A semiconductor structure, comprising:
a substrate;
nanostructures formed over the substrate, wherein the nanostructures comprise channel regions and source/drain regions, and the channel regions comprise a first channel region and a second channel region;
a gate structure vertically sandwiched the channel regions of the nanostructures, wherein a portion of the gate structure is vertically sandwiched between a bottom surface of the first channel region and a top surface of the second channel region; and
a contact wrapping around and vertically sandwiched between the source/drain regions of the nanostructures.
2. The semiconductor structure as claimed in claim 1, further comprising:
inner spacers formed between the nanostructures, wherein the inner spacers are sandwiched between the contact and the gate structure.
3. The semiconductor structure as claimed in claim 2, wherein the inner spacers are in contact with the contact.
4. The semiconductor structure as claimed in claim 1, wherein a bottom surface of the gate structure is substantially aligned with a bottom surface of the contact.
5. The semiconductor structure as claimed in claim 1, further comprising:
silicide layers wrapping around the source/drain regions of the nanostructures.
6. The semiconductor structure as claimed in claim 1, further comprising:
a doped region formed in the substrate under the source/drain regions of the nanostructures.

7. The semiconductor structure as claimed in claim 6, wherein the contact is in direct contact with the doped region in the substrate.

8. The semiconductor structure as claimed in claim 1, wherein a top surface of the contact is higher than a top surface of the gate structure.

9. A semiconductor structure, comprising:

- a substrate;
- nanostructures formed over the substrate and vertically separated from each other, wherein each of the nanostructures comprise:
a first channel region; and
a source/drain region adjacent to the first channel region;
- a gate structure wrapping around the first channel regions of the nanostructures;
- a contact formed over the source/drain regions of the nanostructures;
- first inner spacers vertically sandwiched between the nanostructures and laterally separated the contact from the gate structure; and
a gate spacer formed over a topmost one of the nanostructures, wherein a top surface of the contact is higher than a top surface of the gate spacer.

10. The semiconductor structure as claimed in claim 9, wherein each of the nanostructures comprise:

- a second channel region,
- wherein the source/drain regions of the nanostructures are sandwiched between the second channel regions and the first channel regions of the nanostructures, and the nanostructures continuously extend from the first channel regions to the second channel regions.

11. The semiconductor structure as claimed in claim 10, further comprising:

- second inner spacers vertically sandwiched between the nanostructures,
- wherein the first inner spacers and the second inner spacers are in contact with opposite sides of the contact.

12. The semiconductor structure as claimed in claim 9, wherein the gate spacer vertically overlaps portions of the gate structure.

13. The semiconductor structure as claimed in claim 9, further comprising:

- a substrate source/drain region formed in the substrate, wherein the substrate source/drain region vertically overlaps the source/drain regions of the nanostructures.

14. The semiconductor structure as claimed in claim 13, wherein a portion of the contact is vertically sandwiched between the substrate source/drain region and the source/drain region of a bottommost one of the nanostructures.

15. A semiconductor structure, comprising:

- a substrate;
- nanostructures longitudinally oriented along a first direction and vertically separated from each other, wherein the nanostructures comprise channel regions and doped regions at opposite sides of the channel regions;
- a gate structure wrapping around the channel regions of the nanostructures;
- a contact formed over the doped regions of the nanostructures; and
a gate spacer formed between the contact and the gate structure, wherein the doped regions of the nanostructures partially extend under and vertically overlaps the gate spacer.

16. The semiconductor structure as claimed in claim 15, wherein the contact has curved sidewall portions.

17. The semiconductor structure as claimed in claim 16,
further comprising:
inner spacers covering the curved sidewall portions of the
contact.

18. The semiconductor structure as claimed in claim 15, 5
further comprising:
inner spacers vertically sandwiched between the nano-
structures and laterally sandwiched between the contact
and the gate structure.

19. The semiconductor structure as claimed in claim 18, 10
wherein the gate spacer is wider than the inner spacers in a
cross-sectional view along the first direction.

20. The semiconductor structure as claimed in claim 18,
wherein the contact is in direct contact with a bottommost
one of the inner spacers. 15

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